

Celestial Navigation Plugin for OpenCPN

Introduction

The plugin helps determine your position (Latitude, Longitude) through sextant "altitude" observations of the Sun, Moon, planets, or stars. The plugin can work together with an OpenCPN DR Route (Track), GPS Track, or without a route. The DR track could have been generated by Weather Routing. Celestial Navigation serves as a reliable independent backup when GPS is unavailable, spoofed, or fail during long ocean voyages. It can also be used to supplement GPS as an independent source. It displays a Circle of Position (COP) for each sight. On a small area scaled plot, COP appears as a Line of Position. It calculates Intercept and Azimuth, calculates and displays a FIX from multiple celestial body sights. And will calculate a Running Fix by advancing Lines of Position based on Vessel distance traveled and course between sights.

Notably new to release 2.8.x are the following

- UTC time capture capability. Freeze the UTC clock display at the Sight's "Mark". UTC date and time can be copied into a new or existing sight.
- Sun, Moon, Planet and Star Finder" which can also be used to identify celestial body location and potential 2 and 3 body sights for a fix
- Event Calendar (Sunrise, Sunset, Twilight, etc.)
- An abbreviated Nautical Almanac.
- A Sight evaluation tools useful to assess the quality of a Run of Sights, identify sight outliers, or used to assess your personal sight error
- Coastal Navigation using a Sextant. Vertical Angle of a known Landmark to develop Circle of Position, and Horizontal Angle between 3 landmarks to develop a fix based on angles between the 3 landmarks.
- Circle of Position plotted based on a Bearing of Sunrise or Sunset. A low precision method to estimate position without Sextant.
- Lunar Distancing Lunars Planning tool "Star/planet/moon/sun Finder" geared towards selecting good pair of bodies for a Lunar Sight.
- Collimation. Measure the Sextant's Index arm's accuracy by measuring the angle between 2 stars and comparing to their known angle.

Additional documents

- Celestial Navigation positions accuracy is documented in [Accuracy, Goals, Precision and Testing](#). The last test completed, April 2025.
- Abbreviations, Definitions and Measurements ([Celestial Navigation Definitions](#) page) for terms and definitions used below.
- Training documentation for Lunars: [Lunar Distance Use Case](#).

How to Create and Maintain Sextant Altitude Sights

Topics Covered

1. Create New Altitude Sight (Use Case 1)
2. Duplicate and Edit Sight
3. Fix and Running Fix.
4. Mark Time (new feature 2.8.x).
5. Analyze Sights. Are some of the sights outliers? (new 2.8.x feature)
6. Event and Sun, Moon, Star Finder. (new 2.8.x feature)
7. Planning for the ETA for Meridian Transit using Event Planner (use case 2)
8. Line of Position and Estimated Position without Sextant or GPS.
9. Coastal Sextant Sights. Horizontal Sight. Fix from 3 Landmarks.
10. Coastal Sextant Sights. Vertical Sight.

How to use Celestial Navigation Plugin for Sextant sights.

This How to guide, will show how to use this plugin through 3 use cases. The assumption is that you have overall familiarity of Navigation, Sextant usage, and some Celestial Navigation Theory and Logic. You won't need knowledge on how to do a sight reduction, Cel Nav plugin does that for you. Understanding terminology will be useful. There is another document that discusses Theory and Logic

1. Create an Altitude Sight, Plan sights for a future ETA/Position ("Star finder"), Evaluate Sights taken.
2. Find Position using Coastal Navigation Sextant Sights
3. Find UTC time with a Lunar Sight, including planning for a Lunar. Use the Sextant collimation tool (same methodology as Lunars) to assess your Sextant's accuracy at large angles.

Cel Nav Plugin is based on a Hub and Spoke Model (Illustration 1, next page). Click on the Open CPN Cel Nav Plugin icon, the first form you see is the "Celestial Navigation Sights" form. It's the central hub and shows the sights you've input. Each field (eyes, Type, Body, Time, Measurement, Color) can be sorted. From there you screen navigate to the forms like "Input", "edit", "duplicate". When you close a form, you are returned to the "Sights" form. To exit the plugin, close the "Sights" form.

List of all sights

Type

Body

Time (UTC)

Measurement

Color

Altitude

Kochab

2025-07-04 21:10:34

48° 04.4000'

rgb(33, 255, 6)

Altitude

Antares

2025-07-04 20:45:47

25° 50.3000'

rgb(253, 128, 8)

Altitude

Regulus

2025-07-04 20:39:23

26° 58.6000'

medium violet red

Altitude

Sun

2025-06-06 11:08:53

25° 00.0000'

orange red

Altitude

Saturn

2025-06-05 10:00:00

40° 00.0000'

violet red

Altitude

Moon

2025-06-05 00:00:00

42° 49.9981'

yellow green

Altitude

Sun

2025-06-02 13:48:28

45° 00.0000'

pink

Altitude

Sun

2025-04-23 20:00:00

114° 14.5999'

turquoise

Altitude

Sun

2025-04-23 20:00:00

65° 55.0000'

turquoise

Altitude

Sun

2025-04-23 20:00:00

0° 00.0000'

turquoise

Altitude

Luniter

2025-04-22 03:50:45

34° 49.2000'

medium violet red

New

Fix...

Duplicate

Clock Offset

Edit

Eclipses...

Delete

Generate Almanac...

Delete All

Documentation

Horizon Event...

PDF Documentation

Coastal Sextant...

Sun & Moon...

Lunar Tools...

Analyze Sights...

Hide Sights

Time integrity

Computer local

14:27:53.9 EDT — America/New_York (UTC-04:00)

UTC

18:27:53.9 UTC

GNSS/NMEA UTC

Not available — no valid RMC/ZDA received

System — GNSS

—

System clock

chrony status unavailable

Sight correction

+0 s

Mark time

Copy marked UTC

Clock is live

GNSS comparison is informational; it never changes sight times or Clock Offset.

Time is constantly changing.

Hide Time

Press Hide Sights, to just show the buttons.
When hidden, there is a button to Show Sights

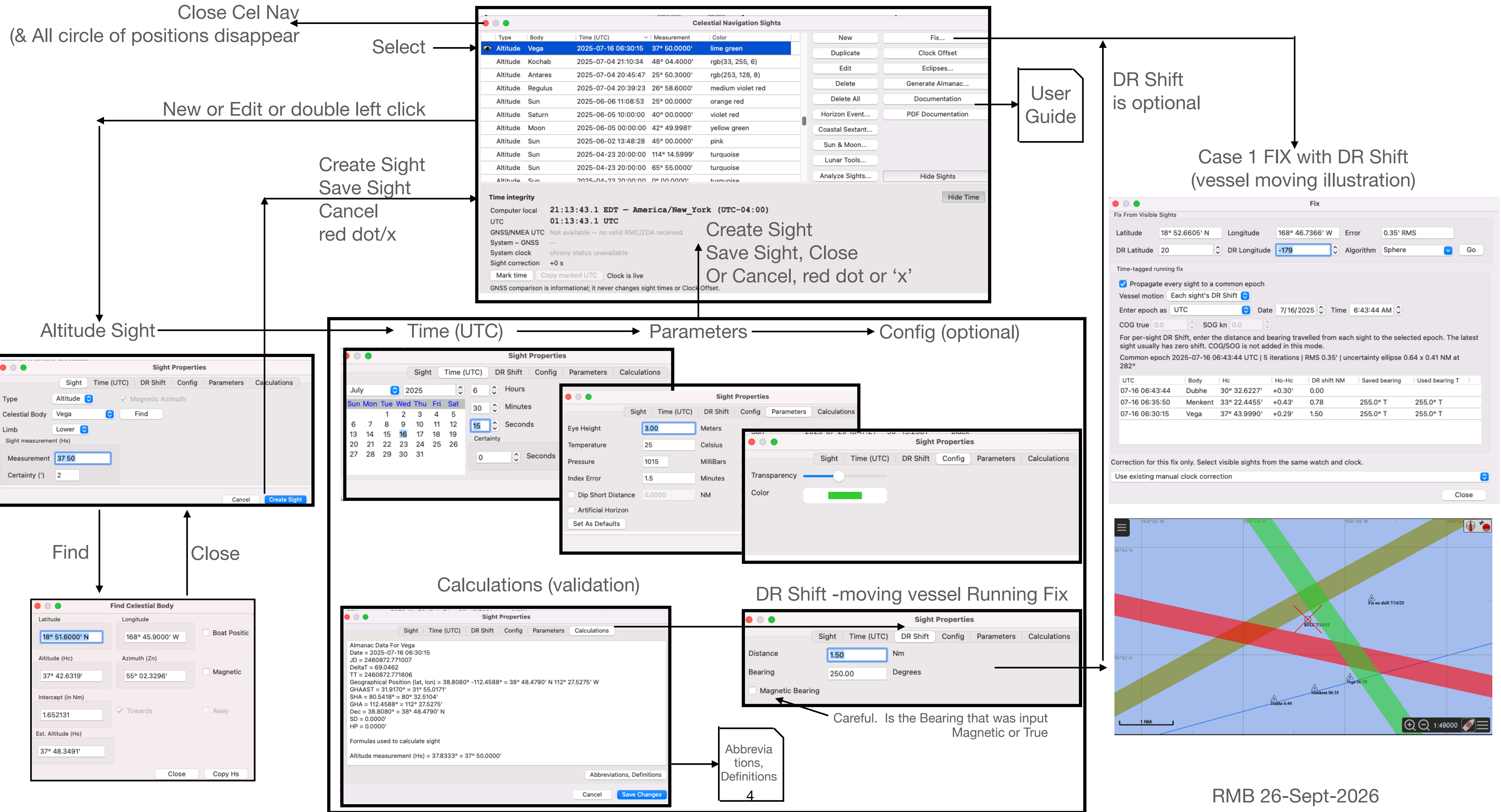
Press Hide Time to hide the Time integrity section
and show more sights. When Time is hidden,
there is a button to Show Time.

The Computer local and UTC time is constantly changing

Mark Time will hold the time's display so you read it. Then
copy and then paste UTC date and time into any Sight. And
then you can release the Hold. The paste no longer available

Illustration 1 Celestial Navigation Plugin Hub and Wheel

24 Sep 2026



We will use the following example to demonstrate How to use the plugin.

Use Case 1 SW of Hawaii UTC 16 Jul 2025

A Cruise SW of Hawaii.

- Course 255°T, Speed 6 knots. No set or drift.
- Height of Eye (HOE) is 3.0 meters.
- UTC July 16, 2025 at ~06:00:00 the ETA for the evening sight.
- Estimated DR is Lat 18° 50N, 168°48'W.

The plan is to do a 3 body sight to establish a fix.

Determined

Temperature 25C,

Pressure 1015 mb,

Index error +1.5 minutes.

First sight is Vega with a run of 3 sights

UTC Date and Time: 2025-07-16 06:30:15.

Avg Hs (altitude) 37° 50'.

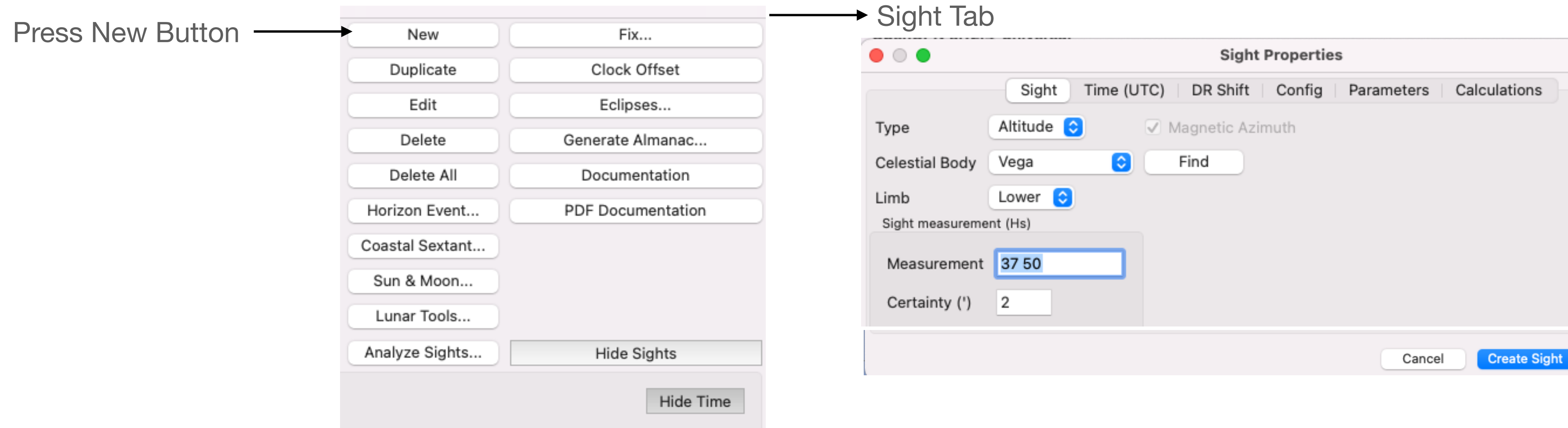
DR position @ 0630: 18°51.6N, 168°45.9W. It was plotted based on the Vessel's log (distance) and Course.

Take a 2nd body "Menkat" about 5 1/2 minutes later (run of 3 sights)

Then a 3rd body "Dubhe" about 8 minutes later (run of 3 sights)

Use Case 1 SW of Hawaii UTC 16 Jul 2025

1. Create Sight: Sight Properties > Sight Tab



Sight Measurement (Hs) is an angular measurement of an astronomical body using a sextant. It can be one of these types:

- Altitude: Angle of the body above the horizon (most common method)
- Lunar: angle from the moon to another body for TIME and assessing Watch error (number of min-sec fast or slow).
- Azimuth: (Future. Some framework is available)

Celestial Body: select the Sun, Moon, Venus, Mars, Jupiter, Saturn, Mercury, and 57 Navigation Stars plus Polaris that are visible in either the Northern or Southern Hemisphere.

Limb: For altitude shots of the Sun and Moon, choose lower limb unless obscured; then choose upper limb.

Sight Measurement (Hs): Enter the sextant reading. 3 angle formats are supported: Degrees minutes, Degrees Minutes Seconds, Decimal Degrees. After input, the angle redisplay is per Open CPN's display settings. Back-sights are indeed supported.

Minutes of certainty determine the width of the plotted circle or line of position COP/LOP. If too small, it is hard to see. 0.1 min. is visible after zooming into a small area chart. 10+ min. visible after zooming out to a large area chart. Try the +/- on Chart Scale

Use Case 1 SW of Hawaii UTC 16 Jul 2025

1. Create Sight: Sight Properties > Time (UTC) Tab

Time (UTC) Tab Type over the defaults

The screenshot shows the 'Sight Properties' dialog box with the 'Time (UTC)' tab selected. The 'Sight' tab is also visible. The calendar shows July 2025 with the 16th selected. The time fields are set to 6 Hours, 30 Minutes, and 15 Seconds. The 'Certainty' field is set to 0 seconds. The 'Create Sight' button is highlighted in blue.

Time (UTC) Tab (Required)

Must be the exact moment of your sight measurement at UTC. Know the time zone you are taking your sights from. Know your watch error and Add (slow) or Subtract (fast) it to your watch time.

Date and Time defaults to the system clock date & time and time zone when a New sight form is opened. Date & Time needs to be the UTC date & time. Example: twilight sight is recorded at 20:25:35 April 22 at UTC-4, also known as ZD+4. The UTC is April 23, 20xx and UTC is 00:25:35. The next day!

Time (Hours, Minutes, Seconds) needs to be UTC after accounting for Watch error. Be aware of Daylight savings time. There is no Daylight Savings time for UT/UTC. A digital watch or quartz watch with UTC and local time is best. Be sure to manually adjust Time to correspond with the exact UTC moment the sight was taken. Example: If your quartz watch is 5 sec Fast, then convert to UTC and subtract 5 sec. If your watch is 1 minute 10 seconds slow, then ADD 1 minute 10 seconds to your UTC.

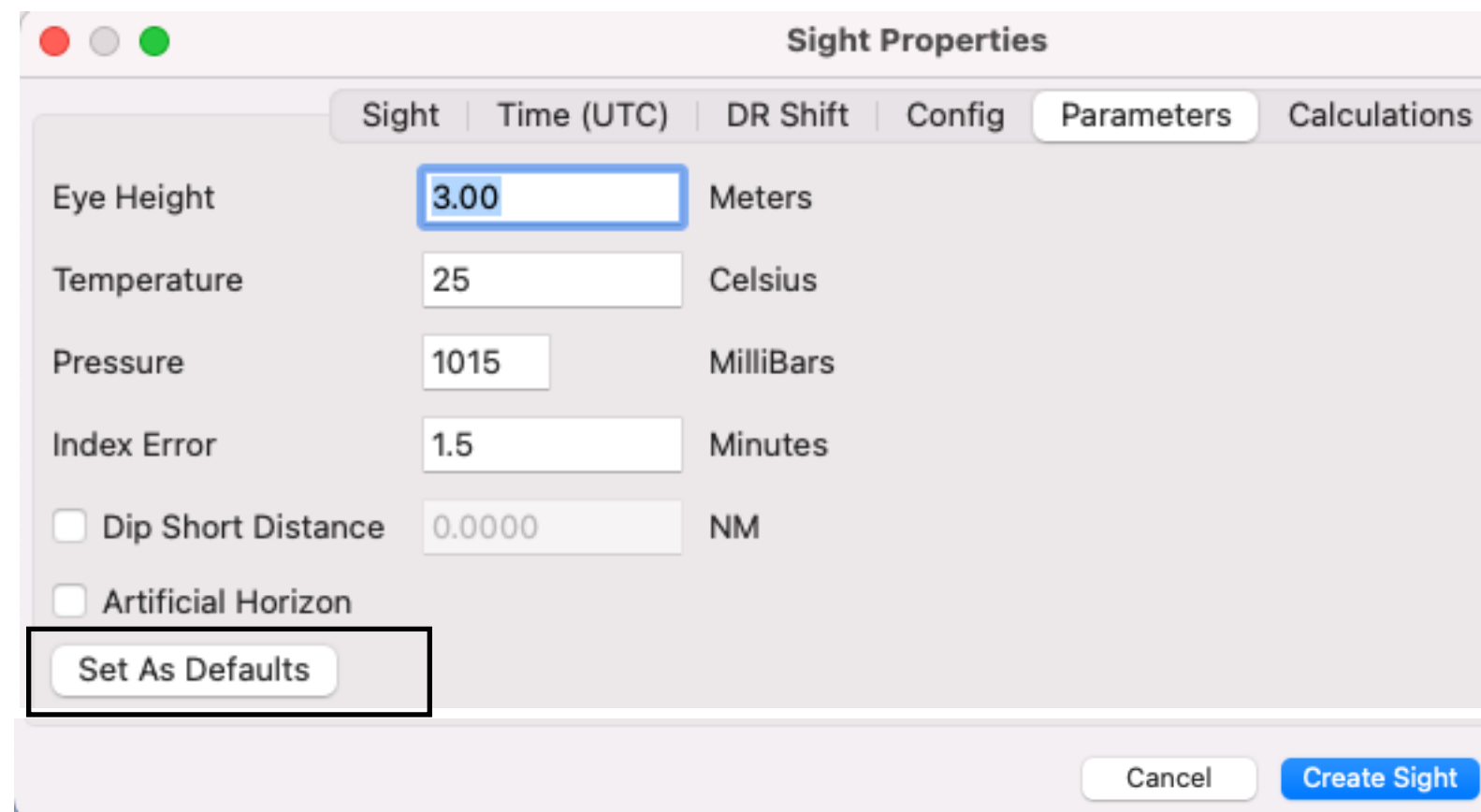
Recording: In preparation, set the stopwatch and record UTC time. When the sight is taken, say “Mark”, so the assistant can record stopwatch time and the sight body and altitude, or if you are solo, try using a recording timer app on your cell phone with a wired microphone. A plot of stopwatch time against Hs can show outlier sights. A new feature for 2.8.5.9 is that you can copy and paste the UTC time identified on the Celestial Sights form. An example will be shown.

Time Certainty: default is 0 seconds which shows a single darker line in the COP. A time Certainty of > 0 shows two darker lines, equivalent to that many seconds apart. For Lunar Sights, Time Certainty is used to bracket your watches's +/- error. The default of 86,400 sec can be changed. A COP or LOP is not shown for Lunar Sights. You can duplicate The lunar sight and EDIT the duplicates to type = altitude and see the COP/LOPs.

Use Case 1 SW of Hawaii UTC 16 Jul 2025

1. Create Sight: Sight Properties > Parameters Tab

Parameters Tab



The screenshot shows the 'Sight Properties' dialog box with the 'Parameters' tab selected. The dialog has a title bar with standard macOS window controls. Below the title bar are five tabs: 'Sight', 'Time (UTC)', 'DR Shift', 'Config', and 'Parameters' (which is active), and 'Calculations'. The 'Parameters' tab contains several input fields and checkboxes. The 'Eye Height' field is set to '3.00' with 'Meters' as the unit. The 'Temperature' field is set to '25' with 'Celsius' as the unit. The 'Pressure' field is set to '1015' with 'MilliBars' as the unit. The 'Index Error' field is set to '1.5' with 'Minutes' as the unit. There are two checkboxes: 'Dip Short Distance' (unchecked) and 'Artificial Horizon' (unchecked). The 'Dip Short Distance' field is set to '0.0000' with 'NM' as the unit. A 'Set As Defaults' button is located at the bottom left of the parameter section. At the bottom right of the dialog are 'Cancel' and 'Create Sight' buttons.

Parameter	Value	Unit
Eye Height	3.00	Meters
Temperature	25	Celsius
Pressure	1015	MilliBars
Index Error	1.5	Minutes
Dip Short Distance	0.0000	NM

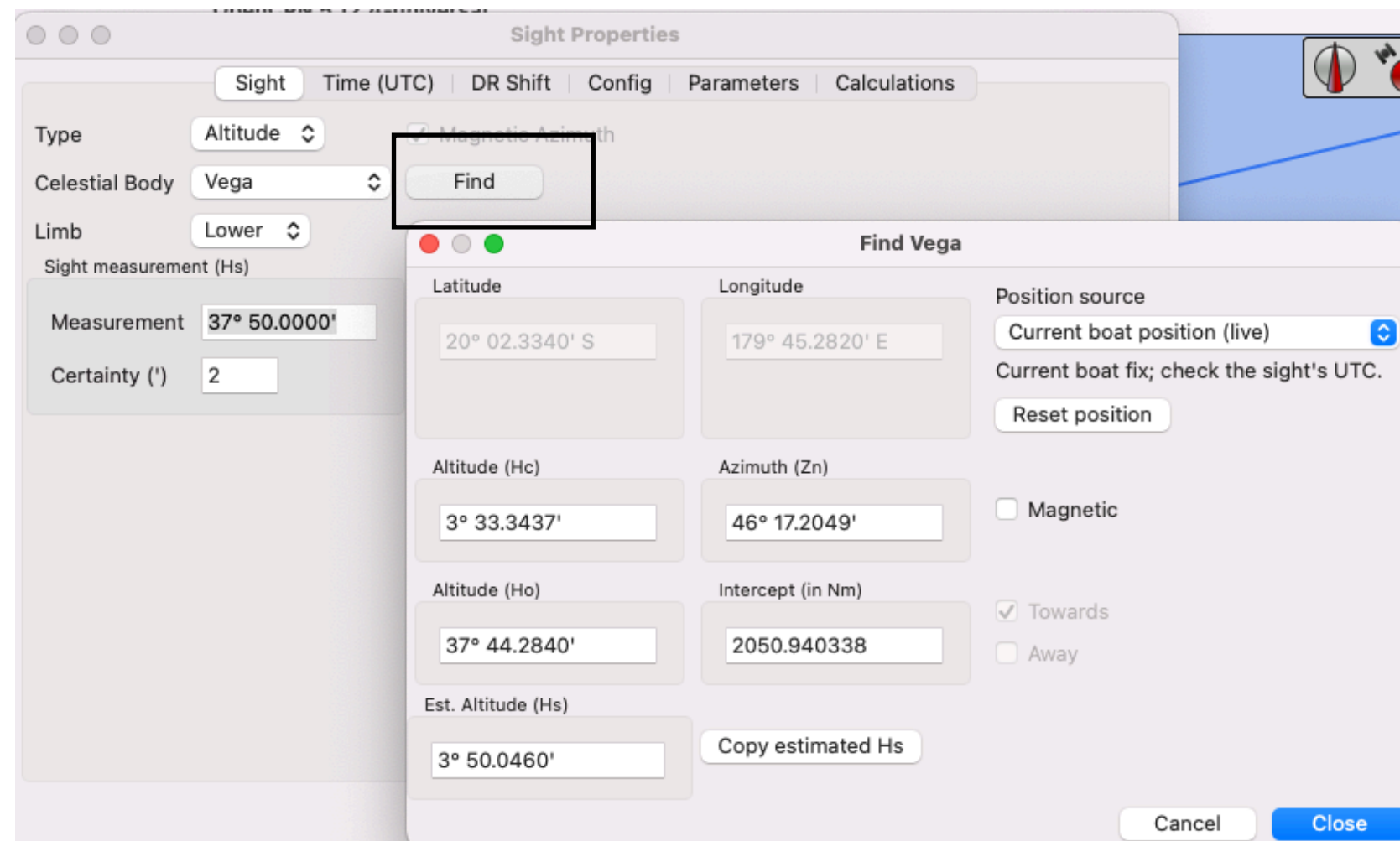
To use the same Parameters for future sights, input parameters and then press, “Set as Defaults” button.

For a natural horizon sight, HOE is the only thing you need.

For Dip short sights, click on the Dip Short radial button, and input the distance in nautical miles from your DR to the shoreline where the celestial body will or was brought down to. Use OpenCPN chart to measure the distance and bearing. To remove Dip short, zero out the distance and unblock the radial button.

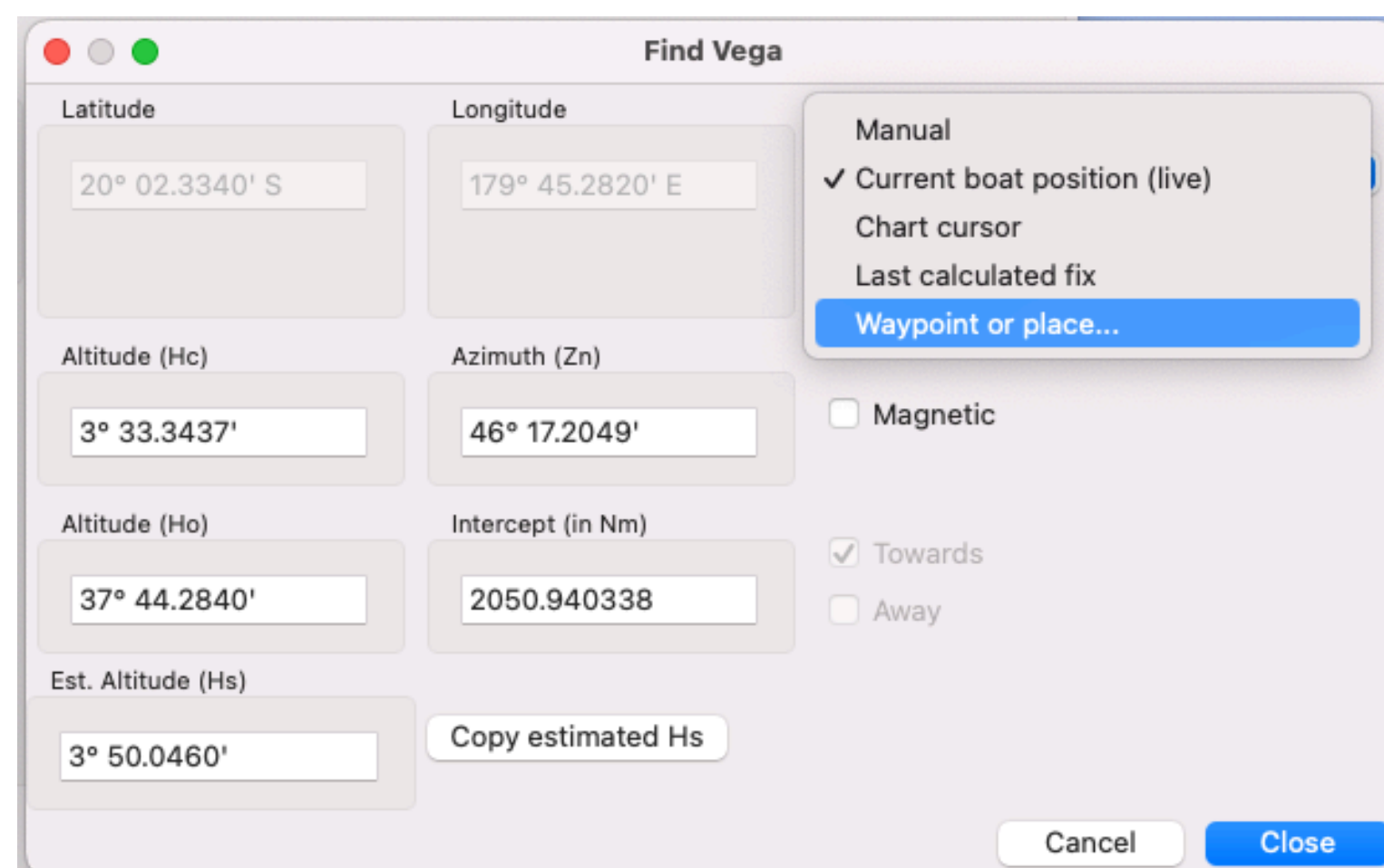
For Artificial Horizon sights, click on or off the radial button.

1. Create Sight: Sight Properties > Sight > Find

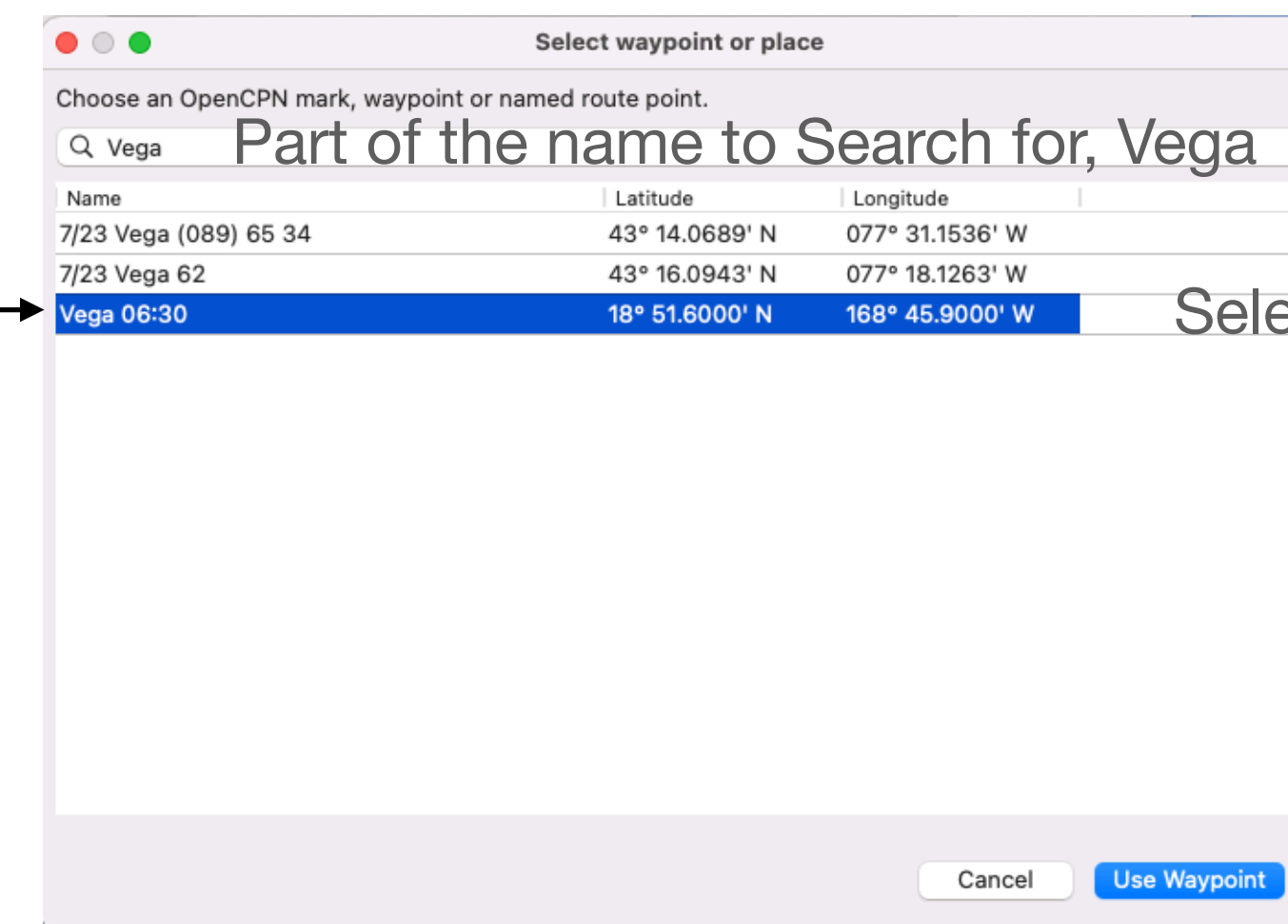


Position source drop down menu
Default value is Boat Position.

Latitude and Longitude defaults to Boat's position. If GPS is attached to OpenCPN, Lat/Lon will be the defaulted but the boat is moving. So, DR will constantly change. You'll need to freeze it for the DR position of the Sight. Then you are done. But, if you don't have GPS, drop a waypoint where the sights were taken. Then on the Find form, You can select that waypoint for the sight's DR position.



From the drop down menu
Select Waypoint



Select Vega 16-30

Use Case 1 SW of Hawaii UTC 16 Jul 2025

1. Create Sight: Sight Properties > Sight > Find (continued)

Caution. If you use manual input for Lat and Lon, and your Open CPN version is 5.12, you must input E or W for longitude. Version 5.14 fixed that bug.

Close on the Find form and
You'll return to Sight tab

FIND. When button is pressed, the sight input data is "reduced" using the Law of Cosines after you've changed DR Latitude & Longitude
Computed Altitude (Hc) the altitude the celestial body should have per Almanac data (calculate form) ***and*** the DR Lat/Lon location is roughly right.
Azimuth (Zn), is the bearing (True or Magnetic) to the body. The Magnetic uses the WMM World Magnetic Model
Altitude Observed Ho Altitude after adjustments are made for IC, Dip, Refraction, Parallax, Temp/Pressure, etc. It is apples to apples to the Computed Altitude (Hc).
Intercept, the distance (in nautical miles) from the vessel's DR position towards ($H_o > H_c$) or away ($H_o < H_c$) from the body's LOP.
Estimated Altitude (Hs) new feature v 2.4.66. An estimate for the sextant sight (Hs). Use the estimate to pre-set Sextant before taking sights. After preset, the body should be just about there against the visible horizon. Very useful for high altitude sights which are difficult to bring down to the horizon by moving index arm/rotating micrometer while observing the body and horizon.
Copy Estimated Hs new feature v 2.4.66. Copies the Estimated Hs into the Hs Measurement field. Useful for Backyard Lunars.

Press "Close" to just close the Find form. You are Returned to "Sight Properties>Sight" tab...Sight not saved yet.

Use Case 1 SW of Hawaii UTC 16 Jul 2025

The screenshot shows a software window titled "Sight Properties" with several tabs: "Sight", "Time (UTC)", "DR Shift", "Config", "Parameters", and "Calculations". The "Sight" tab is active. Inside the tab, there are several input fields and a checkbox. The "Type" field is set to "Altitude". The "Celestial Body" field is set to "Vega", with a "Find" button next to it. The "Limb" field is set to "Lower". Below these, there is a section for "Sight measurement (Hs)" containing a "Measurement" field with the value "37° 50.0000'" and a "Certainty (')" field with the value "2". At the bottom right of the window, there are two buttons: "Cancel" and "Create Sight". The "Create Sight" button is highlighted with a red rectangular box.

You are Returned to “Sight Properties>Sight” tab...

Press Create Sight (or Save Sight) to save your work.

If you press Cancel, the most recent changes are not saved.

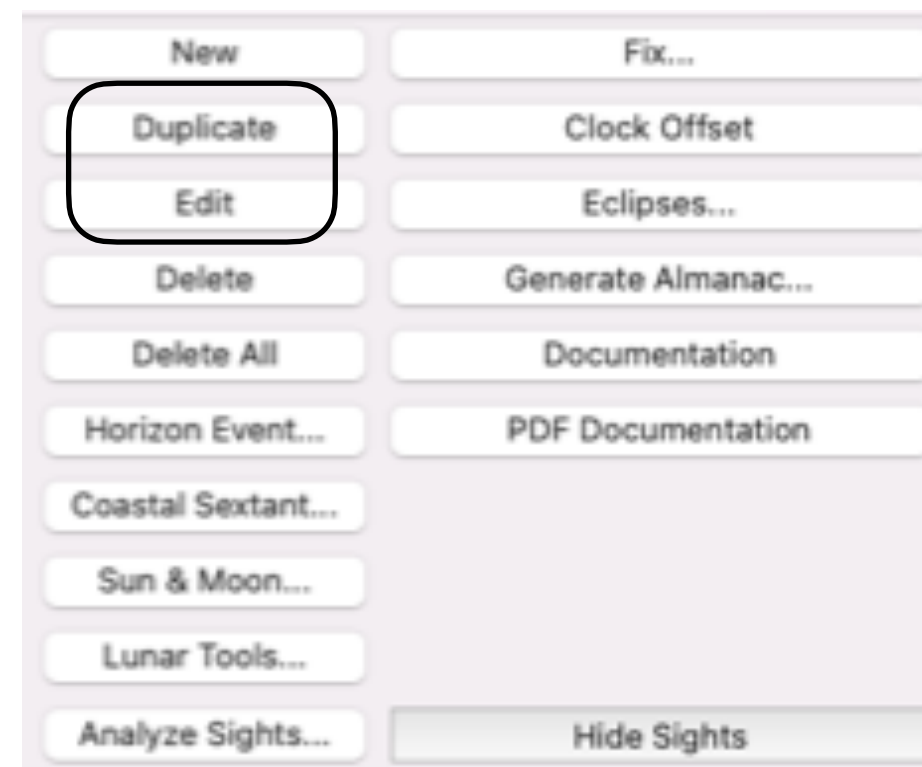
Use Case 1 SW of Hawaii UTC 16 Jul 2025

2. Duplicate and Edit Sights

There are 2 additional sights to input. Use the Duplicate and Edit capability that is available on the Celestial Sights

C 255T S 6.0	UT July 16 2025	Hs	DR	distance (nm)	Bearing °T
Vega	06:30:15	37° 50.0'	18 51.6'N 168° 45.9' W	1.5	255
Menkent	06:35:50	33° 28.8'	18° 51.4N 168° 46.6W	0.78	255
Dubhe	06:43:44	30° 39.0'	18° 51.2N 168° 47.4 W	0	0

A DR was calculated for the 3 sights. Because the time interval is ~ 20 minutes and speed is 6 knots (somewhat slow), a separate DR really isn't critical for many navigators. If doing a Sun-Run-Sun, you'll definitely be tracking a DR. However, we will next demonstrate the full capability of a 3 body fix or a Running Fix. Thus we need a different DR for each sight. Open CPN makes it pretty simple to estimate a DR. If you have a Log as part of your instruments, the distance traveled is right there (no calculation). Then use the Measure Tool to measure the distance traveled along your Course, and then Drop (not insert) a waypoint. If you are doing a Sun (early morning) -Run- then another Sun, there will be a couple hours (distance 12 nm) between the sights. You'll probably want a DR to document your speed and distance traveled on your OCPN DR course plot.



The Duplicate Button is a quick way to copy a sight

- Select Vega, press Duplicate.
- Select Vega again, and press Duplicate.

Now you have 3 sights that are identical to the Vega Sight

Select one of the Vega sights that was duplicated. Select press EDIT. Then change the Body to Menkent Hs for Menkent, and DR for Menkent. Take the other Vega duplicate and do the same for Dubhe. Use the above data to Edit the 2 additional Sights.

Use Case 1 SW of Hawaii UTC 16.Jul 2025

3. Fix or Running Fix (Traditional Cel Nav Method “Each sight’s DR Shift”)

The idea of the running fix is that you Advance the 1st sight to the last sight by the Distance traveled and Bearing (course) from 1st to last sight. So, Vega is 1.5 nm and 255 bearing to Dubhe. Be careful what bearing you meant. True is the default (that’s what is most frequently used offshore), but was your bearing Magnetic? Then you advance the 2nd sight (Menkent) to the last sight Dubhe. Menkent is .78nm bearing 255 degrees to Dubhe. The last sight Dubhe, the Distance and Bearing is 0 for now.

Vega

Sight Properties

SightTime (UTC)DR ShiftConfigParametersCalculations

Distance

1.50

Nm

Bearing

255.00

Degrees

☐ Magnetic Bearing

Menkent

Sight Properties

SightTime (UTC)DR ShiftConfigParametersCalculations

Distance

0.78

Nm

Bearing

255.00

Degrees

☐ Magnetic Bearing

Dubhe

Sight Properties

SightTime (UTC)DR ShiftConfigParametersCalculations

Type

Altitude

☒ Magnetic Azimuth

Celestial Body

Dubhe

Find

Limb

Lower

Sight measurement (Hs)

Measurement

30° 39.0000'

Certainty (')

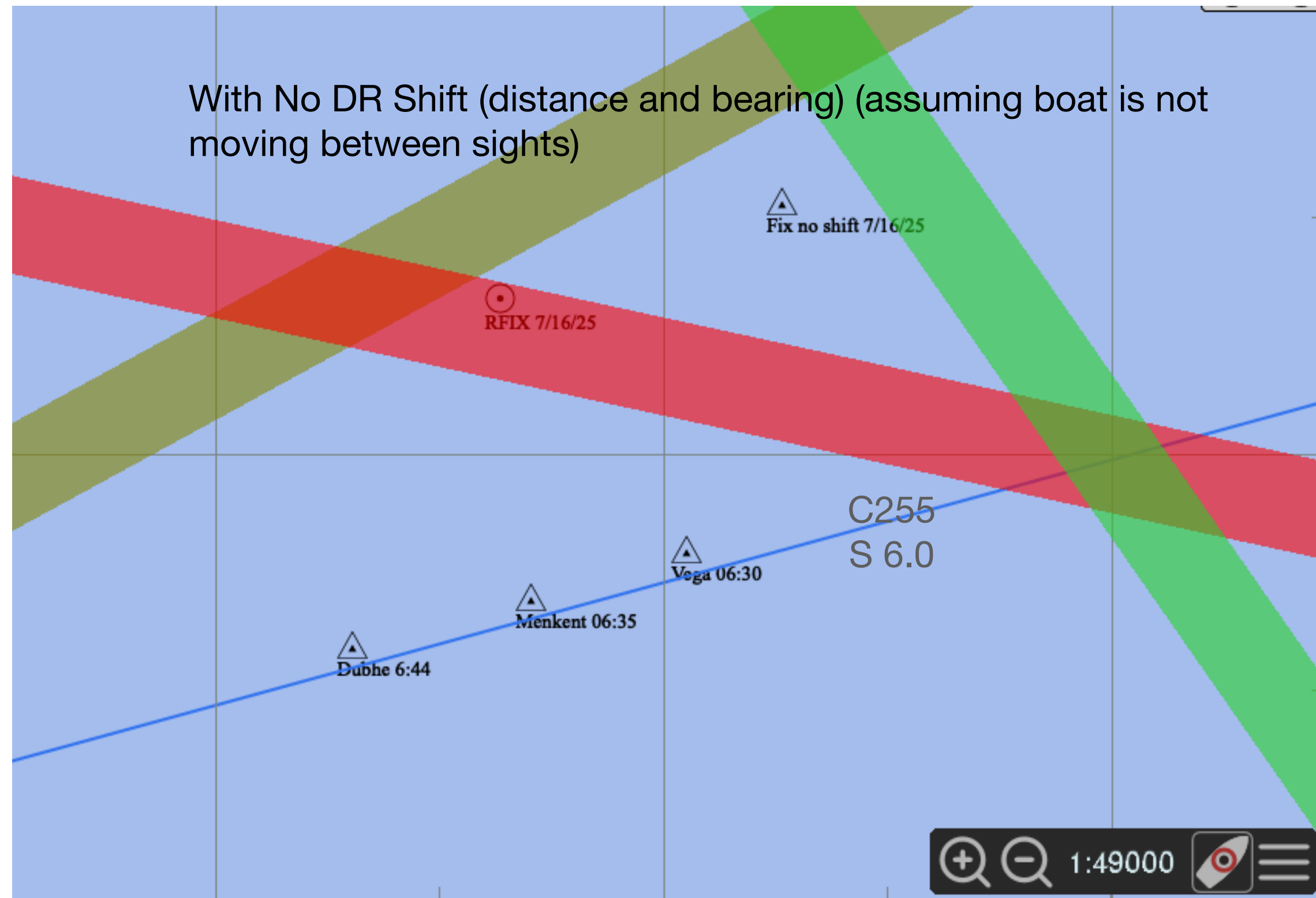
0.20

Cancel

Save Changes

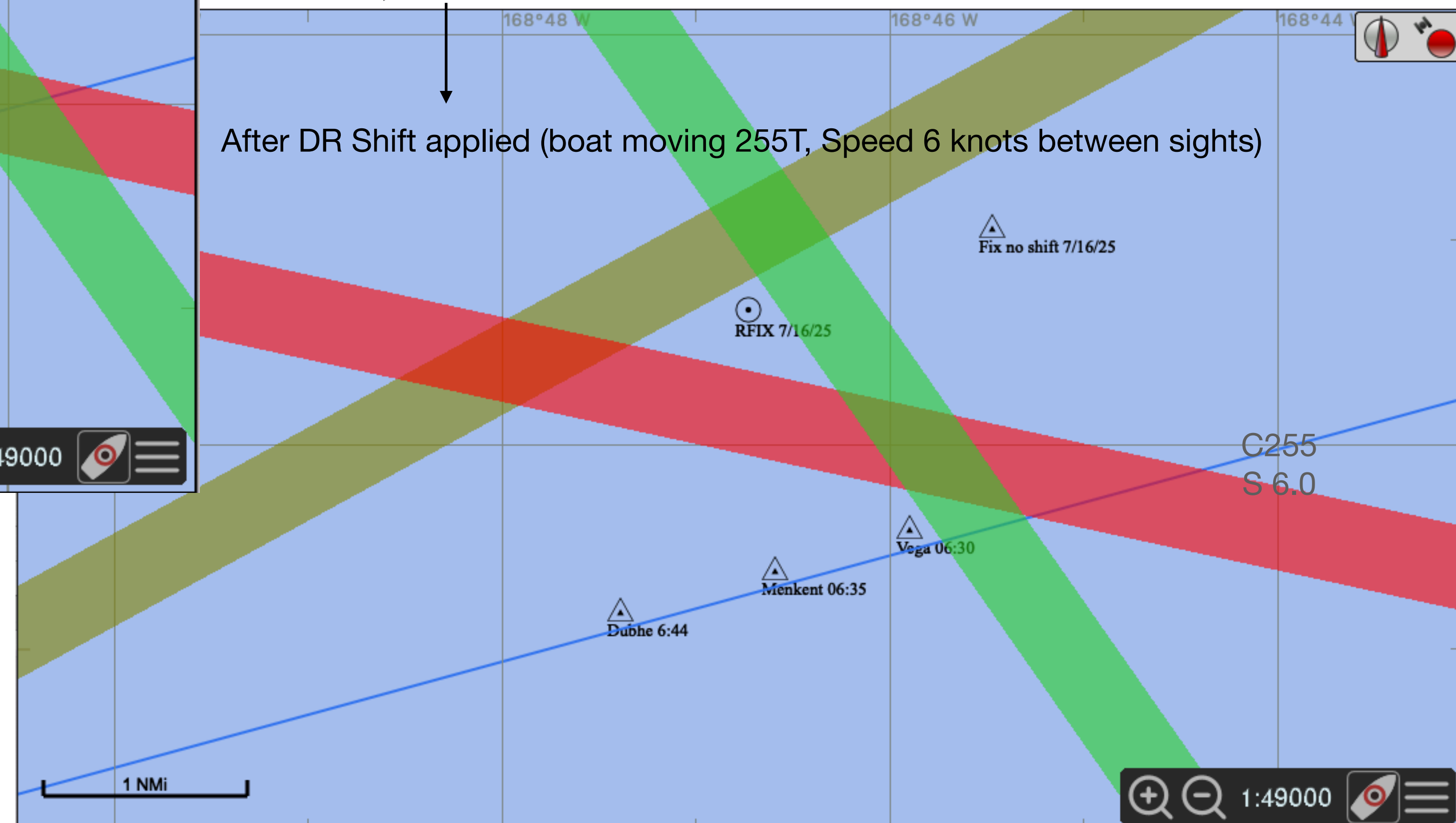
Use Case 1 SW of Hawaii UTC 16 Jul 2025

3. Fix or Running Fix (Traditional Cel Nav Method “Each sight’s DR Shift”)



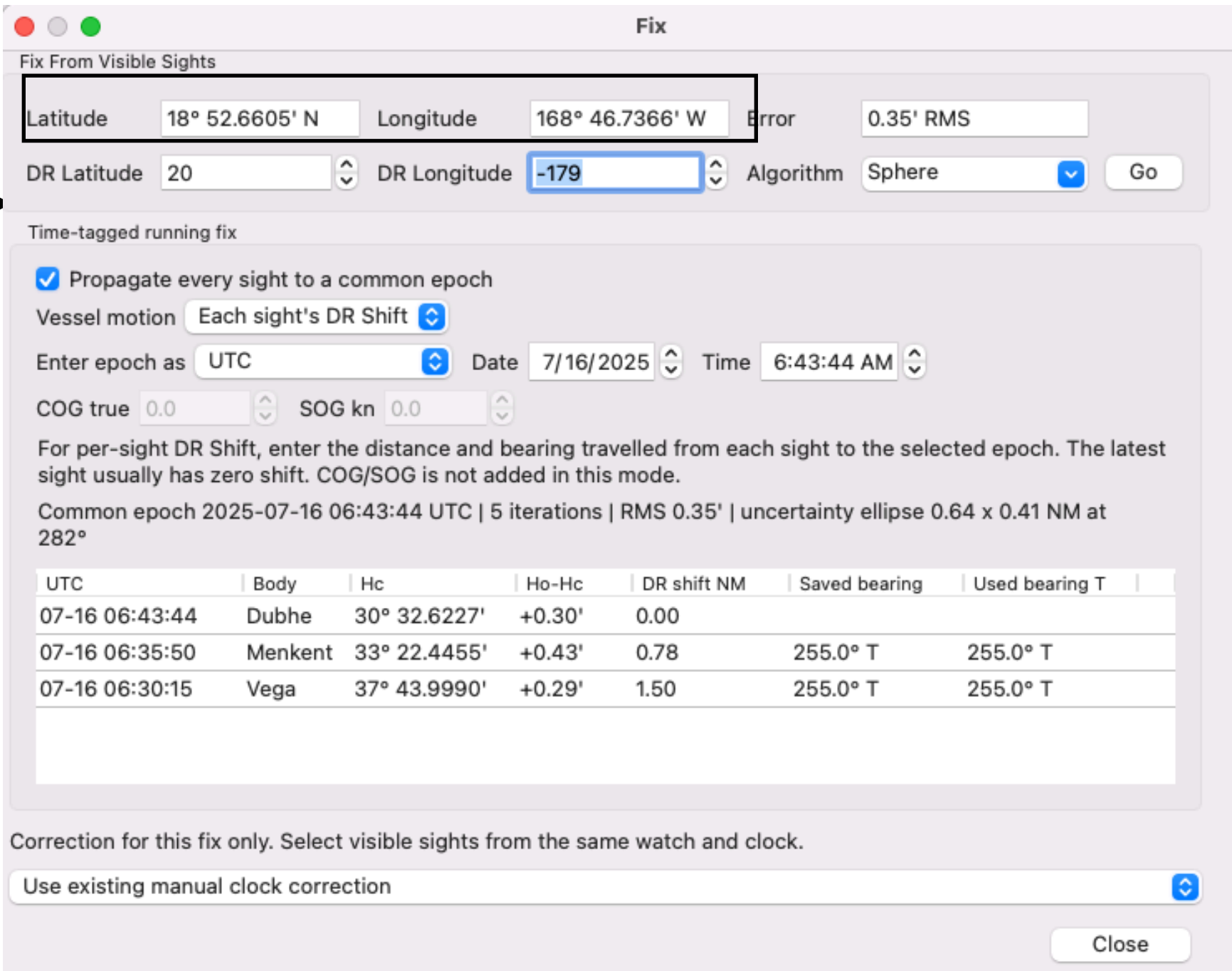
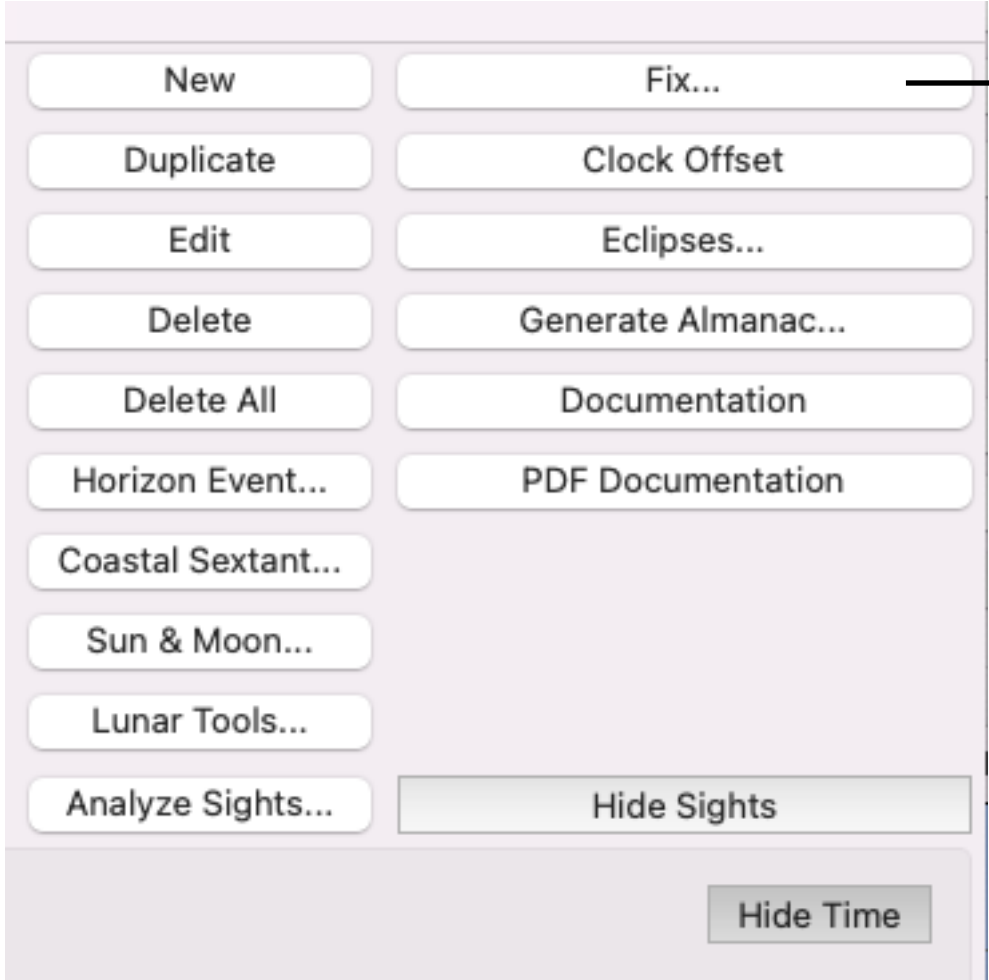
This illustration shows the Circle or lines of Position that immediately occur after creating a sight. They only occur if the “eyes” for the sight is “ON”.

On the left is the COP/LOP before input of DR Shift data. Below, is the COP/LOP redrawn with DR shift data



3. Fix or Running Fix (Traditional Cel Nav Method “Each sight’s DR Shift”)

Create Sight: FIX Form



Fix From Visible Sights

Latitude 18° 52.6605' N Longitude 168° 46.7366' W Error 0.35' RMS

DR Latitude 20 DR Longitude -179 Algorithm Sphere Go

Time-tagged running fix

☒ Propagate every sight to a common epoch

Vessel motion Each sight's DR Shift

Enter epoch as UTC Date 7/16/2025 Time 6:43:44 AM

COG true 0.0 SOG kn 0.0

For per-sight DR Shift, enter the distance and bearing travelled from each sight to the selected epoch. The latest sight usually has zero shift. COG/SOG is not added in this mode.

Common epoch 2025-07-16 06:43:44 UTC | 5 iterations | RMS 0.35' | uncertainty ellipse 0.64 x 0.41 NM at 282°

UTC	Body	Hc	Ho-Hc	DR shift NM	Saved bearing	Used bearing T
07-16 06:43:44	Dubhe	30° 32.6227'	+0.30'	0.00		
07-16 06:35:50	Menkent	33° 22.4455'	+0.43'	0.78	255.0° T	255.0° T
07-16 06:30:15	Vega	37° 43.9990'	+0.29'	1.50	255.0° T	255.0° T

Correction for this fix only. Select visible sights from the same watch and clock.

Use existing manual clock correction

Close

Latitude and Longitude is the Fix (or Running fix) derived by the form. It is using the sights listed on the Fix form. If this is wrong, you need to change the “eyes” on the “Sights” form to include what you require.

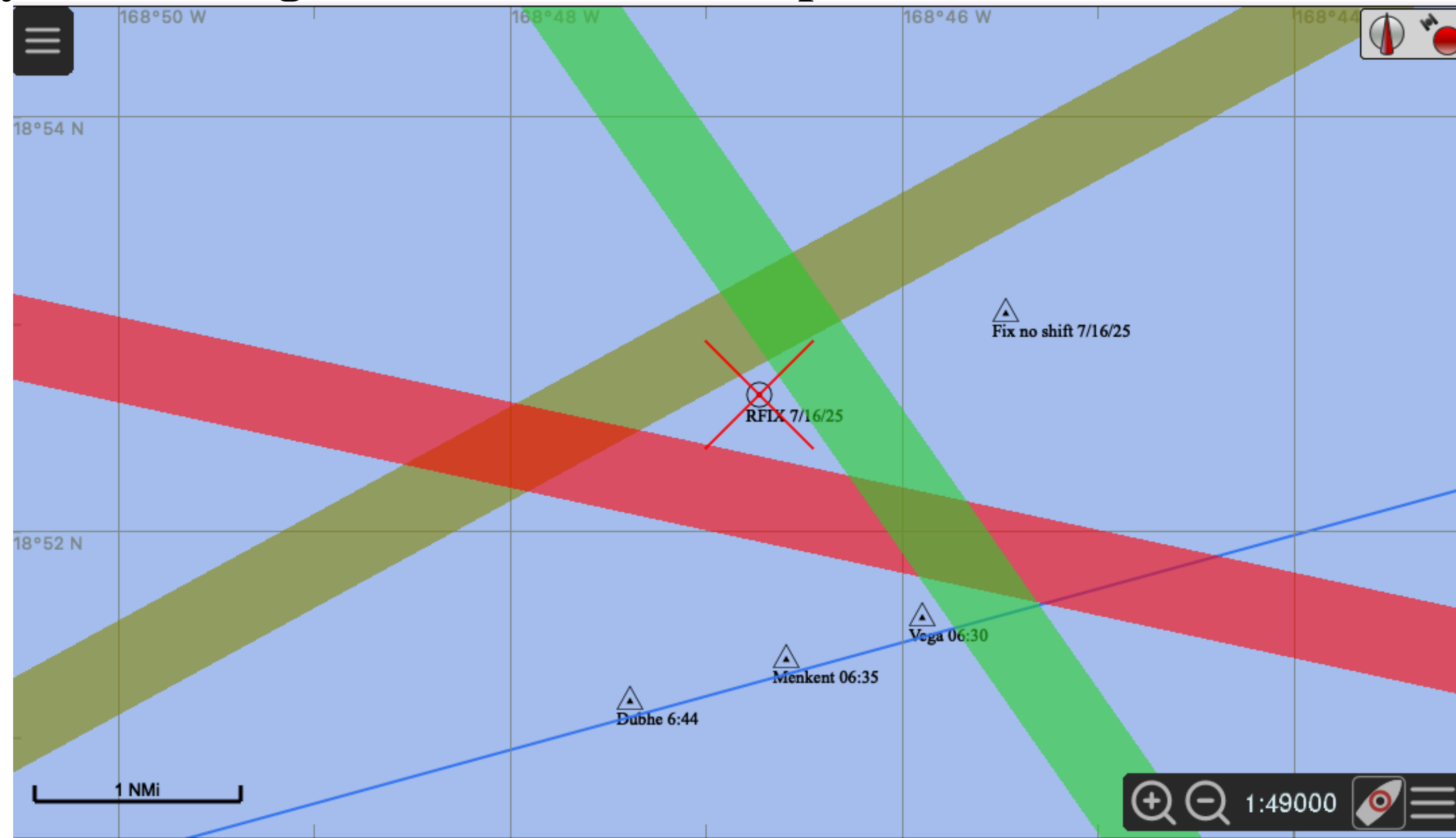
DR Latitude and DR Longitude is starting point to calculate the fix. It needs to be somewhat near the position (e.g. correct hemisphere). Default is “your boat position”. If wrong, type over the Dr Lat and Lo.

Vessel motion defaults to “Each sight’s DR Shift” if DR Shift and Bearing is non zero. The sights below show the DR Shift data.

Vessel motion defaults to “One COG, One SOG” if No DR Shift data is shown. You can input COG and SOG on this form.

3. Fix or Running Fix (Traditional Cel Nav Method “Each sight’s DR Shift”)

COP/LOP display and the Big X while Fix form is open



The Fix form temporarily places a Big Red X at the Fix (Running Fix). When the Fix form is closed, the Big Red X disappears.

If maintaining a DR route, Drop a Mark (say a circle) at the Big X, so you have record of it on the chart. It is up to you if you want to restart the Route from the Fix. General guide line is that you don't if the FIX is close enough to the DR Track / Route.

4. Mark UTC Time of the sight (new for 2.8.8)

A new Cel Nav 2.8.5.9 supports the process where the observer operates the Sextant and calls out Mark. When calling out Mark, the recorder press the Mark button. UTC and Local PC Time clock display is frozen. Then press “Copy marked UTC”. Create a New or Edit and existing sight. Enter the Sight Data, move forward to the Time Tab and press the “Use marked UTC....” to overwrite the UTC Date and Time. You can still edit the UTC Date and Time

Celestial Navigation Sights

Type	Body	Time (UTC)	Measurement	Color
Altitude	Kochab	2025-07-04 21:10:34	48° 04.4000'	rgb(33, 255, 6)
Altitude	Antares	2025-07-04 20:45:47	25° 50.3000'	rgb(253, 128, 8)
Altitude	Regulus	2025-07-04 20:39:23	26° 58.6000'	medium violet red
Altitude	Sun	2025-06-06 11:08:53	25° 00.0000'	orange red
Altitude	Saturn	2025-06-05 10:00:00	40° 00.0000'	violet red
Altitude	Moon	2025-06-05 00:00:00	42° 49.9981'	yellow green
Altitude	Sun	2025-06-02 13:48:28	45° 00.0000'	pink
Altitude	Sun	2025-04-23 20:00:00	114° 14.5999'	turquoise
Altitude	Sun	2025-04-23 20:00:00	65° 55.0000'	turquoise
Altitude	Sun	2025-04-23 20:00:00	0° 00.0000'	turquoise
Altitude	Jupiter	2025-04-22 03:50:45	34° 19.2000'	medium violet red

New

Fix...

Duplicate

Clock Offset

Edit

Eclipses...

Delete

Generate Almanac...

Delete All

Documentation

Horizon Event...

PDF Documentation

Coastal Sextant...

Sun & Moon...

Lunar Tools...

Analyze Sights...

Hide Sights

Time integrity

Computer local14:27:53.9 EDT — America/New_York (UTC-04:00)

UTC18:27:53.9 UTC

GNSS/NMEA UTCNot available — no valid RMC/ZDA received

System — GNSS—

System clockchrony status unavailable

Sight correction+0 s

Mark timeCopy marked UTCClock is live

GNSS comparison is informational; it never changes sight times or Clock Offset.

Time integrity

Computer local14:32:57.8 EDT — America/New_York (UTC-04:00)

UTC18:32:57.8 UTC

GNSS/NMEA UTCNot available — no valid RMC/ZDA received

System — GNSS—

System clockchrony status unavailable

Sight correction+0 s

ReleaseCopy marked UTCLocal and UTC clocks held at the marked instant

GNSS comparison is informational; it never changes sight times or Clock Offset.

Time is constantly changing.

Press “Copy marked UTC” copies the date and time for a sight

Sight Properties

SightTime (UTC)DR ShiftConfigParametersCalculations

September2026

14Hours

32Minutes

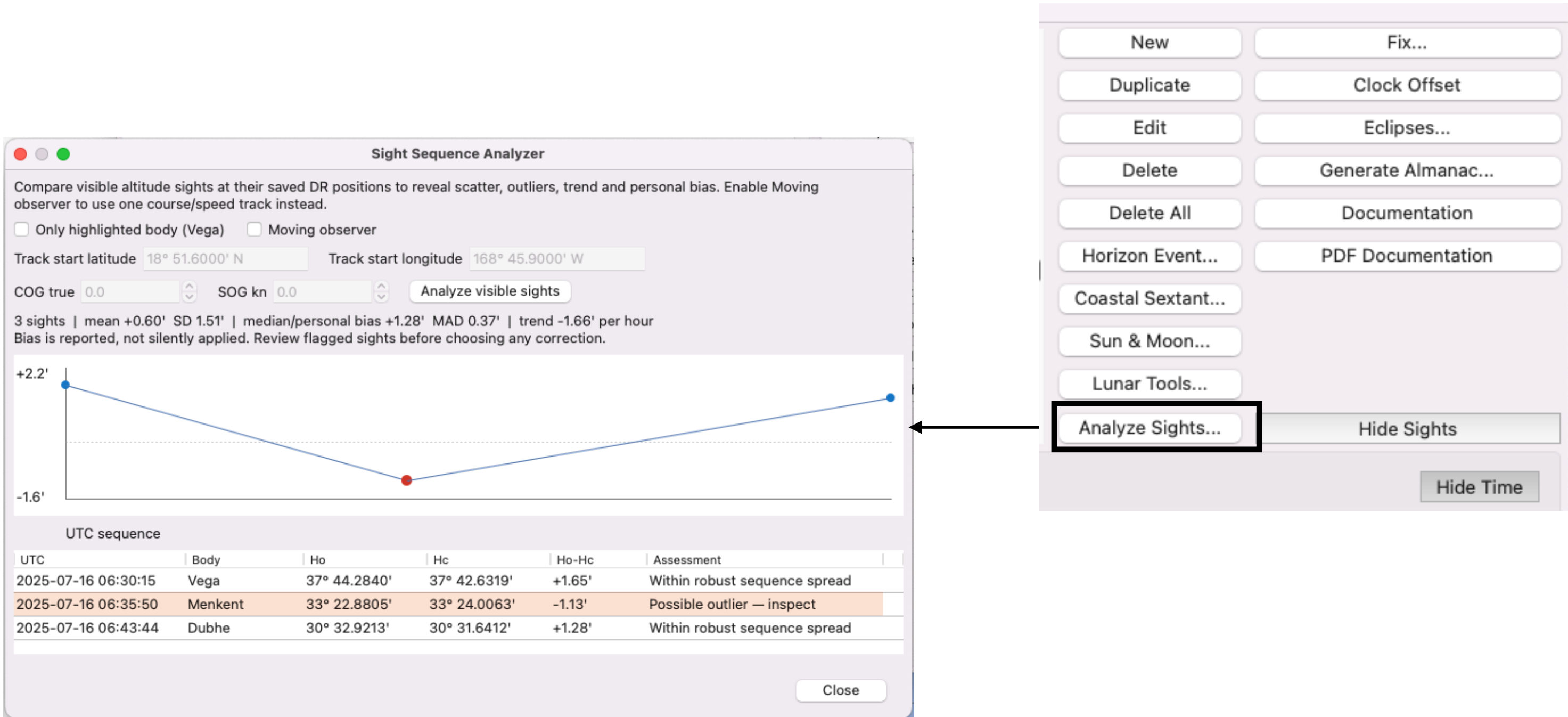
57Seconds

Certainty0Seconds

Use marked UTC: 2026-09-17 18:32:57

5. Analyze sights using Sight Sequence Analyzer (new feature 2.8.x)

This form will analyze every sight with “eyes” ON.
The objective of this tool is to plot Ho-Hc (Y-axis) against UT time. The observer can see “outliers” in a run of sights. If there were a run of 3 sights for each body, there would have been 9 observations which could have been plotted. This tool plots the Sights with “eyes”. If a sight is a true outlier, turn off the “eye” of the outlier. Plot a large sample of sights in the “Sights” list and see your personal error and bias.



6. Sun, Moon, and Sight Planner “Star Finder” (new feature 2.8.x)

Up to now, the Use Case assumed the ETA location to take evening twilight sights and the specific 3 bodies for 3 body fix. But it can be determined by using the “Star Finder”. To use this form,

Inputs

- Select Position:
- Options are Manual input of La and Lo, current boat position, selected Waypoint Position,
- Select Time input:
- UTC, Zone Time (excludes Daylight savings, there is no Daylight Savings offshore), PC time may include Daylight savings of the country
- Display time:
- UTC, Fixed offset (aka Zone time), and Zone offset which is calculated based on Longitude)

Output

Time is displayed using 2 standards. UTC always shown, plus another time standard such as Fixed Offset (Local Zone Time), local PC Time, etc.

Sun, Moon and Sight Planner

Planning context

Position

Waypoint or place...

Latitude

18° 51.6000' N

Longitude

168° 45.9000' W

Time

Manual date/time

Date (UTC)

2025-07-16

Time (UTC)

06:00:00

Enter time as

UTC

Display event times as

Fixed offset

Zone offset (h)

-11.0

Date/time entry

Nautical: YYYY-MM-DD, 24-hour

Auto zone from longitude

Eye height (m)

3.0

Calculate / refresh

Resolved UTC: 2025-07-16 06:00:00 UTC

Time-tagged moving observer

Reference time is the entry above

COG (true)

0.0

SOG (kn)

0.0

Planning context updated; calculations remain fully offline.

Events

Bodies & Best Sights

Almanac

Noon & Polaris

Event	UTC	Selected display time	Bearing true	Observer position
Sunset	2025-07-16 05:56:08	2025-07-15 18:56:08 UTC-11.0	292.9°	18° 51.6000' N, 168° 45.9000' W
Civil dusk	2025-07-16 06:19:30	2025-07-15 19:19:30 UTC-11.0	295.0°	18° 51.6000' N, 168° 45.9000' W
Nautical dusk	2025-07-16 06:47:47	2025-07-15 19:47:47 UTC-11.0	297.7°	18° 51.6000' N, 168° 45.9000' W
Astronomical dusk	2025-07-16 07:16:52	2025-07-15 20:16:52 UTC-11.0	301.0°	18° 51.6000' N, 168° 45.9000' W
Moonrise	2025-07-16 09:43:13	2025-07-15 22:43:13 UTC-11.0	87.6°	18° 51.6000' N, 168° 45.9000' W
Astronomical dawn	2025-07-16 15:25:37	2025-07-16 04:25:37 UTC-11.0	59.1°	18° 51.6000' N, 168° 45.9000' W
Nautical dawn	2025-07-16 15:54:42	2025-07-16 04:54:42 UTC-11.0	62.4°	18° 51.6000' N, 168° 45.9000' W
Moon transit	2025-07-16 15:57:58	2025-07-16 04:57:58 UTC-11.0	180.0°	18° 51.6000' N, 168° 45.9000' W
Civil dawn	2025-07-16 16:22:59	2025-07-16 05:22:59 UTC-11.0	65.1°	18° 51.6000' N, 168° 45.9000' W
Sunrise	2025-07-16 16:46:20	2025-07-16 05:46:20 UTC-11.0	67.1°	18° 51.6000' N, 168° 45.9000' W
Moonset	2025-07-16 22:17:43	2025-07-16 11:17:43 UTC-11.0	276.0°	18° 51.6000' N, 168° 45.9000' W
Local apparent noon	2025-07-16 23:21:13	2025-07-16 12:21:13 UTC-11.0	0.0°	18° 51.6000' N, 168° 45.9000' W

Moon: Waning gibbous, 69.6% illuminated, waning, age 20.3 days; altitude -50° 6000.9448', azimuth 63.7°, Sun separation 113° 07.7427'.

New

Fix...

Duplicate

Clock Offset

Edit

Eclipses...

Delete

Generate Almanac...

Delete All

Documentation

Horizon Event...

PDF Documentation

Coastal Sextant...

Sun & Moon...

Lunar Tools...

Analyze Sights...

Hide Sights

Hide Time

Dusk

Local Apparent Noon is using the GHA method.

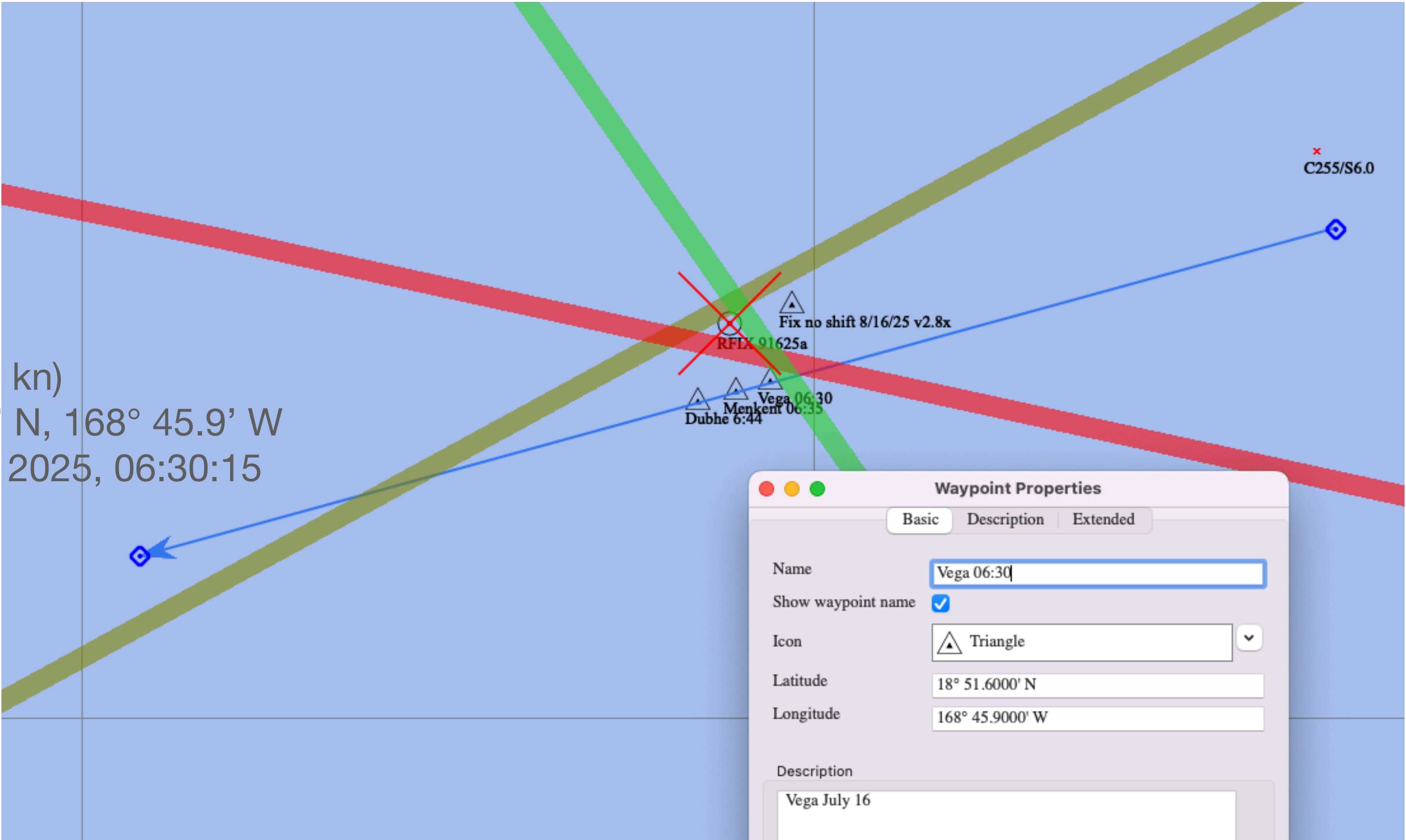
19

RMB 26-Sept-2026

Use Case 2 Estimate ETA (Time and Position) for Sunrise and Meridian Transit

Continue with the Use Case 1, and estimate the ETA (Time and Position) for Sunrise and Meridian Transit

Course 255°T, and Speed (6.0 kn)
Vega Waypoint @ Lat18° 51.6' N, 168° 45.9' W
UTC Date and Time is 16 July 2025, 06:30:15



Use Case 2 Estimate ETA (Time and Position) for Sunrise and Meridian Transit

7. Sun, Moon, and Sight Planner “Star Finder” (new feature 2.8.x)

- 1. Position - Select Vega 06:30 Waypoint
- 2. Time - of the Selected Sight, Vega

Altitude	Sun	2025-07-20 10:00:00	8° 21.3000'
Altitude	Dubhe	2025-07-16 06:43:44	30° 39.0000'
Altitude	Menkent	2025-07-16 06:35:50	33° 28.8000'
Altitude	Vega	2025-07-16 06:30:15	37° 50.0000'
Altitude	Kochab	2025-07-04 21:10:34	48° 04.4000'

Sun, Moon and Sight Planner

Planning context

Position

Waypoint or place...

Latitude

18° 51.6000' N

Longitude

168° 45.9000' W

Time

Selected sight

Date (UTC)

2025-07-16

Time (UTC)

06:30:15

Enter time as

UTC

Display event times as

Fixed offset

Zone offset (h)

-11.0

Date/time entry

Nautical: YYYY-MM-DD, 24-hour

Auto zone from longitude

☒

Eye height (m)

3.0

Calculate / refresh

Resolved UTC:

2025-07-16 06:30:15 UTC

☒ Time-tagged moving observer

Reference time is the entry above

COG (true)

255.0

SOG (kn)

6.0

Planning context updated; calculations remain fully offline.

Events

Bodies & Best Sights

Almanac

Noon & Polaris

Event	UTC	Selected display time	Bearing true	Observer position
Sunset	2025-07-16 05:55:56	2025-07-15 18:55:56 UTC-11.0	292.9°	18° 52.4908' N, 168° 42.4053' W
Civil dusk	2025-07-16 06:19:26	2025-07-15 19:19:26 UTC-11.0	295.0°	18° 51.8809' N, 168° 44.7988' W
Nautical dusk	2025-07-16 06:47:53	2025-07-15 19:47:53 UTC-11.0	297.7°	18° 51.1414' N, 168° 47.6972' W
Astronomical dusk	2025-07-16 07:17:08	2025-07-15 20:17:08 UTC-11.0	301.0°	18° 50.3806' N, 168° 50.6757' W
Moonrise	2025-07-16 09:44:35	2025-07-15 22:44:35 UTC-11.0	87.6°	18° 46.5332' N, 169° 05.6840' W
Astronomical dawn	2025-07-16 15:29:52	2025-07-16 04:29:52 UTC-11.0	59.2°	18° 37.4445' N, 169° 40.7872' W
Nautical dawn	2025-07-16 15:59:06	2025-07-16 04:59:06 UTC-11.0	62.5°	18° 36.6702' N, 169° 43.7557' W
Moon transit	2025-07-16 16:01:58	2025-07-16 05:01:58 UTC-11.0	180.0°	18° 36.5940' N, 169° 44.0474' W
Civil dawn	2025-07-16 16:27:33	2025-07-16 05:27:33 UTC-11.0	65.2°	18° 35.9157' N, 169° 46.6445' W
Sunrise	2025-07-16 16:51:01	2025-07-16 05:51:01 UTC-11.0	67.2°	18° 35.2926' N, 169° 49.0282' W
Moonset	2025-07-16 22:24:14	2025-07-16 11:24:14 UTC-11.0	276.1°	18° 26.3939' N, 170° 22.8299' W
Local apparent noon	2025-07-16 23:28:07	2025-07-16 12:28:07 UTC-11.0	0.0°	18° 24.6765' N, 170° 29.3029' W
Next Last quarter	2025-07-18 00:37:38	2025-07-17 13:37:38 UTC-11.0	—	Geocentric phase
Next New Moon	2025-07-24 19:38:43	2025-07-24 08:38:43 UTC-11.0	—	Geocentric phase
Next First quarter	2025-08-01 12:41:11	2025-08-01 01:41:11 UTC-11.0	—	Geocentric phase
Next Full Moon	2025-08-09 08:21:05	2025-08-08 21:21:05 UTC-11.0	—	Geocentric phase

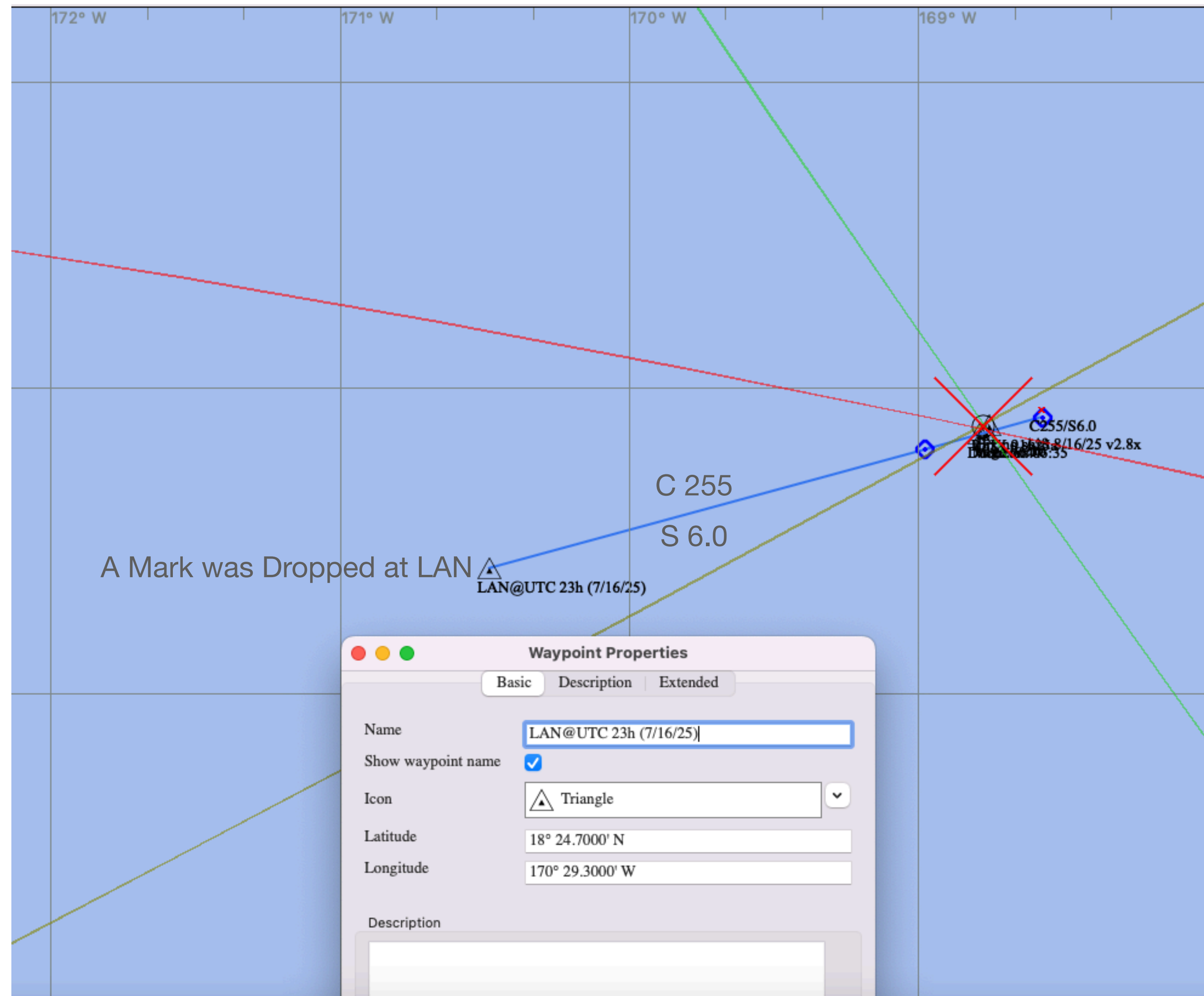
4. Press Calculate / to compute Event Date and Times

3. Course and Speed

The position, UTC, and display time changes For each Event due to COG and SOG inputs

Meridian Transit UTC 16 July 2025 @ 23:28:07
Local Zone Time (UTC-11) is 16 July 2025 @ 12:28:07
ETA Position is N18° 24.7'. W 170° 29.3'

Use Case 2 Estimate ETA (Time and Position) for Sunrise and Meridian Transit



8. Line of Position and Estimated Position without Sextant or GPS.

Sun, Moon and Sight Planner

Planning context

Position

Waypoint or place...

Latitude

18° 51.6000' N

Longitude

168° 45.9000' W

Time

Selected sight

Date (UTC)

2025-07-16

Time (UTC)

06:30:15

Enter time as

UTC

Display event times as

Fixed offset

Zone offset (h)

-11.0

Date/time entry

Nautical: YYYY-MM-DD, 24-hour

Auto zone from longitude

Eye height (m)

3.0

Calculate / refresh

Resolved UTC: 2025-07-16 06:30:15 UTC

Time-tagged moving observer

Reference time is the entry above

COG (true)

255.0

SOG (kn)

6.0

Planning context updated; calculations remain fully offline.

Events

Bodies & Best Sights

Almanac

Noon & Polaris

Event	UTC	Selected display time	Bearing true	Observer position
Sunset	2025-07-16 05:55:56	2025-07-15 18:55:56 UTC-11.0	292.9°	18° 52.4908' N, 168° 42.4053' W
Civil dusk	2025-07-16 06:19:26	2025-07-15 19:19:26 UTC-11.0	295.0°	18° 51.8809' N, 168° 44.7988' W
Nautical dusk	2025-07-16 06:47:53	2025-07-15 19:47:53 UTC-11.0	297.7°	18° 51.1414' N, 168° 47.6972' W
Astronomical dusk	2025-07-16 07:17:08	2025-07-15 20:17:08 UTC-11.0	301.0°	18° 50.3806' N, 168° 50.6757' W
Moonrise	2025-07-16 09:44:35	2025-07-15 22:44:35 UTC-11.0	87.6°	18° 46.5332' N, 169° 05.6840' W
Astronomical dawn	2025-07-16 15:29:52	2025-07-16 04:29:52 UTC-11.0	59.2°	18° 37.4445' N, 169° 40.7872' W
Nautical dawn	2025-07-16 15:59:06	2025-07-16 04:59:06 UTC-11.0	62.5°	18° 36.6702' N, 169° 43.7557' W
Moon transit	2025-07-16 16:01:58	2025-07-16 05:01:58 UTC-11.0	180.0°	18° 36.5940' N, 169° 44.0474' W
Civil dawn	2025-07-16 16:27:33	2025-07-16 05:27:33 UTC-11.0	65.2°	18° 35.9157' N, 169° 46.6445' W
Sunrise	2025-07-16 16:51:01	2025-07-16 05:51:01 UTC-11.0	67.2°	18° 35.2926' N, 169° 49.0282' W
Moonset	2025-07-16 22:24:14	2025-07-16 11:24:14 UTC-11.0	276.1°	18° 26.3939' N, 170° 22.8299' W
Local apparent noon	2025-07-16 23:28:07	2025-07-16 12:28:07 UTC-11.0	0.0°	18° 24.6765' N, 170° 29.3029' W
Next Last quarter	2025-07-18 00:37:38	2025-07-17 13:37:38 UTC-11.0	—	Geocentric phase
Next New Moon	2025-07-24 19:38:43	2025-07-24 08:38:43 UTC-11.0	—	Geocentric phase
Next First quarter	2025-08-01 12:41:11	2025-08-01 01:41:11 UTC-11.0	—	Geocentric phase
Next Full Moon	2025-08-09 08:21:05	2025-08-08 21:21:05 UTC-11.0	—	Geocentric phase

Sunset

Sunrise

8. Line of Position and Estimated Position without Sextant or GPS.

Use Case 2 Horizon Event...Sunrise

Horizon Event

Record first upper-limb appearance at sunrise or final upper-limb disappearance at sunset. The result is an approximate navigation constraint, not a sextant-quality fix.

Observation — always visible

Event

Sunrise — first upper limb

July2025

Sun	Mon	Tue	Wed	Thu	Fri	Sat
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

UTC date

UTC time (24-hour)

16:51:00

Capture current UTC

Time uncertainty

10.0

seconds

Time source

System UTC capture

Current system timing: chrony status unavailable

Bearing, horizon conditions and the result are below; scroll this lower section when necessary.

Bearing (optional)

☒

Include a bearing to show possible positions along the event LOP

Bearing reference

True bearing

Observed bearing

65.00

degrees

Magnetic variation (E + / W -)

0.00

degrees

Compass deviation (E + / W -)

0.00

degrees

Bearing uncertainty (+/-)

3.0

degrees

Effective bearing

65.00° true (65.00° true +0.00° variation +0.00° deviation)

Horizon conditions

Height of eye

3.00

metres

Air temperature

11.0

C

Pressure

1,010.0

hPa

Horizon quality

Clear sea horizon

Horizon/refraction uncertainty (+/-)

0.3

arcminutes

Cancel

Save Changes

New

Fix...

Duplicate

Clock Offset

Edit

Eclipses...

Delete

Generate Almanac...

Delete All

Documentation

Horizon Event...

PDF Documentation

Coastal Sextant...

Sun & Moon...

Lunar Tools...

Analyze Sights...

Hide Sights

Hide Time

It is difficult for average offshore cruiser to identify Time and bearing of Sun UL/LL Sunrise or Sunset. On the other hand, professionals have better tools that give much better results.

The Sunrise observed at 16:51:00. Bearing 65°

Note the uncertainties used in this example.

- Time uncertainty : 10 seconds (skill of navigator. Much better if using Gyro with Azimuth ring which has a prism)
- Bearing uncertainty : 5 degrees (hockey puck compass..Gyro is better)
- Refraction Uncertainty : 0.3 arc minutes.

8. Line of Position and Estimated Position without Sextant or GPS.

Celestial Navigation Sights

Type	Body	Time (UTC)	Measurement	Color
Sunrise	Sun	2025-07-16 16:51:00	65.0° T	maroon
Altitude	Dubhe	2025-07-16 06:43:44	30° 39.0000'	rgb(128, 128, 4)
Altitude	Menkent	2025-07-16 06:35:50	33° 28.8000'	rgb(251, 2, 7)
Altitude	Vega	2025-07-16 06:30:15	37° 50.0000'	lime green
Sunset	Sun	2025-07-16 05:55:56	293.0° T	red
Altitude	Kochab	2025-07-04 21:10:34	48° 04.4000'	rgb(33, 255, 6)

New

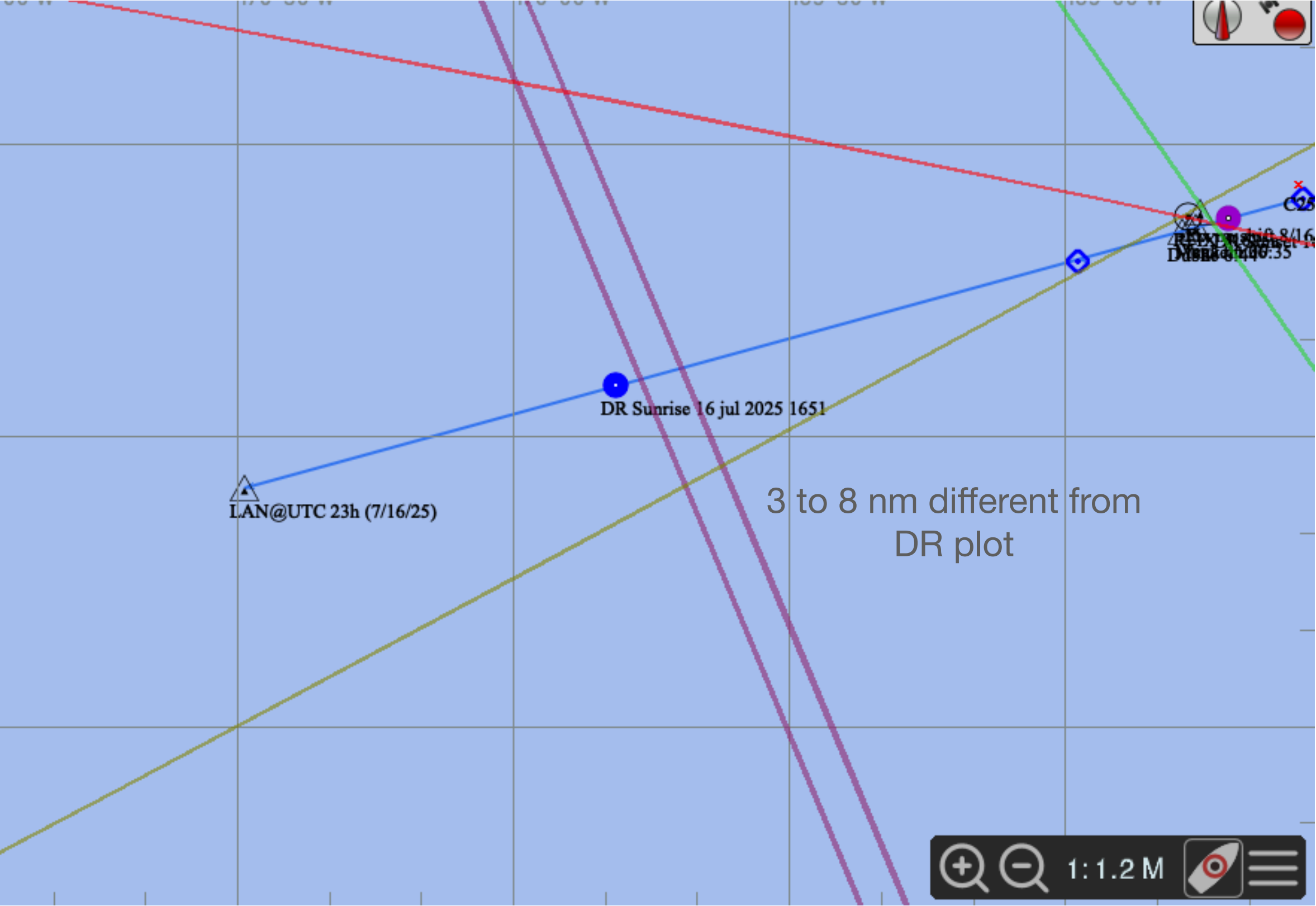
Duplicate

Edit

Delete

Delete All

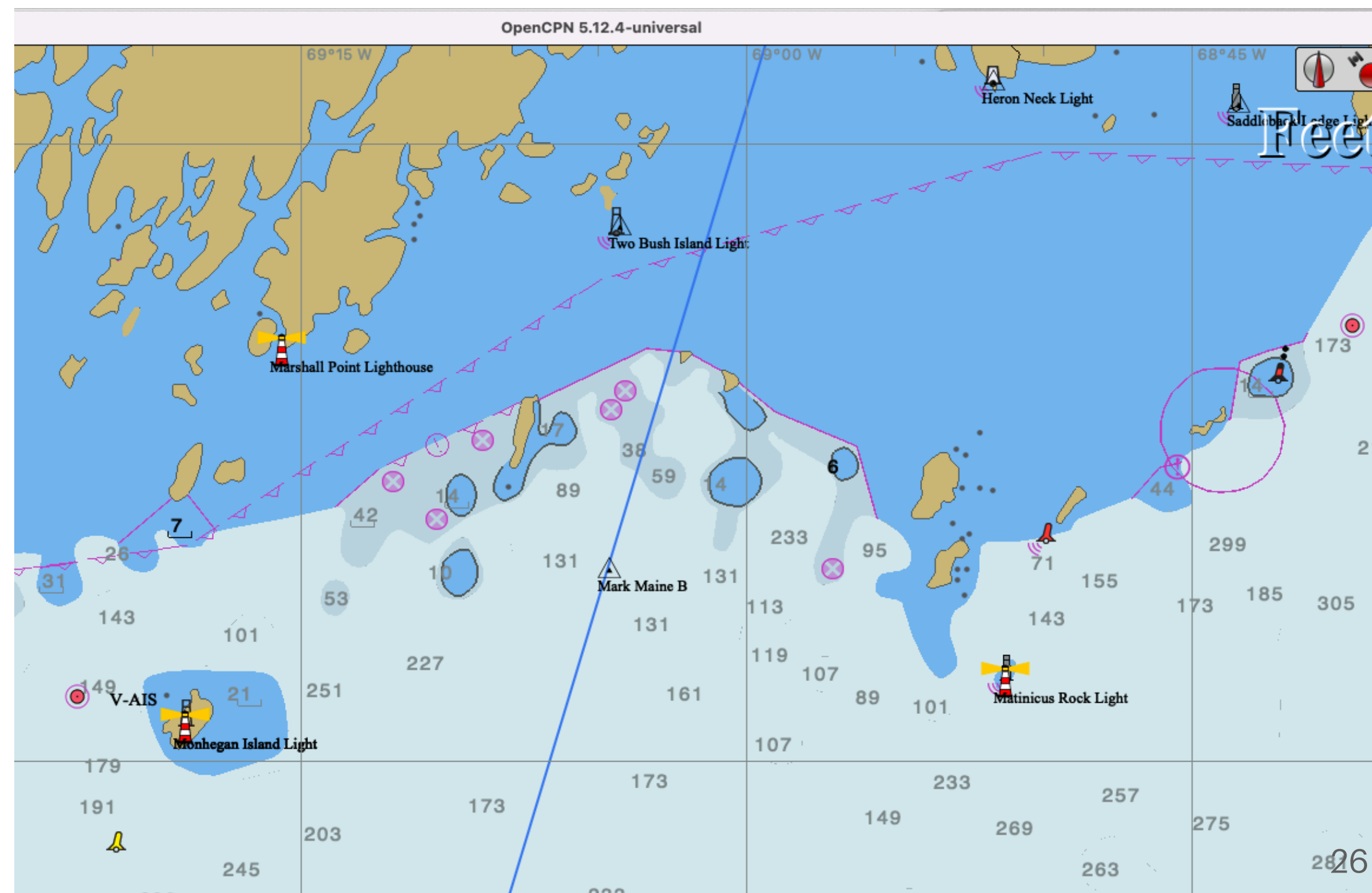
Horizon Event...



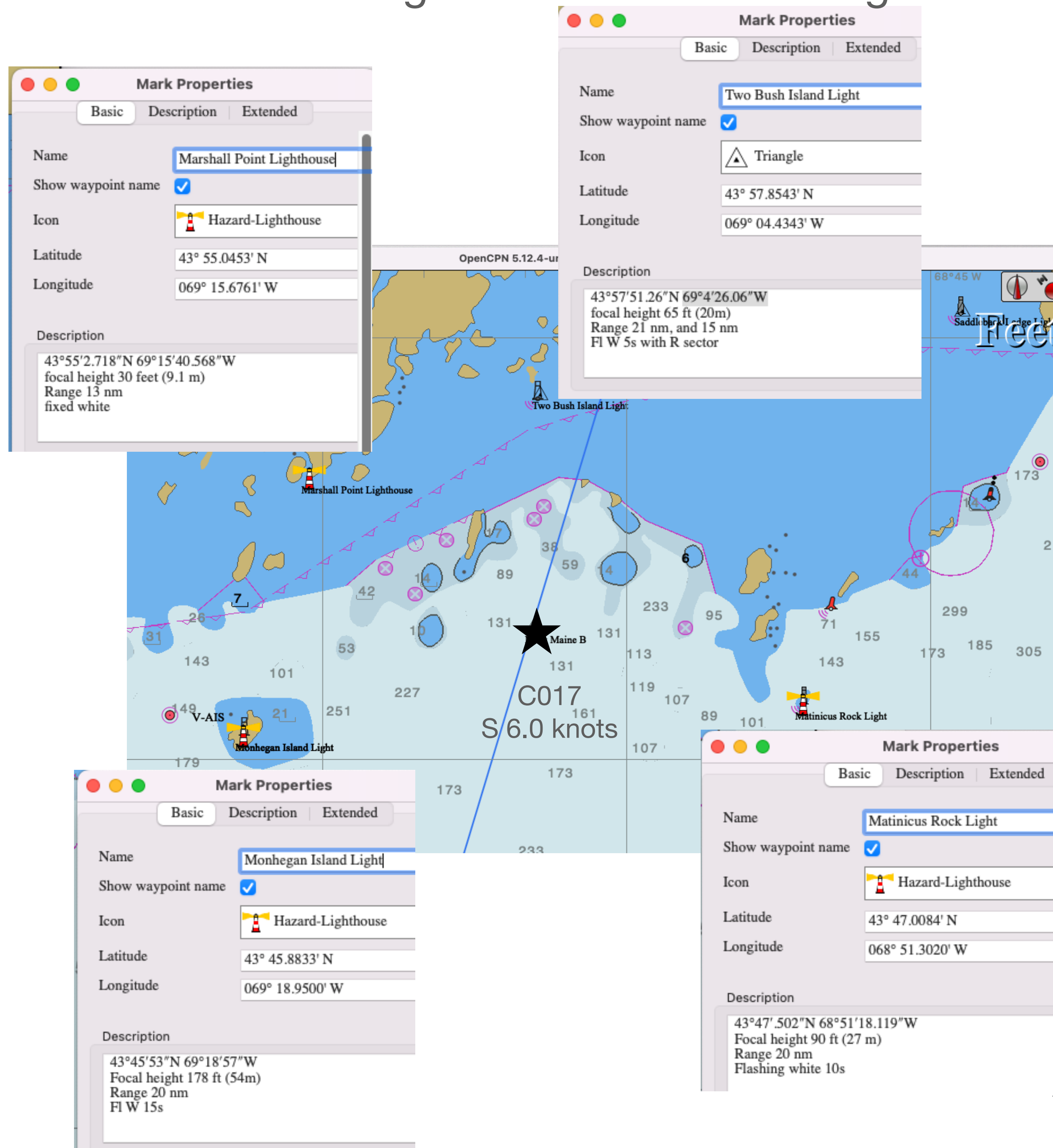
9. Position using Horizontal Sextant Sights of Landmarks or Waypoints

- A useful capability for Offshore Navigation when approaching the Coast.
- Before approach, while studying the Coastal Chart identify potential sights
- Drop DR for the Landmarks (e.g. Lighthouses) in OpenCPN.
 - Position,
 - Light pattern,
 - Light Range,
 - Height (from Light to Sea)
- Because ENC charts lack sufficient “landmark” detail in Maine.
- Besides Sextant, Radar can also be used

Use Case : Arriving Rockland Harbor Maine



9. Position using Horizontal Sextant Sights of Landmarks or Waypoints



9. Position using Horizontal Sextant Sights of Landmarks or Waypoints

Basic

Description

Extended

Name

Two Bush Island Light

Show waypoint name

☑

Icon

△ Triangle

Latitude

43° 57.8543' N

Longitude

069° 04.4343' W

Description

43°57'51.26"N 69°4'26.06"W
focal height 65 ft (20m)
Range 21 nm, and 15 nm
Fl W 5s with R sector

43°57'51.26"N 69°4'26.06"W
focal height 65 ft (20m)
Range 21 nm, and 15 nm
Fl W 5s with R sector

Two Bush Island Light

Basic

Description

Extended

Name

Monhegan Island Light

Show waypoint name

☑

Icon

⚠ Hazard-Lighthouse

Latitude

43° 45.8833' N

Longitude

069° 18.9500' W

Description

43°45'53"N 69°18'57"W
Focal height 178 ft (54m)
Range 20 nm
Fl W 15s

Basic

Description

Extended

Name

Matinicus Rock Light

Show waypoint name

☑

Icon

⚠ Hazard-Lighthouse

Latitude

43° 47.0084' N

Longitude

068° 51.3020' W

Description

43°47'50.2"N 68°51'18.119"W
Focal height 90 ft (27 m)
Range 20 nm
Flashing white 10s

Coastal sextant navigation

These are coastal sextant methods. A vertical angle gives distance from an object; two horizontal angles can give a fix. Enter what you actually observed; calculations never change the raw readings.

Vertical angle / distance

Horizontal angles / fix

Enter three charted objects in left-centre-right visual order and the two included angles. The approximate position helps select a solution; check ambiguity independently. Formal uncertainty excludes landmark-coordinate error, misidentification and instrument biases. Small residuals do not establish position accuracy.

Left object latitude

43° 45.8833' N

angle

Left object longitude

069° 18.9500' W

angle

Centre object latitude

43° 57.8543' N

angle

Centre object longitude

069° 04.4343' W

angle

Right object latitude

43° 47.0084' N

angle

Right object longitude

068° 51.3020' W

angle

Left-centre measurement

111° 00.0000'

(Sextant Measurement)

degrees

Centre-right measurement

107° 00.0000'

(Sextant Measurement)

degrees

Index error (on the arc +)

-0.15

arcmin

Angle uncertainty (1-sigma)

0.2

arcmin

Approximate vessel latitude

43° 49.7000' N

(DR)

angle

Approximate vessel longitude

069° 04.6000' W

(DR)

angle

Select left waypoint...

(Selected Monhegan Light)

Select centre waypoint...

(Selected Two Bush Light)

Select right waypoint...

(Selected Matinicus Light)

Sequential angle readings

☐ Advance the vessel between the two HSA readings

Second-angle interval from first

0

seconds

COG true

0.0

degrees

SOG

0.0

kn

A further refinement is to
Input Course and Speed

Use boat position

Solve and plot HSA fix

Solve and plot landmark fix

Fix 43° 49.9215' N, 069° 04.4356' W; residuals -0.0000' / -0.0000'; formal angular-input uncertainty (1-sigma) 0.001 NM; condition 2.1. Not total fix accuracy.

Vertical and horizontal plots can stay on the chart together. Solving replaces only that tab's plot. Closing retains entries and plots; Clear chart plots removes both plots. New / clear observation resets both tabs.

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New / clear observation

Clear chart plots

Close

New

Fix...

Duplicate

Clock Offset

Edit

Eclipses...

Delete

Generate Almanac...

Delete All

Documentation

Horizon Event...

PDF Documentation

Coastal Sextant...

Sun & Moon...

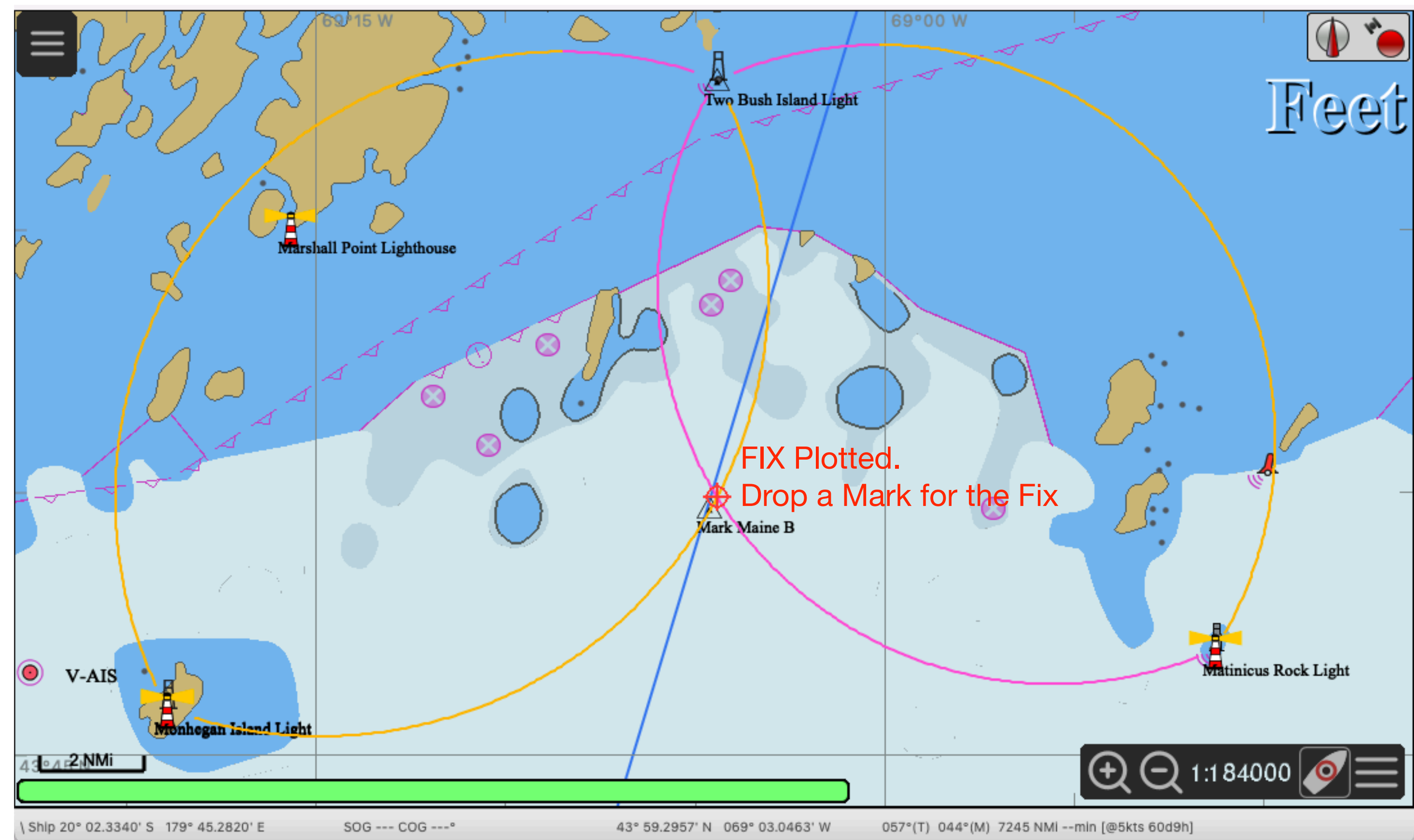
Lunar Tools...

Analyze Sights...

Hide Sights

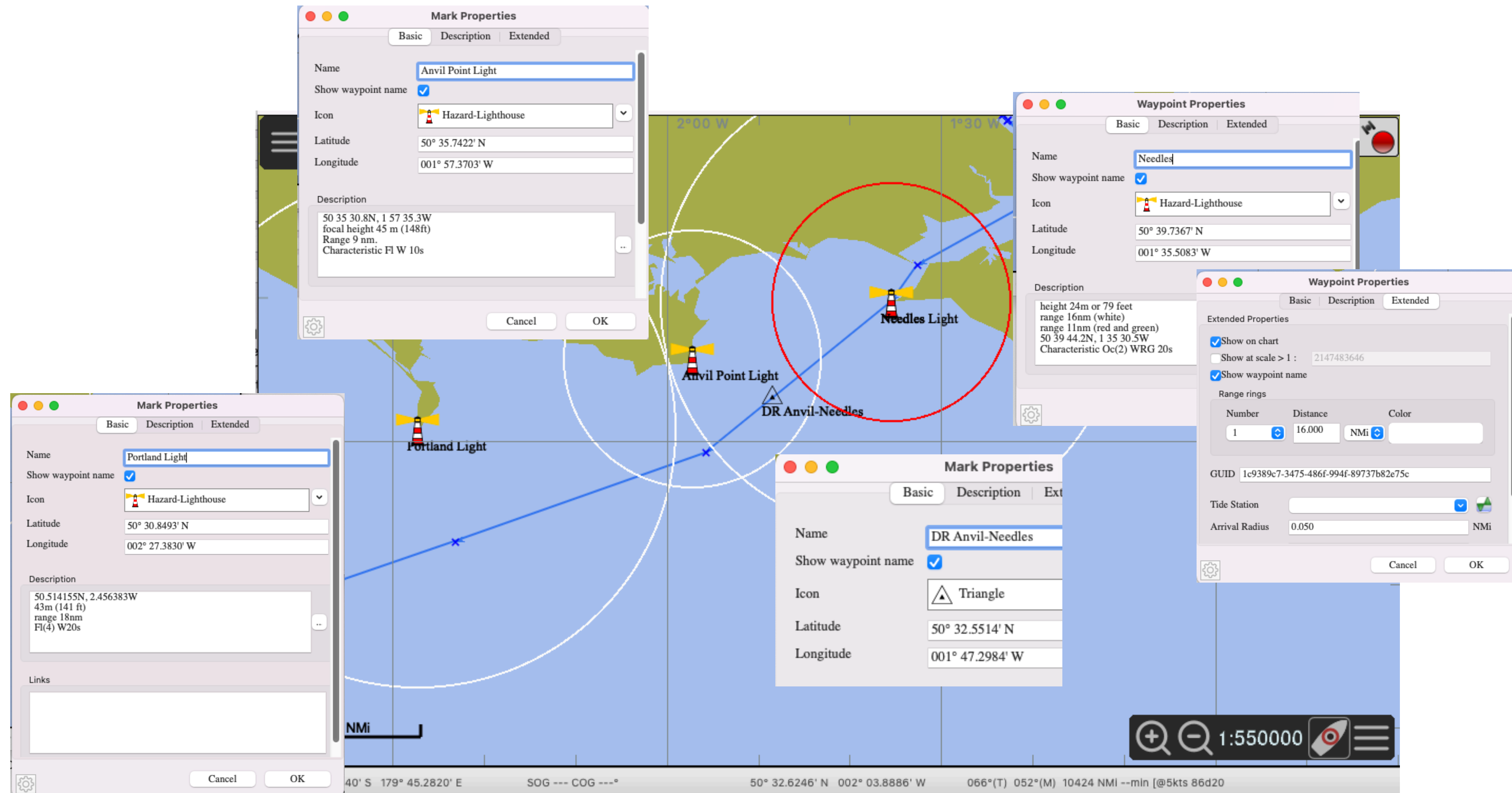
Hide Time

9. Position using Horizontal Sextant Sights of Landmarks or Waypoints



10.0. Coastal Sextant Sights. Vertical Sight.

Use Case : Departed Southampton, UK. Passed Needles Light.
On a DR Route C230, Speed 6.0 knots.
Dropped a Mark “DR Anvil-Needles” SE Anvil Point Light



10.0. Coastal Sextant Sights. Vertical Sight.

New

Duplicate

Edit

Delete

Delete All

Horizon Event...

Coastal Sextant...

Sun & Moon...

Lunar Tools...

Analyze Sights...

Coastal sextant navigation

These are coastal sextant methods. A vertical angle gives distance from an object; two horizontal angles can give a fix. Enter what you actually observed; calculations never change the raw readings.

Vertical angle / distance

Horizontal angles / fix

Observation

Sea horizon to top

Target latitude

50° 35.7000' N

angle

Target longitude

001° 57.4000' W

angle

Observed vertical angle

0° 10.0000'

angle or decimal degrees

Index error (on the arc +)

-0.15

arcmin

Charted top height

45

m above height datum

Water level above height datum

0.0

m (tide included)

Height of eye

3.0

m

Select target waypoint...

Estimated observer horizon: 3.66 NM; target geographic visibility: 17.83 NM. Standard refraction estimate, not a guarantee of visibility or a light's luminous range. Waterline reaches the horizon at an index-corrected waterline-to-top angle of 22.82 arcmin (no dip correction). Sea-horizon mode subtracts index error and dip; the resulting angle may be zero or negative. Do not substitute a waterline reading.

Optional bearing fix

☒ Include observed bearing from vessel to target

Record the bearing at effectively the same time as the vertical angle, or reduce it to that position epoch before entry.

Bearing reference

True

degrees

Observed bearing

290

degrees

Magnetic variation (E + / W -)

0.0

degrees

Compass deviation (E + / W -)

0.0

degrees

Use WMM at current boat/time

Variation and deviation are ignored for True bearings.

Calculate and plot range

Range 7.596 NM; corrected angle 0° 07.1051'; effective target height 45.00 m.

Warning: This Bowditch mode assumes standard terrestrial refraction; unusual atmospheric gradients can move the visible horizon

Bearing/range estimate: 50° 33.0947' N, 001° 46.1734' W (true bearing 290.00°).

Vertical and horizontal plots can stay on the chart together. Solving replaces only that tab's plot. Closing retains entries and plots; Clear chart plots removes both plots. New / clear observation resets both tabs.

New / clear observation

Clear chart plots

Close

Mark Properties

BasicDescriptionExtended

Name

Anvil Point Light

Show waypoint name

☒

Icon

Hazard-Lighthouse

Latitude

50° 35.7422' N

Longitude

001° 57.3703' W

Description

50 35 30.8N, 1 57 35.3W
focal height 45 m (148ft)
Range 9 nm.
Characteristic Fl W 10s

Cancel

OK

Map

1°00 W

Measure Tool shows 7.5nm from Anvil Point Light

1:548000

Lunar Sights to derive UTC and estimate Longitude

Objectives : How to input and interpret Lunar Distance, “Lunars”

Topics:

- 1. Lunars Objective - To Tell Time at Greenwich Meridian (UTC in Cel Nav Plugin forms)**
- 2. After Lunar Distancing Time, what was next.**
- 3. Accuracy Achieved**
- 4. Relevance Today**
- 5. Process Steps. Onboard ship or On land and your backyard.**
- 6. How to Input Lunar Sights into Celestial Navigation, an example**
- 7. Sensitivity Analysis**
- 8. Analyze your Lunar for position (optional)**
- 9. Which Celestial Bodies paired with Moon make great Lunars sights.**
- 10. Planning for Lunar Sights.**
- 11. Pre-setup sextant.**
- 12. Practice doing Backyard Lunars. A tip to improve Sextant skills.**
- 13. Tips to improve your Lunar accuracy**
- 14. Theory and Logic**

1. Lunar Distancing Objective and Background

- Objective of Lunars was to determine TIME at a Baseline Meridian, Greenwich. The baseline time is now UT/UTC.
- Ancient Greeks and Babylonians theorized the Moon could be used for Universal Time and then observer could determine Longitude. In theory, anyone on Earth could tell the same event's baseline meridian time by measuring the Moon's "distance" to another celestial body. The "distance", was an angle called Lunar Distance. And then after TIME was determined for the baseline meridian, Longitude could be calculated using the observer's local time.
- 16th century mathematicians and astronomers advanced Spherical Triangles (and Navigator's triangle) to the point where the math could be used to tell time by measuring the distance of the Moon to the Sun, planets, and stars.
- It takes the moon 27.32166 days to complete a full orbit around the earth respect to the background stars (Moon Sidereal month). Thus the Moon moves Eastward. Relative to the background celestial bodies, almost 13.2° per day. This translates to 32.94 arc-min per hr. or 0.55 arc-min per min. From that, 0.2 arc-min Marine Sextant measurement accuracy will be ~22 sec., and 0.1 arc-min measurement accuracy will be ~11 sec.
- You need a high quality Marine sextant to achieve .1' measurement accuracy. At the equator, 4 seconds in time is 1 nm error. And 11 seconds in time is approximately ~3 nm error. That's the limit for Lunar derived position accuracy a marine sextant.
- The maritime sextant was designed to meet the need to measure Lunar Distance. Double reflecting to measure 120° angles. Higher magnification to determine more precisely when the celestial body touches the Moon (today 7x are common). Shades over the Horizon mirror because you might be observing the Sun using the Horizon mirror. Eventually a vernier was added next to micrometer drum to more precisely distinguish 0.1 - 0.2 min. of arc.
- Meanwhile, mathematician and astronomers worked on the position of the celestial bodies problem. Isaac Newton theory of gravity was crucial to initially calculating the Moon and other celestial body positions. But, it took more time to figure it out with sufficient accuracy. In 1752-1762, Johann Tobias Mayer calculated Lunar Distance tables and altitude corrections. His widow was awarded, after his death, some Longitude Prize monies. In 1767, Nevil Maskelyne included Lunar Distance tables in the first Nautical Almanac.

1. Lunar Distancing Objective and Background (continued)

- Then, John Harrison's H1 Chronometer was demonstrated 1736. And then successfully tested on a voyage to Jamaica in 1761. He won the largest share of the Longitude prize. An accurate chronometer would be set for Greenwich Time which then was used to determine the ship's Longitude by comparing Local Time to Greenwich Time.
- In 1802, Nathaniel Bowditch's American Practical Navigator contained Lunar Distance Tables and simplified procedures to clear the Lunar distances. This made the method more accessible to everyday navigators. He was an advocate for both Chronometer and Lunars. Don't rely on one method!
- The traditional practice was to make note in the ship log the Lunar's derived time and then use judgement and other sources to assess chronometer error rate (Fast or Slow). The common practice, even today, is that no one directly changes the chronometer time. And for sight reduction, chronometer time with error rate (fast or slow) is referenced. The log is not touched up.
- You'd think that shortly after 1761 with John Harrison's Chronometer, Lunar Distancing method would die-off. But, chronometers weren't widely available because an entire industry needed to be created to mass produce affordable chronometers.
- Captain Cook's first Pacific voyage (1768 - 1771) didn't have a chronometer and he charted much of the Pacific aided by Lunar Distancing as one of his tools. Captain Bligh (1789) used Lunar Distancing. Navigators on multi-year voyages would use Lunars to double check their chronometer if they had one. After all, during a long voyage, even a chronometer builds up a significant error.
- Lewis and Clark had a chronometer, sextant, artificial horizon equipment when they surveyed the Louisiana purchase (1804-6). John C. Fremont on his first expedition to California (1842) took a high quality sextant. They must have used a sextant for position by Artificial Horizon when on land. Lunars could also have been part of their "tool kit", but AH is easier than Lunars.
- Up to the 1850s, many ships lacked chronometers. Too few, too expensive. Chronometer improvements were made; adoption was slow, but inevitable. Eventually Lunars fell out of use. Bowditch Practical American Navigator reduced emphasis on Lunars in 1882. Tables were dropped in 1912. Nautical Almanac last published Lunar tables in 1907.

2. After Lunar Distancing, what was next?

- After computing time (Greenwich Time,...GMT....UT/UTC), calculate Longitude. Calculate Latitude using a Meridian Transit or Polar sights. Re-evaluate chronometer's error (number of min-sec fast or slow) for future sights.

3. How accurate was Lunars in the 1700s?

- The Longitude Prize was awarded if you could demonstrate a solution better than $.5^{\circ}$ longitude accuracy. Harrison was the big winner. And Tobias Mayer won some prize money for his work on Lunars. The $.5^{\circ}$ was not the estimate for accuracy; it was a threshold or goal for the award. If statistical analysis is found to identify historic accuracy, that would be beneficial.
- Ship Logs included Lunars, Altitude Sights, and DR positions. The ship logs are today's evidence that Lunars was trusted.
- For over 100 years after chronometers were introduced, navigators found it to be a valuable tool.

3b. How accurate is it Today?

- With a good quality sextant, well adjusted, and used with skill, the goal for your Lunar Distance observation should be consistently $\pm 0.2'$ accuracy. The limitation is your skill given the stability of the vessel. It is easier on land.

4. Is Lunars relevant today?

- Lunar gives the celestial navigator the opportunity to practice using their sextant. The more practice the better you become. Not only for Lunars, but for all sights. Practice makes perfect. Lunars can be done anywhere. At home in your backyard, near the water, on the water. You can do it during twilight, during the day, in the middle of night. Do it any time the Moon is visible, day or night. And of course, off-shore. The Celestial Navigation Plugin was designed to support practicing Lunar sights, UT/UTC determination, "in the backyard" and middle of the ocean. And, it is not tethered to the internet.
- Do you want to verify your sextant's arc error and develop calibration data based on observation? Lunar method can help. Point your sextant to 2 bright stars and measure the Angle difference! Cel Nav Plugin includes such a calibration tool.
- OpenCPN's Celestial Navigation's plugin for Lunars is quick and easy. Input the sight and get immediate results. Change data and instantly see the TIME result change.

5. What are the Process steps to do Lunars on board Ship?

You should be doing 2 altitude sights of one body, 4 lunar sights, 2 altitude sights of the Moon. Sequence matters.

1. Measure One (1) Altitude Sight of the Moon.
2. Measure One (1) Altitude Sight of the Celestial Body.
3. Measure Four (4) Lunar Distance Angle of the Moon to your selected Celestial Body.
4. Hold the Sextant at an angle. The horizon mirror point to one object, and index mirror pointed to the other. Make sure the mirror that points to the sun has shades on to protect your eyes. Measure “edge” to “edge”.
5. Measure One (1) Altitude Sight of the Celestial Body. Note the reverse order!
6. Measure One (1) Altitude Sight of the Moon.
7. Then average the Lunar Distance Angle and UTC Time. Average the Moon Altitude Sight and UTC. Average the Celestial Body Altitude Sight and UTC. After averaging, the 3 sets of sights should have roughly the same UTC.
8. Input the average measurements into the Celestial Navigation Sights form. Verify Index Error, Height of Eye, Temp/Barometer (minor impact) in the Parameters form.
9. Input your DR Position (Open CPN’s “Boat position” is the default, which should be changed to freeze it. This is found in the Find form.
10. Input the average Ship’s UTC in the Cel Nav Time (UTC) form.
11. Press the “TIME” Button in the Sights form to see the results.

5b. Process Steps to do Lunars on Land (continued)

One of the positives about learning Lunars is that you can do Lunar sights in your backyard without measuring an altitude sight.

- Do it during the day with the Sun and Moon.
- Do it in the middle of the night when you can't see the visible horizon.
- We'll next describe how to do it.

5c. The Process Steps to do Lunars in your backyard

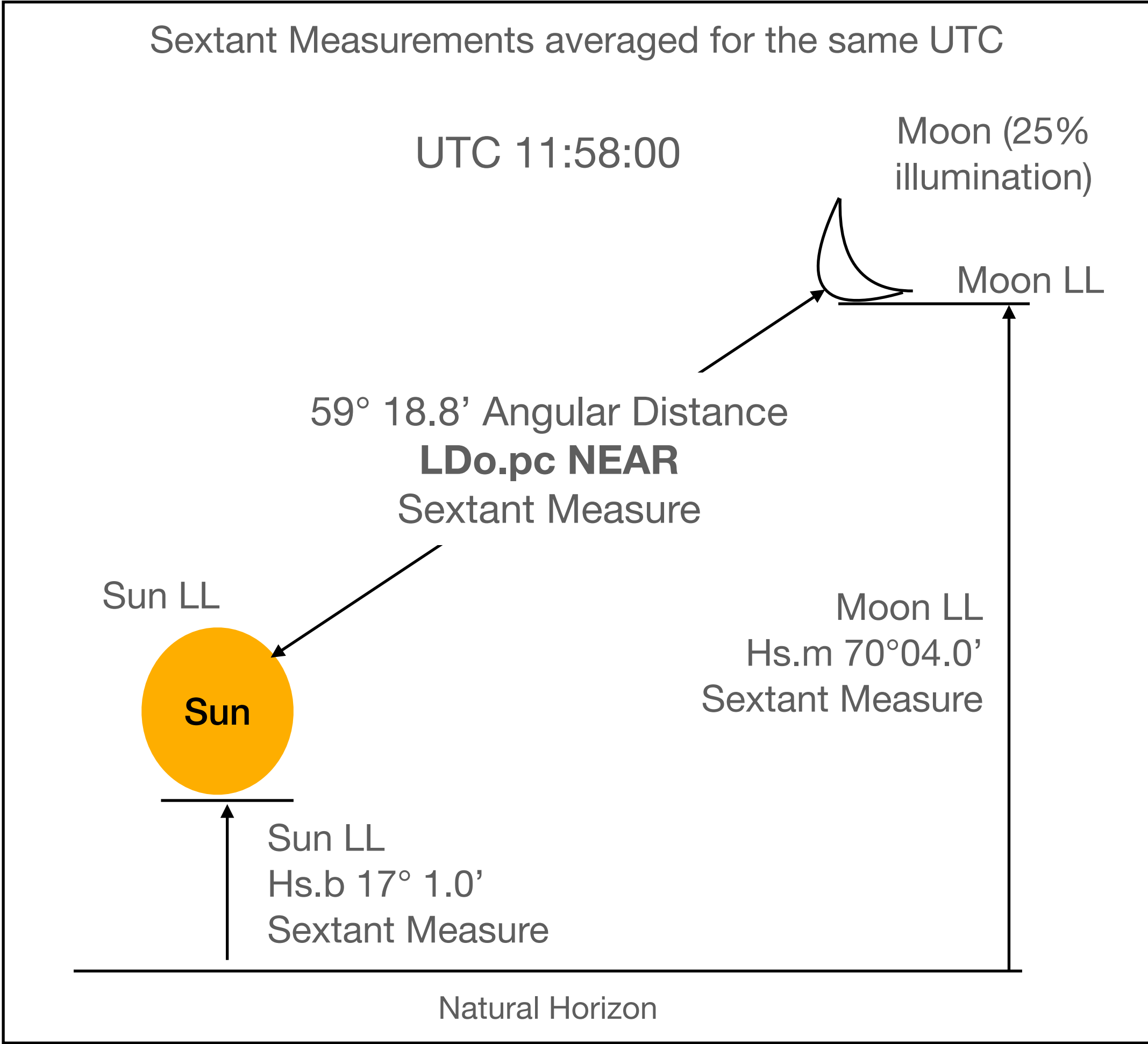
You can eliminate Steps 1, 2, 5, and 6. You need to do Step 7 and 8 for the 3+ sets of sights. When inputting Altitude Sights for the Moon and for the Celestial Body, use Celestial Nav plugin's Estimated (Hs) Altitude and for UTC time, use the Lunar average UTC. But you still need to input your Latitude and Longitude so that the Altitude (Hs) of the Moon and Celestial Body will calculate. Lunar Distance calculation relies on the Parallax and Refraction which is derived from the 2 bodies. There is no DIP in Lunar Measurement. The Moon and it's pair needs to be above the horizon for the computation.

In Use Case 1 we'll demonstrate how to input Lunars into the Cel Nav Plugin, and we'll indicate what to do for a "Backyard" Lunar.

Lunar Distance Use Case 1 “Sun-Moon” Aug 18, 2025 11:58

The following is an example of a Sun-Moon Lunar Sight. Sun-Moon Lunars were known to be one of the best picks for UTC time and longitude. Historically, the Moon at 50% illumination was considered the best for various reasons. Through this example, we’ll demonstrate the Celestial Navigation Plugin features. And after that, we’ll show how to determine good Lunar Candidates.

6. How to input this Lunar Sight



Observer Location (DR): N43°16.1', W 076° 58.8'

Date: Aug 18, 2025 (at UTC)

UTC: 11:58:00 (at UTC) an estimated time

Height of Eye (HOE): 2.44m (8 ft).

IE (Index Error): -0.8'

Temperature: 17 in °C (63° F)

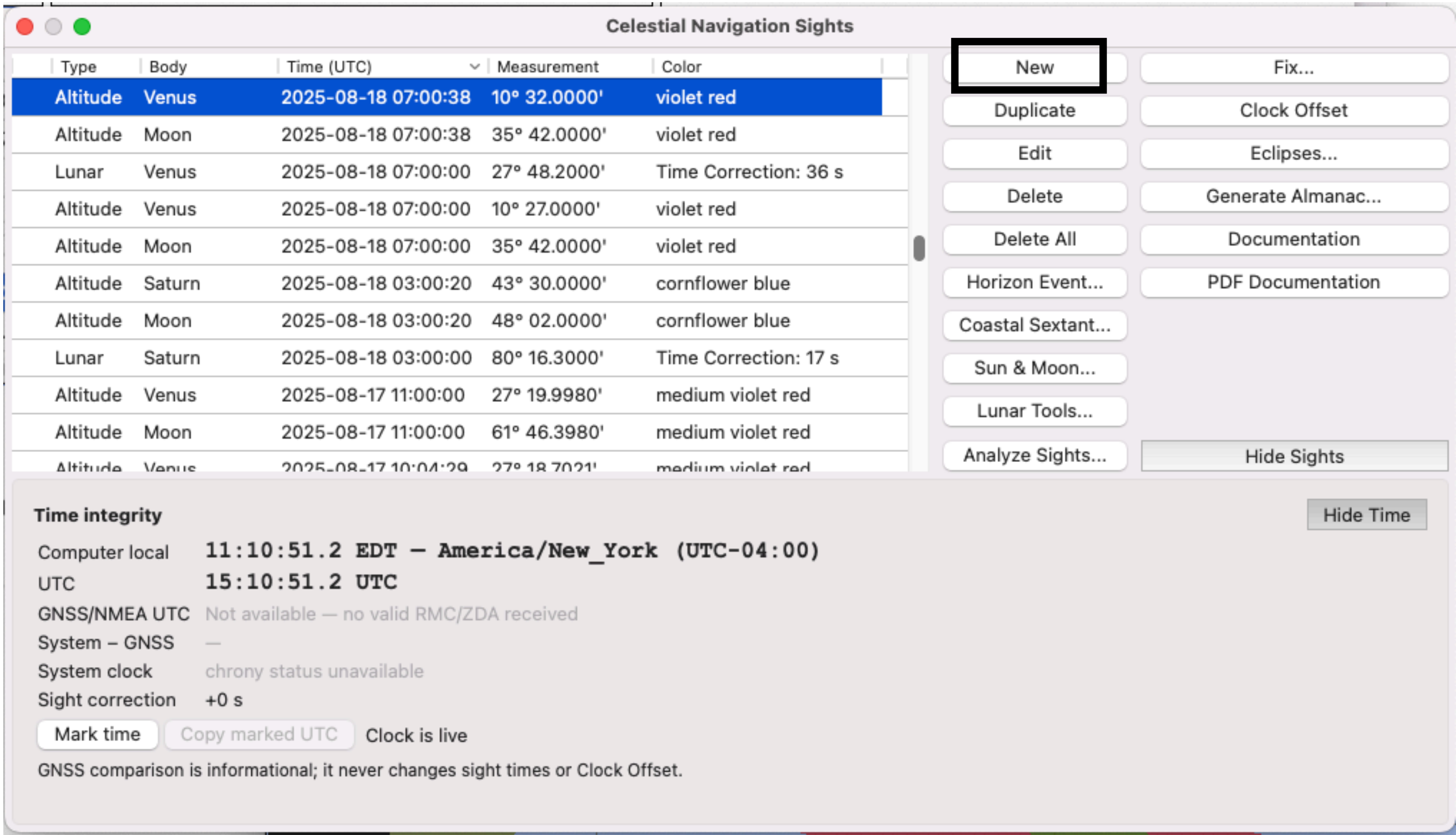
Pressure: 1013 in mb (29.92” Hg)

Lunar Distance Use Case 1 “Sun-Moon” Aug 18, 2025 11:58

6. Input Lunar Sights into Celestial Navigation Plugin

Select Celestial Navigation Icon from Open CPN Menu

The Celestial Navigation opens with Celestial Navigation “Sights” displaying existing sights



To input a new sight, select one of 2 options.

New or

Duplicate. If Duplicate, it will duplicate the highlighted sight and then Edit the Duplicated Sight. You can duplicate an Altitude sight and change it to a Lunar Sight.

6. Input Lunar Sights into Celestial Navigation Plugin

Input the Measurements

For Lunars, there are 3 angle measurements to input. The angle between the Moon and the Celestial Body, the Altitude of the Moon (w/Limb) to the Horizon. The Altitude of the Body (only the Sun has a Limb) to the Horizon.

Sight Properties Form.

The screenshot shows the 'Sight Properties' dialog box in OpenCPN. The 'Sight' tab is selected. The 'Type' is set to 'Lunar'. The 'Celestial Body' is set to 'Sun'. The 'Limb' is set to 'Near'. The 'Lunar distance (LDOpc)' is set to '59° 18.8000'' and the 'Certainty (')' is set to '0.20'. There are also sections for the 'Moon' and 'Body', each with fields for 'Altitude', 'Limb', 'Distance contact', and 'Altitude uncertainty'. The 'Moon' section has a 'Find' button. The 'Body' section has a 'Find' button. At the bottom are 'Cancel' and 'Save Changes' buttons.

Type: select Lunar.

Celestial Body: select Sun

- Also available are Planets,
- 57 Navigation Stars and Polaris. 9 Stars known as Lunar Stars (see page 14 for more about a good selection).

Select Limb: select Near or Far

- Refer to the next illustration for examples of Near or Far.
- The body is **Near** when the closest edge of a celestial body (Sun, Star, Planet) is brought to the nearest visible edge of the moon.
- The body is **Far** when bringing the nearest edge of the body across the moon’s face to the far side of the moon.

Measurement. Input Lunar Distance Angle between the Moon and the Celestial Body.

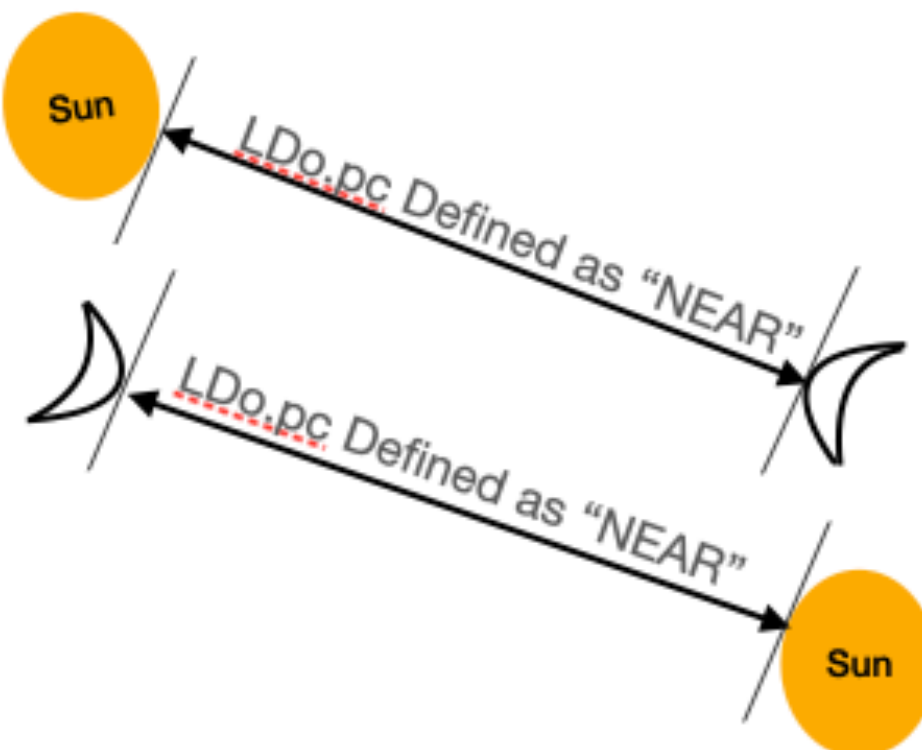
New Feature (v 2.4.66) - Sextant Measurements can be input in any of 3 OpenCPN supported “angles”. Decimal Degrees, Degrees-Min, and Degrees-Min-Sec. Redisplayed per OpenCPN display preferences

Certainty or Altitude uncertainty (new. v 2.8.x). Measurement accuracy. Used to predict position uncertainty. You’ll achieve better certainty with Lunar distance than Altitude Measurements because Lunar distance doesn’t have Dip which is a large source of inaccuracy,

Near versus Far

When you measure a Lunar Distance (Moon to the other body), you need to identify whether the other Body (Sun in Use Case 1) is meeting the Moon by touching the Near Side or the Far Side. Here are examples that illustrate. And corrections used.

Supported or Not Supported by Lunars

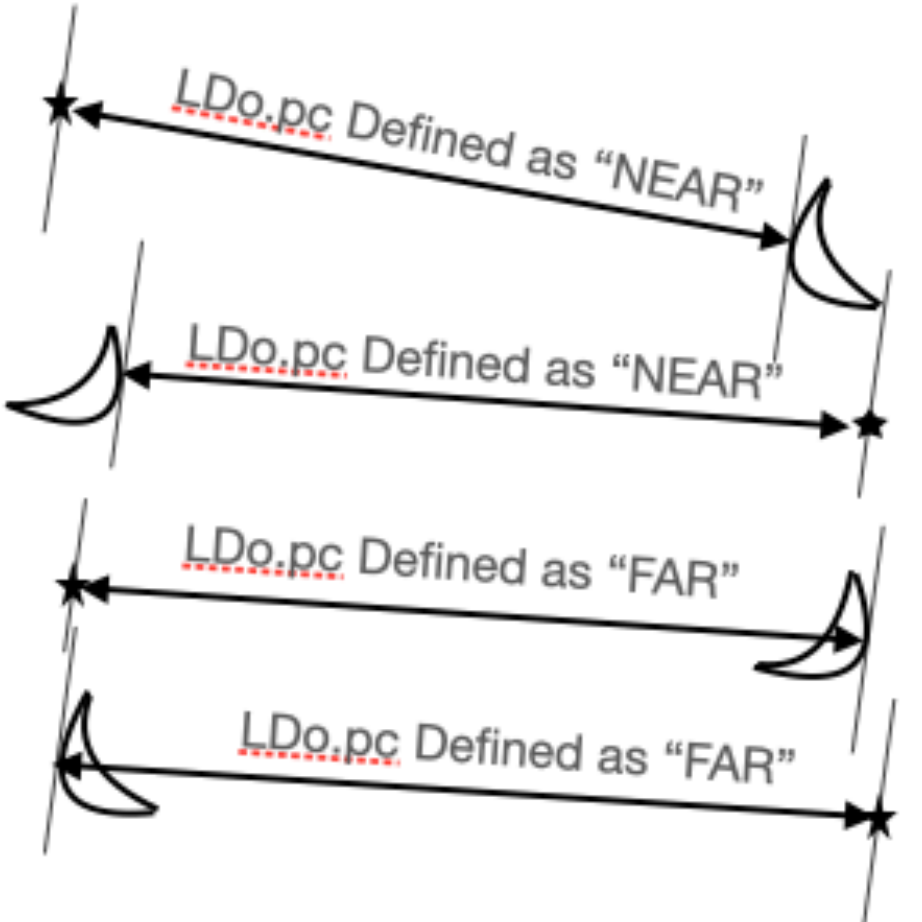


Sun Cases

Horns of the Moon are perpendicular to and point away from the Sun

NEAR equation for Sun

$LDo = LDo.pc \text{ (incl IC) + SDs PLUS SDm}$
Note: LDo has no dip adjustment



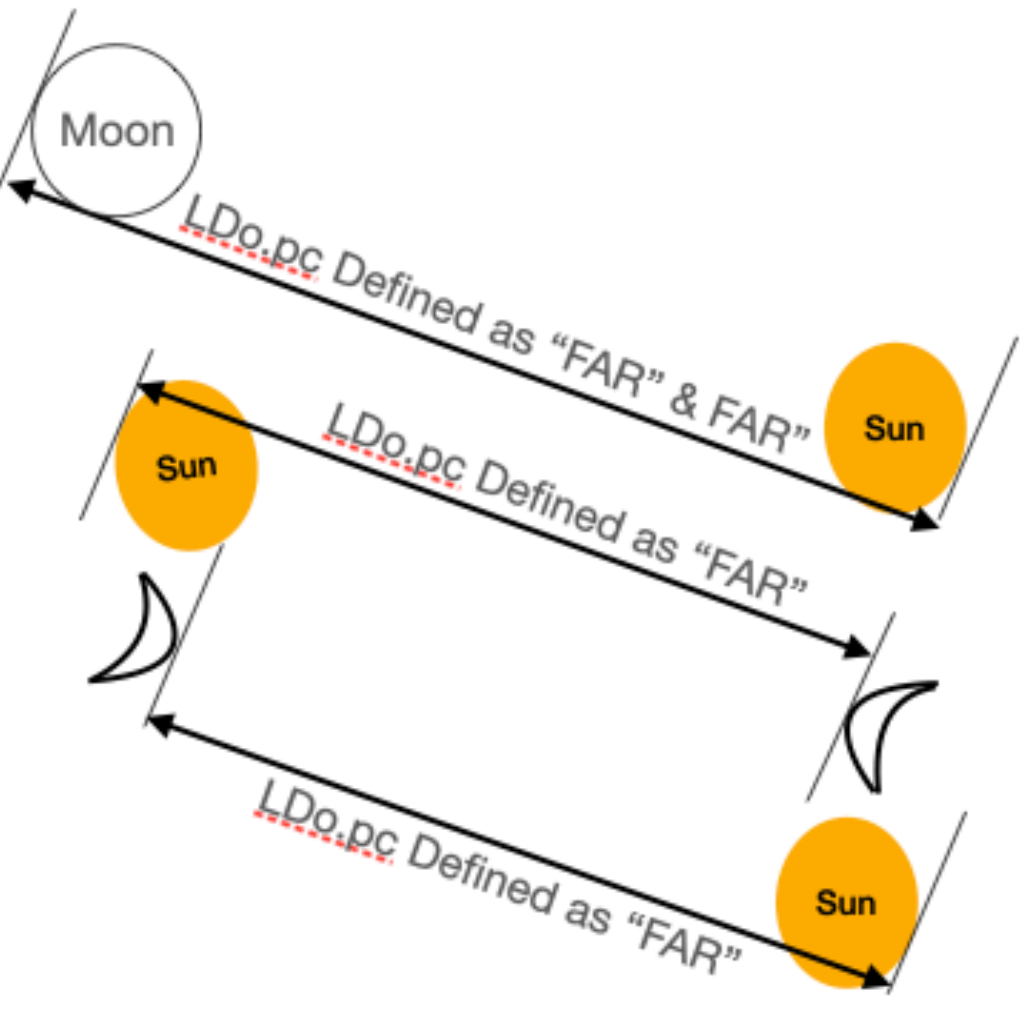
Star and Planet Cases

NEAR equation

$LDo = LDo.pc \text{ (incl IC) PLUS SDm}$
(Assume $SD=0$ for planets)
Note: LDo has no dip adjustment

FAR equation

$LDo = LDo.pc \text{ (incl IC) Minus SDm}$
(Assume $SD=0$ for planets)
Note: LDo has no dip adjustment



Sun Cases (not supported)

Horns of the Moon are perpendicular to and point away from the Sun

FAR Edge of Moon to FAR edge of Sun

$LDo = LDo.pc \text{ (incl IC) MINUS SDs MINUS SDm}$
Reason: Only can happen for Full Moon (1 day/mo)
Full Moon is 180° distant from Sun.
No marine sextant can measure more than 133 - 135°

Near Edge Moon to FAR edge Sun (supported...new v2.8.x)

Sight Challenge: Near Edge of Moon, “eclipse” the Sun to just touch the Far Edge of the Sun. In the age of lunars, it wasn’t done. The edge wasn’t sharp. **If that’s what you are doing, select Far limb because Near limb is the default.**

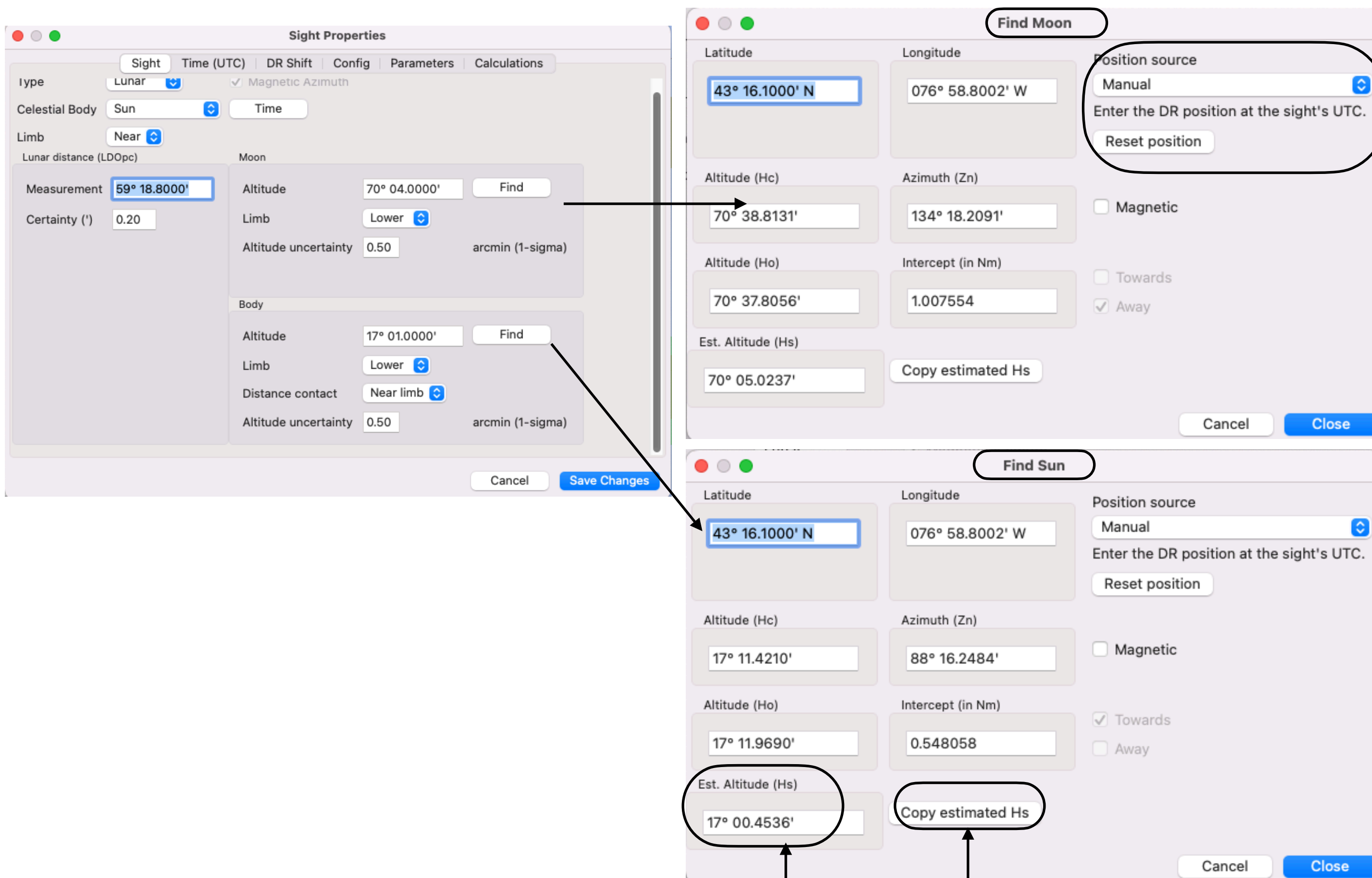
Body			Find
Altitude	17° 01.0000'		
Limb	Lower		☒ Near limb ☐ Centre ☐ Far limb
Distance contact	Near limb		
Altitude uncertainty	0.50		arcmin (1-sigma)

* Distance contact for Far limb for Sun only

Lunar Distance Use Case 1 “Sun-Moon” Aug 18, 2025 11:58

6. Input Lunar Sights into Celestial Navigation Plugin

Find Celestial Body to input Lat/Longitude & for Backyard Lunars



Backyard Lunars enabled by
Est. Altitude and Copy Hs

Est. Altitude (Hs) can be used to preset
sextant

Latitude and Longitude defaults to “Boat Position”. New Feature (v 2.8.x), Position can be grabbed from a Waypoint and other options. This eliminates typing La and Lo. In OCPN “v 5.14” Examples of valid inputs -76 58.8, -76, -76.9812345, -76 58 48 (°-min-sec), -76°58.8’, 76°58.8W, W76 58.8. For Open CPN v 5.12 for Lo, you must identify W or E (a OCPN bug fixed with v 5.14). For La, +/- can be used instead of N/S

Est. Altitude (Hs). New Feature (v 2.4.66). Cel Nav estimates an Hs by considering altitude adjustments (IE, Dip, SD/Limb, Refraction, Parallax, Augmentation). It serves three purposes.

1. Validate your sight measurement.
2. Plan your sights by pre-setting sextant index arm and to record sight, over type it.
3. Enables you to do Lunars in the backyard or when you can’t see the horizon. Copy estimated Hs and that’s it.

Press **Copy estimated Hs** button to copy it into Altitude Sight field.

Caution Hs estimate is within +/- 0.2’ altitude from 1.5° to 90. It is not calculated for negative angles or when the body (Hc) is below the horizon.

6. Input Lunar Sights into Celestial Navigation Plugin

Sight Properties

SightTime (UTC)DR ShiftConfigParametersCalculations

August202511Hours

58Minutes

0Seconds

Certainty

5400Total UTC search span (seconds)

Sequential lunar observation

Entered times: nominal UTC

All readings use the same time basis. Existing manual Clock Offset is added once. Results can check that UTC or solve an additional correction without changing the readings.

☐The three angles were recorded at different times

Lunar distance recorded timeUse the date/time controls above (reference epoch)

Moon altitude recorded time (24 h)11:58:00

Body altitude recorded time (24 h)11:58:00

☐Advance vessel between readings

COG true0.0

SOG kn0.00

Cancel

Save Changes

Time (UTC)

Input your chronometer’s time in Coordinated UTC like normal Altitude Sights.

Certainty has always been part of the Cel Nav plugin, a very nice feature. For altitude sights, the Circle of Position line width is the Measurement uncertainty. The 2 thin lines drawn down the middle of the COP plot denote Time Certainty. The Lunar solution (v 2.4.66 and v 2.8x) does not produce a Circle of Position plot.

For lunar sights, this field sets the total interval searched for UTC. A new sight defaults to 86,400 seconds, searching ±12 hours around the entered watch reading. Normally leave this unchanged. Older saved sights may retain a smaller span. Unlike the older interpolation calculation, narrowing this interval does not refine the calculation: it excludes potential solutions outside the selected interval.

If the old Lunar sight doesn’t solve, expand the certainty to 86,400 seconds or more.

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RMB 26 Sept, 2026

6. Input Lunar Sights into Celestial Navigation Plugin

Sight Properties

Sight Time (UTC) DR Shift Config Parameters Calculations

August 2025 11 Hours 58 Minutes 0 Seconds

Calendar: Sun Mon Tue Wed Thu Fri Sat
1 2
3 4 5 6 7 8 9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31

Certainty: 5400 Total UTC search span (seconds)

Sequential lunar observation

☒ Entered times: nominal UTC
Entered times: recorded watch readings

Results can check that UTC or solve an additional correction without changing the readings.

☐ The three angles were recorded at different times

Lunar distance recorded time Use the date/time controls above (reference epoch)

Moon altitude recorded time (24 h) 11 : 58 : 0

Body altitude recorded time (24 h) 11 : 58 : 0

☐ Advance vessel between readings COG true 0.0 SOG kn 0.00

Cancel Save Changes

Time (UTC).....continued

Sequential lunar observation (new feature 2.8.x)

Entered times : nominal UTC is the default setting. In this option, you are using an average UTC Time (and DR Position) for the 2 Altitude sights and Lunar Sights, And the UTC average to approximately the same time. This is the assumption of the Use Case 1.

Entered times: recorded watch readings.

On the other hand, if the Altitude Sights have significantly different time, input separate times for the Altitude Sights. A Good example, would be 1 Altitude Sight for Body, 1 for Moon, and 1 Lunar Distance. Input the times them individually.

And if the vessel is also moving during the sights, “Advance vessel between readings” and input COG and SOG. Try it, see the impact.

The plugin supports both processes.

Lunar Distance Use Case 1 “Sun-Moon” Aug 18, 2025 11:58

6. Input Lunar Sights into Celestial Navigation Plugin

To See the Lunar Distance Calculated UTC Time, Press TIME in Sight Properties Form

RESULTS

The WGS84 model solves UTC and position from the lunar distance and two altitudes, including limb contact, refraction and parallax. Dip corrects horizon altitudes only. Recorded measurements are retained.

Recover UTC from the lunar distance

2 possible UTC solutions were found. The default uses proximity to DR when available, otherwise proximity to entered UTC. Check the DR and all

Saved DR used to rank solution branches: 43° 16.1000' N, 076° 58.8002' W

UTC candidate	Additional clock correction	LD cleared	Local rate	UTC uncertainty
2025-08-18 11:56:35.200	-84.8 s	60° 06.1780'	-33.07"/h (-0.551'/min)	21.8 s (1-sigma)
2025-08-18 11:56:36.694	-83.3 s	60° 06.1646'	-33.07"/h (-0.551'/min)	21.8 s (1-sigma)

(More data below)

UTC candidate is Date and Time derived from Lunar Distance Cleared. Time is determined for Lunar Distance Cleared. And for version 2.8.x, 2 candidates are identified. Both of which are near the same time. If you want to use it for watch verification, pick one. Your decision.

Additional Clock correction The difference between UTC that was input for the sight and the UTC derived from Lunar Distance Cleared. The best Marine Sextant with “perfect” measurements will hit the marine sextant limit of 11 to 12 seconds time accuracy. You’ll do no better.

Lunar Distance Cleared. Primary result of Lunars. Similar meaning as Hc (Altitude Computed). It includes altitude adjustments: Index Correction, Limb, Refraction, Parallax, Augmentation. No Dip correction because horizon is not involved.

Local Rate (arc min/hr or arc-min/min). Speed of the Lunar Distance Change. Rate in arc-min per min of time is useful to pre-set your sextant for next sight. After taking a Lunar sight, change your index arm by .5 arc-min and wait a minute for the body to touch the moon again.

UTC uncertainty. Calculated from Measurement certainty/uncertainty of the 3 sights. In this use case, the forms show 0.2’ for Lunar Certainty, and 0.5’ for Altitude uncertainty. Use the Certainty based on your experience. Try sensitivity analysis. Input “0” for the Measurement certainty/uncertainties and UTC uncertainty drops to zero. That’s not realistic. Change Altitude uncertainty from 0.5’ to 2.0’ without changing Lunar distance Measurement Certainty. Notice Altitude uncertainty moves the UTC uncertainty needle by half a second of time. Set back the Altitude uncertainty to 0.5 and Lunar Measurement Certainty changes from 0.2 to 0.8. That’s by a factor of 4x.

6. Input Lunar Sights into Celestial Navigation Plugin

For Lunar Distance Calculated UTC Time, Press TIME in Sight Properties Form

UTC candidate	Additional clock correction	LD cleared	Local rate	UTC uncertainty
2025-08-18 11:56:35.200	-84.8 s	60° 06.1780'	-33.07/h (-0.551/min)	21.8 s (1-sigma)
2025-08-18 11:56:36.694	-83.3 s	60° 06.1646'	-33.07/h (-0.551/min)	21.8 s (1-sigma)

Candidate	Latitude	Longitude	From saved DR	Position uncertainty (horizontal RMS)
1 (nearest DR)	43° 18.8620' N	076° 36.9750' W	16.1 NM	4.05 NM (1-sigma)

Branch 1 geometry: Moon Zn 134.5°; Sun Zn 88.3°; azimuth separation (Δ Zn) 46.2°; effective crossing 46.2°. Moderate geometry. The retained

The lunar distance determines the constant watch offset. The two accompanying corrected altitudes can also intersect to give position and longitude; a rough DR/hemisphere chooses between the two mathematical intersections. All results use a WGS84 ellipsoid. Position uncertainty is horizontal RMS, not a 68% confidence-circle radius. Systematic errors are excluded.

Save lunar solution Close

The bottom half of form now shown

There is only 1 DR Shown in this example. There might be 2.

Select the candidate closest to the DR

Longitude - Historically, Longitude was the objective (after TIME). In 2.8.x, Latitude and Longitude is derived using the Moon and the Body. The 46° spread of the 2 body fix is shown in description. A spread between 60° to 120° is a valid Fix. The narrow spread might still be in the ballpark for Longitude. The accuracy of Lunars with the best Sextant and perfect to near perfect measurement is roughly 10-20 seconds, maybe 3nm. Historically, Navigators used a meridian transit to determine an accurate Latitude but not Longitude. Because the Sun's "hang time" at noon, made Longitude not accurate. They trusted their DR skills between the Meridian Transit Sight and the Lunar Sight. And if the Sun-Moon Lunar was a relatively short time from the Meridian Transit, They determined Longitude based on an assumed Latitude which they maintained by DR. Today, when we do Lunars, Lunar experts stop at UTC time because we have additional capabilities. But from the 18th to mid 19th century, Longitude was the reason for doing Lunars.

Distance from the DR - 16.1 nm. Position is calculated from DR position (La, Lo) to 2 body fix position (La, Lo).

Not a good Lunar measurement! Lunars are most sensitive to LD measurement. This bad sight was used to illustrate Lunar Distance Measurement Error needs to be minimized. Good news is that the Moon and Sun outlines are incredibly clear. With a lot of practice and reasonable ship stability, you should achieve 0.2 arc-min Lunar Distance measurement error. With averaging, maybe 0.1'

7. Sensitivity Analysis (for time)

The 59°18.8’ Lunar measurement wasn’t very accurate. To demonstrate through Sensitivity analysis, change the LD sight from 59°18.8’ to 59°18.0’. That’s almost an arc-min difference. Select the Lunar you just did, and use Duplicate. Then Edit the duplicate Lunar. Change the LD Measurement to 59°18.0’. It is an almost perfect LD measurement. Longitude error will be .1 arc min, despite the 2 body sight has insufficient spread for a fix.

Next change the LD Measurement to 59°18.2 (0.2’ error). A very good lunar sight, not perfect. Clock correction is -19s (the 2 estimates rounded). UTC Time is earlier by 19s. 19s is about 4.75 minutes of Longitude (15°/hr, .25’/sec) based on Sun time.

The following illustrates the Lunar measurement change from 18.8’ to 18.2’. And the “FIX” from Lunars dropped on the chart. And use of the Measure tool to determine distance from DR to Lunar’s Fix

Sight Properties

RESULT

Sight

Time (UTC)

DR Shift

Config

Parameters

Calculations

Type

Lunar

Magnetic Azimuth

Celestial Body

Sun

Time

Limb

Near

Lunar distance (LDOpc)

Measurement

59° 18.2000'

Certainty (')

0.20

Moon

Altitude

70° 04.0000'

Find

Limb

Lower

Altitude uncertainty

0.50

arcmin (1-sigma)

Body

Altitude

17° 01.0000'

Find

Limb

Lower

Distance contact

Near limb

Altitude uncertainty

0.50

arcmin (1-sigma)

Cancel

Save Changes

Lunar-distance UTC recovery

Results

Calculations

The WGS84 model solves UTC and position from the lunar distance and two altitudes, including limb contact, refraction and parallax. Dip corrects horizon altitudes only. Recorded measurements are retained.

Recover UTC from the lunar distance

2 possible UTC solutions were found. The default uses proximity to DR when available, otherwise proximity to entered UTC. Check the DR and all

Saved DR used to rank solution branches: 43° 16.1000' N, 076° 58.8002' W

UTC candidate	Additional clock correction	LD cleared	Local rate	UTC uncertainty
2025-08-18 11:57:40.503	-19.5 s	60° 05.5888'	-33.07/h (-0.551'/min)	21.8 s (1-sigma)
2025-08-18 11:57:41.997	-18.0 s	60° 05.5753'	-33.07/h (-0.551'/min)	21.8 s (1-sigma)

Position from the two measured altitudes

Candidate	Latitude	Longitude	From saved DR	Position uncertainty (horizontal RMS)
1 (nearest DR)	43° 18.3980' N	076° 53.2708' W	4.6 NM	4.05 NM (1-sigma)

Branch 1 geometry: Moon Zn 134.4°; Sun Zn 88.3°; azimuth separation (ΔZn) 46.1°; effective crossing 46.1°. Moderate geometry. The retained

The lunar distance determines the constant watch offset. The two accompanying corrected altitudes can also intersect to give position and longitude; a rough DR/hemisphere chooses between the two mathematical intersections. All results use a WGS84 ellipsoid. Position uncertainty is horizontal RMS, not a 68% confidence-circle radius. Systematic errors are excluded.

Save lunar solution

Close

There are 2 options here to choose. UTC from Lunar Distance is preferred. The other is a fallback.

Lunar Distance Use Case 1 “Sun-Moon” Aug 18, 2025 11:58

8. Alternative your Lunar for position (OPTIONAL)

The Lunar Sight does not plot 2 circle of positions.

To wish to visualize the impact of the UTC change, copy the Lunar sight and turn them into 2 Altitude Sights. The 2 altitude sights will plot circle of positions.

Do this by the following. Select a Lunar sight (the Lunar with 18.2' measurement), Click on the Eyes. Then press Duplicate 2 times to create 2 additional sights (see below). Select one of the duplicates, Press EDIT and then update Type to Altitude change Celestial Body, Limb, Measurement, and even Certainty corresponding to the Lunar's "Body". Change UTC time to 11:57:41 which is the Lunar corrected time. Then do the same for the other duplicate.

EDIT Type (Altitude), Body, Limb, Measurement, Certainty

The screenshot displays the 'Celestial Navigation Sights' application. The main window contains a table of sightings and a sidebar with action buttons. The table has columns for Type, Body, Time (UTC), Measurement, and Color. The third row, representing a Lunar sight of the Sun, is selected. The sidebar buttons include 'New', 'Duplicate', 'Edit', 'Delete', and 'Delete All'. The 'Duplicate' button is highlighted with a black rectangle. Two arrows originate from this button: one points to a 'Duplicate Sight' dialog box for the Sun, and the other points to a 'Duplicate Sight' dialog box for the Moon. The Sun dialog shows a measurement of 17° 01.0000' and a certainty of .5. The Moon dialog shows a measurement of 70° 04.0000' and a certainty of 0.50. To the right, a 'Sight Properties' window is open, showing a calendar for August 2025 with the 18th selected, and time fields for Hours (11), Minutes (57), and Seconds (41), along with a Certainty field set to 0.

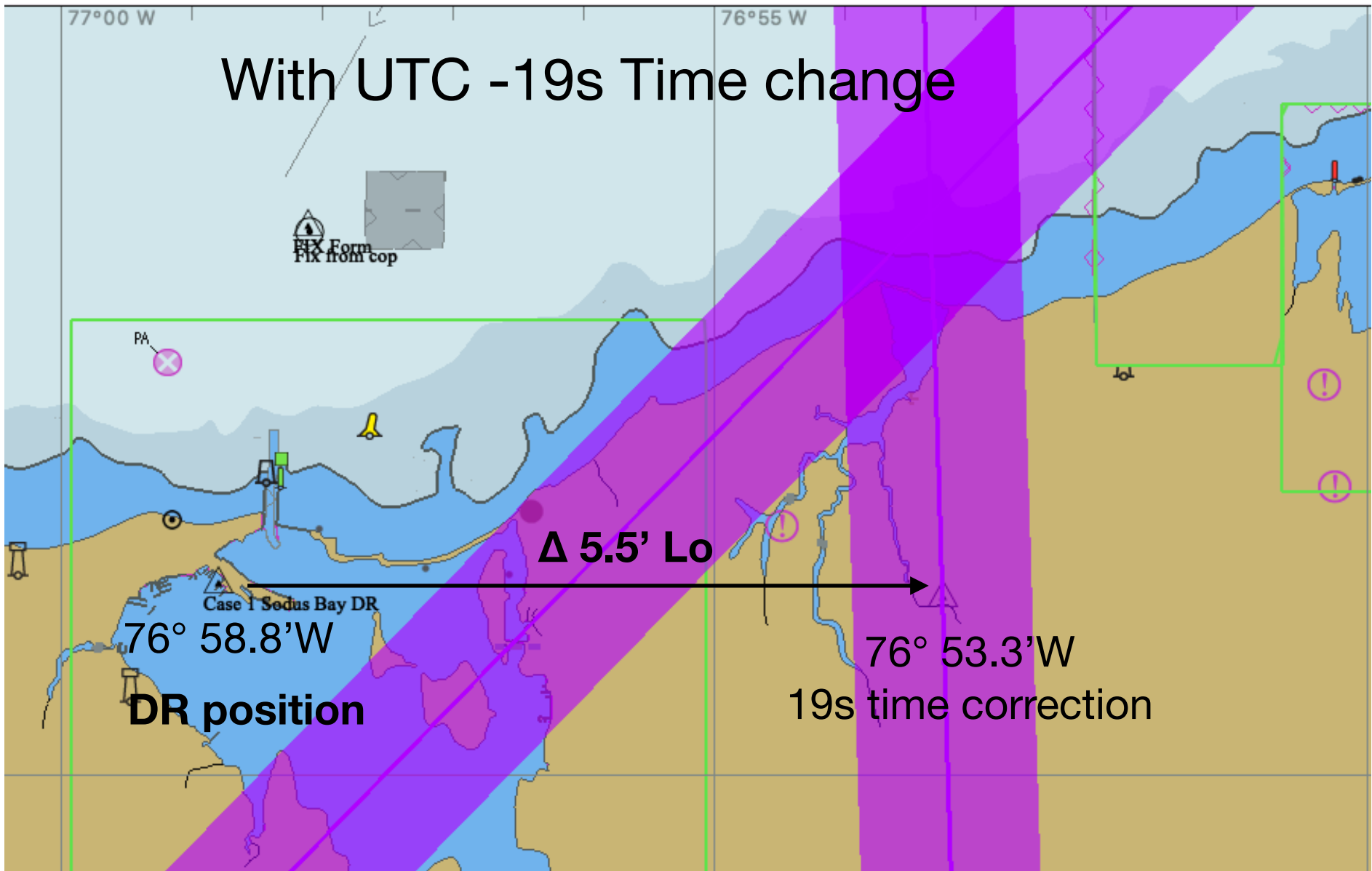
RMB 26 Sept, 2026

Lunar Distance Use Case 1 “Sun-Moon” Aug 18, 2025 11:58

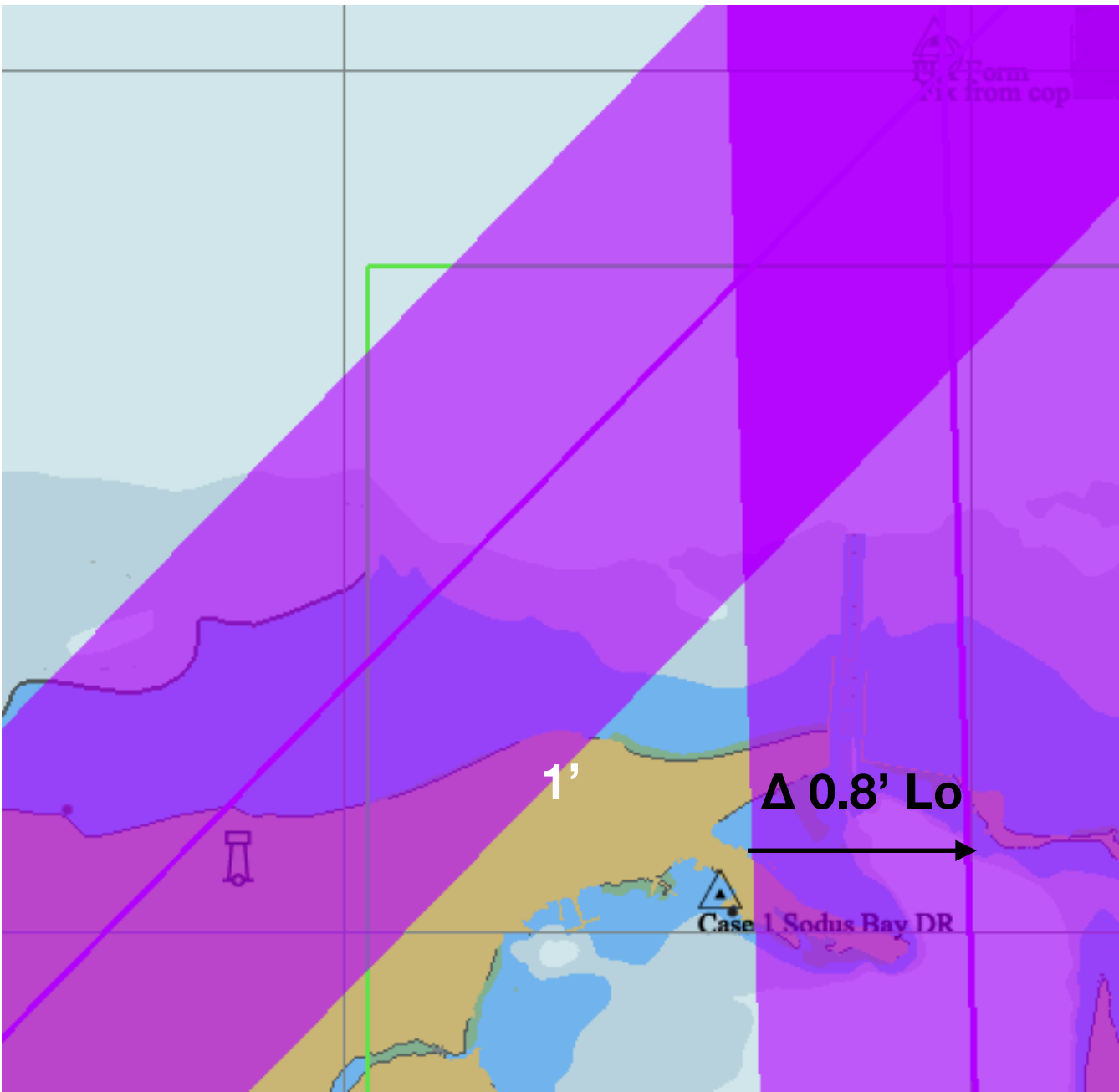
8. Analyze your Lunar for position (OPTIONAL)

Celestial Navigation Sights					
Type	Body	Time (UTC)	Measurement	Color	
Lunar	Sun	2025-08-18 11:58:00	59° 18.2000'	Time Correction: -19 s	
Altitude	Sun	2025-08-18 11:57:41	17° 01.0000'	purple	
Altitude	Moon	2025-08-18 11:57:41	70° 04.0000'	purple	

Circle of Position Plot



19 second UTC change results in 5.5' Lo change



Without UTC Time change

9. Which Celestial Bodies paired with the Moon make a great Lunar?

Following cited by experienced Lunar Navigators. No hard and fast rule to pick the best one.

#1 Sun and 50% illuminated “Half Moon”

- The most frequent Lunars pair. The boundary between Sun and Moon is exceedingly sharp/high contrast which is a big advantage for accuracy. It doesn’t literally have to be half Moon. The example was 25% illum. Use judgement.
- A 1st or 3rd quarter Moon can calculate time. Position (2.8x solution) might be weak because the 2 body fix doesn’t have a 60-120° spread typical of valid 2 body fixes. The spread is better when using a 50% illuminated Moon.

#2 Planets (Venus, Mars, Jupiter, Saturn) and Moon

- Venus, Mars, Jupiter, Saturn travel along the Ecliptic plane. They make for a good pair with the Moon.
- In the Evening and through Twilight Morning and Evening, there is almost always at least 1 visible major planet.

#3 Bright Stars near the Ecliptic Plane with the Moon. Sharp!

- There were 9 Lunar bright stars known as “Lunar Stars” Bright, close to the Ecliptic Plane, spread 360° East to West, and North to South. Lunar Distances were pre-calculated in the Nautical Almanac and Bowditch American Practical Navigator. There are a few others. Scored lower in the Lunar criteria. Vega and Deneb isn’t on the list. That’s because they aren’t remotely close to the Ecliptic. They’d be poor selections compared to the “9” to pair with moon to tell TIME. But perfectly fine to measure distance.

<u>Lunar Stars</u>	<u>SHA</u>	<u>Dec.</u>	<u>Others</u>	<u>SHA</u>	<u>Dec.</u>
Aldebaran	291°	N17°	Arcturus	146°	N19
Altair	027°	N 9°	Enif	034°	N10°
Antares	112°	S27°	Nunki	075°	S26 ^a
Fomalhaut	015°	S30°	Procyon	244°	N5°
Hamal	328°	N24°	Riegel	281°	S8°
Markab	013°	N15°	Sirius	248°	S17°
Pollux	243°	N28°			
Regulus	208°	N12°			
Spica	158°	S11°			

SHA or Sidereal Hour Angle. Shows 360° spread
DEC or Declination. Shows North vs South Hemisphere stars

9. Which Celestial Bodies paired with the Moon make a great Lunar?

#4 Selection and what LD measurement to target

- Outdoors, pick a bright sharp star/planet on the Moon's trajectory/path, "Ecliptic". They are often among the best Lunar pair.
- If Lunar Distance is too small, the Lunar Distance can be "Cleared" if done carefully and time might be ok. But, Position (Lat or Lon) might not be so good depending on how narrow the spread is between the two bodies.
- If the Lunar Distance hourly change rate is very small, then LD Cleared can be good, but time will be less accurate.
- If large (i.e. $> 100^\circ$), it gets harder to sight the body with the moon. Pre-set the Sextant and practice helps.
- A good LD Range is $60^\circ - 100^\circ$. Historically, a 90° spread when doing Half Moon-Sun Lunar was considered ideal.
- Moon or Celestial Body Altitude higher than horizon ($>5^\circ$ to 10°) to avoid anomalous atmospheric refraction.

#5 Full or almost Full Moon and a Star. Does Moon's glow wash out dim stars or make the touchpoint unsharp or unclear? Measuring Far Edge of the Sun to the Near Edge of the Moon is not going to give you good results compared to Near Edge of the Sun to the Near Edge of the Moon. The clearer and more contrast for the boundary layer, the better the measurement. That's why the standard Lunar Practice is Near Edge of the Moon to the Near Edge of the Sun.

**There is no universally accepted rule. Selection depends on your objective: LD
Clear accuracy, Time accuracy, Latitude/Longitude.
Use your "Seaman eyes"! Try it and see what works for you.**

In Use Case 1, it was easy to find the Sun and Moon in the sky! But at Twilight, what should be selected to pair with the Moon?

New to release 2.8.x, a Star Finder and Lunar Tools are available. They will be demonstrated next.

10. Planning for a Lunar Sight

Use Case 2 Jupiter-Moon UTC Dec 9, 2025 0430

New Feature (v 2.8.x). Use “Sun, Moon and Sight Planner” for a few Lunar Pair options

Input
Lat and Lon Position
UTC Date and Time

Display event time.
UTC, PC clock,
or Zone Time (No DST)

Position

Manual

Latitude

43° 57.0000' N

Longitude

078° 09.0000' W

Time

Manual date/time

Date (UTC)

2025-12-09

Time (UTC)

04:30:00

Enter time as

UTC

Display event times as

Fixed offset

Zone offset (h)

-5.0

Date/time entry

Nautical: YYYY-MM-DD, 24-hour

Auto zone from longitude

☒

Eye height (m)

3.0

Calculate / refresh

Resolved UTC: 2025-12-09 04:30:00 UTC

☒ Time-tagged moving observer

Reference time is the entry above

COG (true)

0.0

SOG (kn)

0.0

Planning context updated; calculations remain fully offline.

Events

Bodies & Best Sights

Almanac

Noon & Polaris

☒ Limit recommendations by Hc

Minimum

5.0

Maximum

85.0

°

All catalogued bodies above the horizon remain listed; these limits affect recommended pairs and triads only.

Body	Hc	Zn true	GHA	Dec	Mag	Sci
Jupiter	44° 25.1570'	103.4°	29° 50.5944'	21° 29.9645' N	-2.0	93
Rigel	36° 51.6560'	165.8°	66° 43.4126'	08° 10.2447' S	0.1	80
Moon	26° 35.2530'	89.2°	7° 30.0093'	18° 36.6495' N	-12.7	78
Procyon	33° 03.5029'	118.6°	30° 30.3851'	05° 09.5111' N	0.3	72
Alnilam	42° 24.8972'	156.9°	61° 17.2393'	01° 11.0736' S	1.7	70
Betelgeuse	48° 52.2064'	146.2°	56° 31.5461'	07° 24.7429' N	0.5	68
Alpheratz	37° 07.5619'	277.0°	143° 14.5153'	29° 14.2369' N	2.1	67
Sirius	21° 47.2436'	144.7°	44° 05.8603'	16° 45.0213' S	-1.5	67
Pollux	47° 48.4262'	95.4°	28° 56.6998'	27° 57.7181' N	1.1	65
Polaris	44° 31.9764'	359.7°	98° 52.4542'	89° 22.5772' N	2.0	65
Dubhe	33° 55.5841'	34.5°	339° 20.6667'	61° 36.3499' N	1.8	64
Menkar	45° 54.7865'	211.9°	99° 45.6836'	04° 11.5594' N	2.5	59
Bellatrix	50° 26.3760'	157.6°	64° 02.3415'	06° 22.4305' N	1.6	58
Kochab	29° 11.5427'	7.6°	283° 01.4274'	74° 02.6627' N	2.1	54
Schedar	52° 40.3806'	310.5°	135° 10.5673'	56° 41.0879' N	2.2	50

Create selected sight...

Recommended pair / triad	Score	Geometry
Moon / Polaris	84	Crossing angle 90°
Jupiter / Kochab	82	Crossing angle 96°
Rigel / Moon	82	Crossing angle 77°
Jupiter / Polaris	81	Crossing angle 104°

Sky plot magnitude

3 or brighter

☐ Show below horizon

Sun yellow; Moon blue; planets red; grey stars = daylight; hollow = below horizon

New

Fix...

Duplicate

Clock Offset

Edit

Eclipses...

Delete

Generate Almanac...

Delete All

Documentation

Horizon Event...

PDF Documentation

Coastal Sextant...

Sun & Moon...

Lunar Tools...

Analyze Sights...

Hide Sights

Hide Time

Sky plot magnitude

1 or brighter

☐ Show below horizon

Sun yellow; Moon blue; planets red; grey stars = daylight; hollow = below horizon

10. Planning for a Lunar Sight

Use Case 2 Jupiter-Moon UTC Dec 9, 2025 0430

New Feature (2.8.x) Lunar Tool's Lunar Planner identifies best pair.

New

Fix...

Duplicate

Clock Offset

Edit

Eclipses...

Delete

Generate Almanac...

Delete All

Documentation

Horizon Event...

PDF Documentation

Coastal Sextant...

Sun & Moon...

Lunar Tools...

Analyze Sights...

Hide Sights

Hide Time

Lunar
Candidates
Jupiter was best
overall (excluding
Longitude
considerations)

Lunar Distance and Sextant Tools										
UTC date/time entry Nautical: YYYY-MM-DD, 24-hour										
Lunar sequence Lunar planner Sextant check Advanced										
Pairs are ordered by 0.1' time (shortest first), with pairs below the geometric horizon last. The time is the approximate UTC change corresponding to 0.1' of lunar distance. Visibility and instrument range require your judgement.										
Latitude 43° 57.0000' N					Longitude 078° 09.0000' W					
UTC date 2025-12-09					UTC time 04:30:00					
Body	Below horizon	Distance	Rate	0.1' time	Moon altitude	Body altitude	Moon Zn true	Body Zn true	Moon illum.	Magnitude
Elnath		51° 59.2404'	+34.4'/h	10.5 s	26° 35.2530'	70° 48.5909'	89.2°	138.2°	74.8%	1.6
Regulus		15° 19.6806'	-34.3'/h	10.5 s	26° 35.2530'	11° 49.2781'	89.2°	84.9°	74.8%	1.4
Hamal		96° 25.3662'	+34.2'/h	10.5 s	26° 35.2530'	54° 39.9210'	89.2°	246.5°	74.8%	2.0
Saturn		139° 44.8384'	+34.2'/h	10.5 s	26° 35.2530'	10° 42.3485'	89.2°	253.6°	74.8%	0.5
Pollux		21° 45.4544'	+34.0'/h	10.6 s	26° 35.2530'	47° 48.4262'	89.2°	95.4°	74.8%	1.1
Jupiter		21° 10.0392'	+34.0'/h	10.6 s	26° 35.2530'	44° 25.1570'	89.2°	103.4°	74.8%	-2.0
Aldebaran		65° 12.8769'	+33.7'/h	10.7 s	26° 35.2530'	62° 34.1031'	89.2°	176.2°	74.8%	0.9
Menkar		90° 47.9932'	+32.9'/h	11.0 s	26° 35.2530'	45° 54.7865'	89.2°	211.9°	74.8%	2.5
Capella		55° 00.9680'	+31.8'/h	11.3 s	26° 35.2530'	81° 10.7918'	89.2°	72.2°	74.8%	0.1
Alpheratz		115° 52.4816'	+31.0'/h	11.6 s	26° 35.2530'	37° 07.5619'	89.2°	277.0°	74.8%	2.1
Mirfak		73° 43.5068'	+30.8'/h	11.7 s	26° 35.2530'	77° 35.3340'	89.2°	304.5°	74.8%	1.8
Bellatrix		56° 18.4391'	+30.7'/h	11.7 s	26° 35.2530'	50° 26.3760'	89.2°	157.6°	74.8%	1.6
Markab		136° 02.9322'	+30.6'/h	11.8 s	26° 35.2530'	16° 59.1695'	89.2°	275.2°	74.8%	2.5

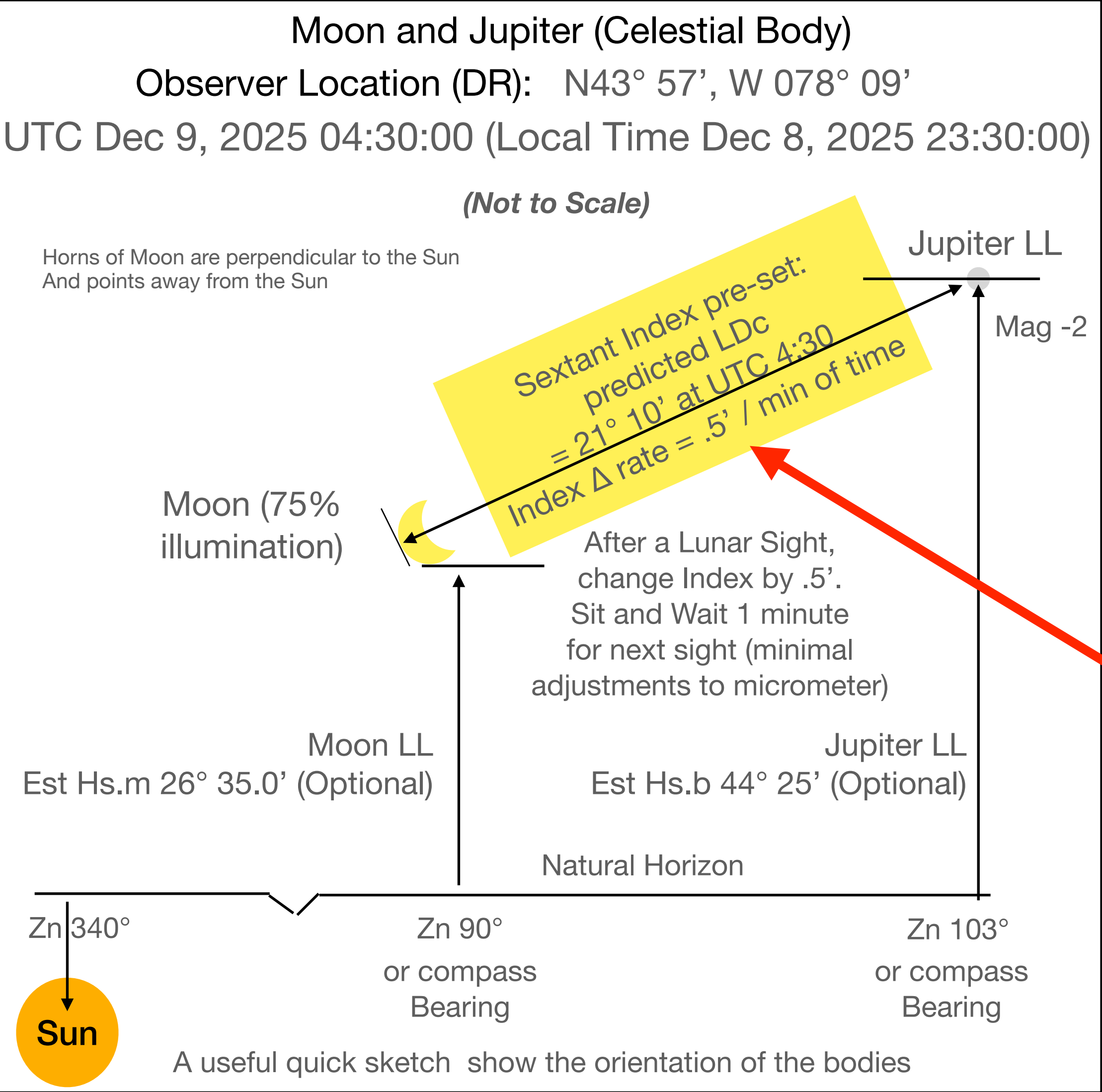
LD Distance (Jupiter @ 21° 10' is ok for Time, narrow spread for longitude is negative, Aldebaran spread @ 65° is better. Saturn @139° too far)
UT Time Accuracy (Jupiter's 10.5 sec is great, Deneb's 38.6s is poor for TIME)
Magnitude (Jupiter's -2.0 is bright, Menkar's 2.5 is dim)
Pollux versus Jupiter. They are close to each other. Both along the ecliptic. Jupiter was much brighter, appears first, and then grabs the attention in the night sky,

Alphard	27° 40.1779'	-17.3'/h	20.8 s	26° 35.2530'	4° 42.3240'	89.2°	107.0°	74.8%	2.0
Acamar	104° 22.9105'	+17.3'/h	20.8 s	26° 35.2530'	3° 23.8903'	89.2°	197.2°	74.8%	2.9
Sirius	50° 27.6698'	+15.1'/h	23.8 s	26° 35.2530'	21° 47.2436'	89.2°	144.7°	74.8%	-1.5
Polaris	71° 24.3164'	+13.1'/h	27.6 s	26° 35.2530'	44° 31.9764'	89.2°	359.7°	74.8%	2.0
Alkaid	62° 21.5931'	-12.8'/h	28.2 s	26° 35.2530'	9° 35.9938'	89.2°	25.4°	74.8%	1.9
Deneb	115° 38.4228'	+9.3'/h	38.6 s	26° 35.2530'	15° 21.0792'	89.2°	319.5°	74.8%	1.2

Key Message - Go outside and see for yourself. The best Lunar Pair is usually obvious from the sky (cloud cover). And that night, Jupiter was fantastic and was the obvious choice if you were interested in Time.

10 Planning for Lunar Sights

Use Case 2 Lunar Distance for Moon and Jupiter



- * Lunar Pair: Moon and Jupiter (Celestial Body)
 - * UTC Date: Dec 9, 2025 (at UTC), (Local Dec 8, 2025)
 - * UTC: 04:30:00 (UTC) estimated sight start time
- Observer Location (DR): N43° 57', W 078° 09'
- Height of Eye (HOE): 2.1m (7 ft).
- IE (Index Error): -1.5'
- Temperature: -12 °C (10.4°F)
- Pressure: 1027 in mb (30.3 inches)

Step 1 - Create a New Lunar Sight with minimum data
Step 2 - Lunar Distance and Sextant Tools, Lunar Planner

Lunar Distance and Sextant Tools

UTC date/time entry Nautical: YYYY-MM-DD, 24-hour

Lunar sequence Lunar planner Sextant check Advanced

Pairs are ordered by 0.1' time (shortest first), with pairs below the geometric horizon last. The time is the approximate UTC change corresponding to 0.1' of lunar distance. Visibility and instrument range require your judgement.

Latitude 43° 57.0000' N Longitude 078° 09.0000' W

UTC date 2025-12-09 UTC time 04:30:00

Body	Below horizon	Distance	Rate	0.1' time	Moon altitude	Body altitude	Moon Zn true	Body Zn true	Moon illum.	Magnitude
Elnath		51° 59.2404'	+34.4°/h	10.5 s	26° 35.2530'	70° 48.5909'	89.2°	138.2°	74.8%	1.6
Regulus		15° 19.6806'	-34.3°/h	10.5 s	26° 35.2530'	11° 49.2781'	89.2°	84.9°	74.8%	1.4
Hamal		96° 25.3662'	+34.2°/h	10.5 s	26° 35.2530'	54° 39.9210'	89.2°	246.5°	74.8%	2.0
Saturn		139° 44.8384'	+34.2°/h	10.5 s	26° 35.2530'	10° 42.3485'	89.2°	253.6°	74.8%	0.5
Pollux		21° 45.4544'	+34.0°/h	10.6 s	26° 35.2530'	47° 48.4262'	89.2°	95.4°	74.8%	1.1
Jupiter		21° 10.0392'	+34.0°/h	10.6 s	26° 35.2530'	44° 25.1570'	89.2°	103.4°	74.8%	-2.0
Aldebaran		65° 12.8769'	+33.7°/h	10.7 s	26° 35.2530'	62° 34.1031'	89.2°	176.2°	74.8%	0.9
Menkar		90° 47.9932'	+32.9°/h	11.0 s	26° 35.2530'	45° 54.7865'	89.2°	211.9°	74.8%	2.5
Capella		55° 00.9680'	+31.8°/h	11.3 s	26° 35.2530'	81° 10.7918'	89.2°	72.2°	74.8%	0.1
Alpheratz		115° 52.4816'	+31.0°/h	11.6 s	26° 35.2530'	37° 07.5619'	89.2°	277.0°	74.8%	2.1
Mirfak		73° 43.5068'	+30.8°/h	11.7 s	26° 35.2530'	77° 35.3340'	89.2°	304.5°	74.8%	1.8
Bellatrix		56° 18.4391'	+30.7°/h	11.7 s	26° 35.2530'	50° 26.3760'	89.2°	157.6°	74.8%	1.6
Markab		136° 02.9322'	+30.6°/h	11.8 s	26° 35.2530'	16° 59.1695'	89.2°	275.2°	74.8%	2.5

11 Pre-Setup your Sextant.

Use Case 2 Lunar Distance for Moon and Jupiter

Preset your Sextant to 22°10' for the LD. You can press the Time to get a more refined pre-setup. The Moon-Jupiter intercept should be near by a Predicted LD.

Sight Properties

Type: Lunar

Celestial Body: Jupiter

Limb: Far

Measurement: 22° 10.0000'

Certainty: 1.00

Moon

Altitude: 0

Limb: Lower

Altitude uncertainty: 0.20

Body

Altitude: 0

Limb: Lower

Altitude uncertainty: 0.20

Pre-set your Sextant to 25° 30'. The moon will be near the visible Horizon

Find Moon

Latitude: 43° 57.0000' N

Longitude: 078° 09.0000' W

Position source: Manual

Altitude (Hc): 26° 35.2530'

Azimuth (Zn): 89° 14.5323'

Altitude (Ho): 0° 38.4841'

Intercept (in Nm): 1556.768838

Est. Altitude (Hs): 25° 29.9248'

Copy estimated Hs

Sight Properties

Time (UTC)

December 4, 2025

4:30:00

Total UTC search span (seconds): 31536

Sequential lunar observation

Entered times: nominal UTC

All readings use the same time basis. Existing manual Clock Offset is added once. Results can check that UTC or solve an additional correction without changing the readings.

The three angles were recorded at different times

Lunar distance recorded time: 4:35:8

Moon altitude recorded time (24 h): 4:35:8

Body altitude recorded time (24 h): 4:35:8

Advance vessel between readings: COG true 0.0 SOG kn 0.00

Pre-set your Sextant to 44° 27'. The Jupiter will be near the visible Horizon

12 Backyard Lunars. You don't need to know Altitude of the 2 bodies

Use Case 2 Lunar Distance for Moon and Jupiter

Preset your Sextant to 21°10' for the LD. You can press the Time to get a more refined pre-setup. The Moon-Jupiter intercept should be near by

Sight Properties

Sight

Time (UTC)

DR Shift

Config

Parameters

Calculations

Type

Lunar

Celestial Body

Jupiter

Limb

Far

Lunar distance (LDOpc)

Measurement

22° 10.0000'

Certainty (')

1.00

Moon

Altitude

0

Limb

Lower

Altitude uncertainty

0.20

Body

Altitude

0

Limb

Lower

Distance contact

Near limb

Altitude uncertainty

0.20

Find Moon

Latitude

43° 57.0000' N

Longitude

078° 09.0000' W

Altitude (Hc)

26° 35.2530'

Altitude (Ho)

0° 38.4841'

Est. Altitude (Hs)

25° 29.9248'

Intercept (in Nm)

1556.768838

Intercept (in Nm)

2700.932895

Est. Altitude (Hs)

44° 27.3097'

Position source

Manual

Enter the DR position at the sight's UTC.

Reset position

on at the sight's UTC.

Cancel

Close

Sight Properties

Sight

Time (UTC)

DR Shift

Config

Parameters

Calculations

December

2025

4

Hours

30

Minutes

0

Seconds

31536

Total UTC search span (seconds)

Sequential lunar observation

Entered times: nominal UTC

All readings use the same time basis. Existing manual Clock Offset is added once. Results can check that UTC or solve an additional correction without changing the readings.

The three angles were recorded at different times

Lunar distance recorded time

Use the date/time controls above (reference epoch)

Moon altitude recorded time (24 h)

4

:

35

:

8

Body altitude recorded time (24 h)

4

:

35

:

8

Advance vessel between readings

COG true

0.0

SOG kn

0.00

Automatic Copy

Automatic Copy

Press copy estimated Hs, and the Est Altitude (hs) will copy into the Altitude.

Pre-set your Sextant to 44° 27'. The Jupiter will be near the visible Horizon

25

RMB 26 Sept, 2026

11. Tips to improve Lunar sights Accuracy?

- A high quality Sextant. Collimated through the entire index arm arc-range. Minimize all adjustable Sextant Errors. They should be as close to 0 as possible. If you don't know it, check collimation using the Lunar Tools form (new 2.8.x).
- Your initial goal is to consistently achieve better than $\pm 0.3'$ LDo measurement accuracy.
- Experienced Navigators have achieve between $\pm 0.1'$ to $\pm 0.2'$ LDo measurement accuracy. 0.1 is achieved by averaging Lunar Sights. The sight error should be between $3'$ to $6'$ Longitude.

Easier to achieve LDo measurement accuracy than Altitude sight measurement accuracy. Because the LDo measurement has no DIP! The problem with DIP is the uncertainty in the Height of Eye due to your vessel bobbing up and down, timing the swells when sighting the horizon, and unpredictable nature of anomalous refraction.

None of this DIP uncertainty exists when measuring Lunar Distance LDo. There is no DIP!

- You should shoot 2 Altitude Sights for Lunars. The purpose is to determine Parallax, Refraction, and Limb corrections. And for Lunars, these altitude sights don't need to be "spot on".

The 2 Altitude Sights can afterwards be used for a 2-body fix. Your Altitude Sight goal should initially be $1'$ or less "sight error". Sextant professionals consistently get no worse than $0.5'$ sight error. That gives an idea for improvement potential. Dip correction uncertainty is the cause of some of the "Sight error", and the error depends on ship size and stability. So there is judgement to what goal is achievable.

Combine that with a Polar sight during twilight, a Sun Sight (sun-run-sun), Meridian Transit, you'll have a Running Fix with 3 bodies.

- In the 1700's, Celestial mechanics was good enough for the times, but not perfect. Now the Commercial Edition of the Nautical Almanac is accurate mostly $0.0'$ error with occasional $0.1'$ error. And U.S. Navy Observatory Celestial Data is better than that. In a 2024 study of the Celestial Nav Plugin Altitude sights versus Commercial Edition of the Nautical Almanac, the Plugin was ± 0.1 (@1 standard deviation) of the Nautical Almanac. And at least 50% of the time, it matched the Almanac.
- The Use Case 1 Lunar measurement was initially off by $0.6'$. Room for improvement! Try the following tips.

11. Tips to improve Lunar Accuracy

- A high quality metal sextant with good inspection report. No arc errors throughout the angle range you'll measure.
- The higher the magnification, the better. 7X is readily available.
- Well Adjusted. Perpendicularity, then side and index error. Goal is as close to 0.0 as possible. These errors impact Lunar accuracy.
- The very best Lunar Distance observation. Altitude sight accuracy is not as important for UTC Time.
- Practice-Practice-Practice. Ship Navigators took sights almost daily. That's why they are so good.
- Select a celestial body that is traveling along the same trajectory plane as the Moon.
 - Start off with Sun, it gives the best result.
 - Then try Planets because they travel close to the Moon's trajectory.
 - Then "Lunar Stars". They were bright and close to the ecliptic (some closer than others).
- Don't select a celestial body just because it is near the Moon. There are often better candidates.
- Consider a body who's LD angle is between 60° to 100°. 90° was considered optimal with the Sun. Smaller than 60° is easier. Pre-set Sextant make larger angles possible. Sometimes, larger LDo fixes problems when the body isn't on the Moon travel path. Larger than 100° measurements are harder to do, but possible. Did your sextant inspection report include angles > 90°? Cel Nav Plugin has a Collimation form in Lunar Tools to assess your Sextant's collimation.
- Don't select a star that is far off the moon's path or far off the ecliptic (e.g. Vega) and expect good Time results.
- Ensure the sharpest focus and high contrast. You want to see a clear outline of the Moon and body.
- Hold the sextant frame with both hands to avoid wobbling. Do small and few micrometer adjustments. Presetting helps
- Make sure the body is aligned (an imaginary line) to the center of the Field of View. That way, the sextant frame is aligned to the plane of the bodies. You'll be rocking your sextant to ensure the touch is tangent.
- Preset the sextant to the expected Lunar Distance. Then wait for the body to touch the Moon.
- In the Lunars chapter of Bowditch's American Practical Navigator, the editors recommended that the center of the body (planet, star) touches the moon. This indicates that the navigators used sextants with good magnification.
- Try not to do "live" adjustments to the micrometer. Pre-set sextant to the predicted LD, then patiently wait for the "touch" to occur, and then MARK. Then turn the micrometer a little ahead by the Δ of LD arc-min/min of time and wait for the next MARK.

11. Tips to improve Lunar Accuracy (continued)

- Bowdich chapter on Lunars has a lot of practical advice.
- Average the run of Sights. Or Plot the sights taken and choose an angle measurement/time pair on the best fit line

How to Choose the Sight data from a Run of sights using a Plot of sights.

- Slide 5 shows how to shoot a run of sights for Lunars.
- And calculate Average Time and Average Measurement for the 3 groups: Lunar Sight, Moon Sight, and Celestial Body Sight. And then you'd input the average data into the Celestial Nav plugin forms. But, what if the Average time isn't roughly the same?
- The alternative is to plot the individual Sight data and fit 3 lines: 1 line through the Lunar sights, 1 line through the Moon sights, and 1 line through the Celestial Body sight. It is best done on 1 plotting sheet.
 - The Altitude / Lunar Measurements the y-axis. The elapsed minutes the x-Axis.
 - Visually draw 3 lines to fit the 3 sets of data. Or use a spreadsheet to fit it for you.
 - Visually, select somewhere along the X-axis and draw a vertical line from that point.
 - Where vertical line intersects the 3 plotted lines, is the measurement data for x-axis UTC Time.

How to pre-setup your sextant index to make it easier to do a Lunar Sight?

- It is sometimes difficult to bring the Celestial Body across to the Moon. A simpler approach is to predict the LD angle for a given time and Lunar Pair, and set your sextant to that LD angle.
- Then, sight the Moon. With a little rotation of your sextant, you'll see the Celestial object near the moon.

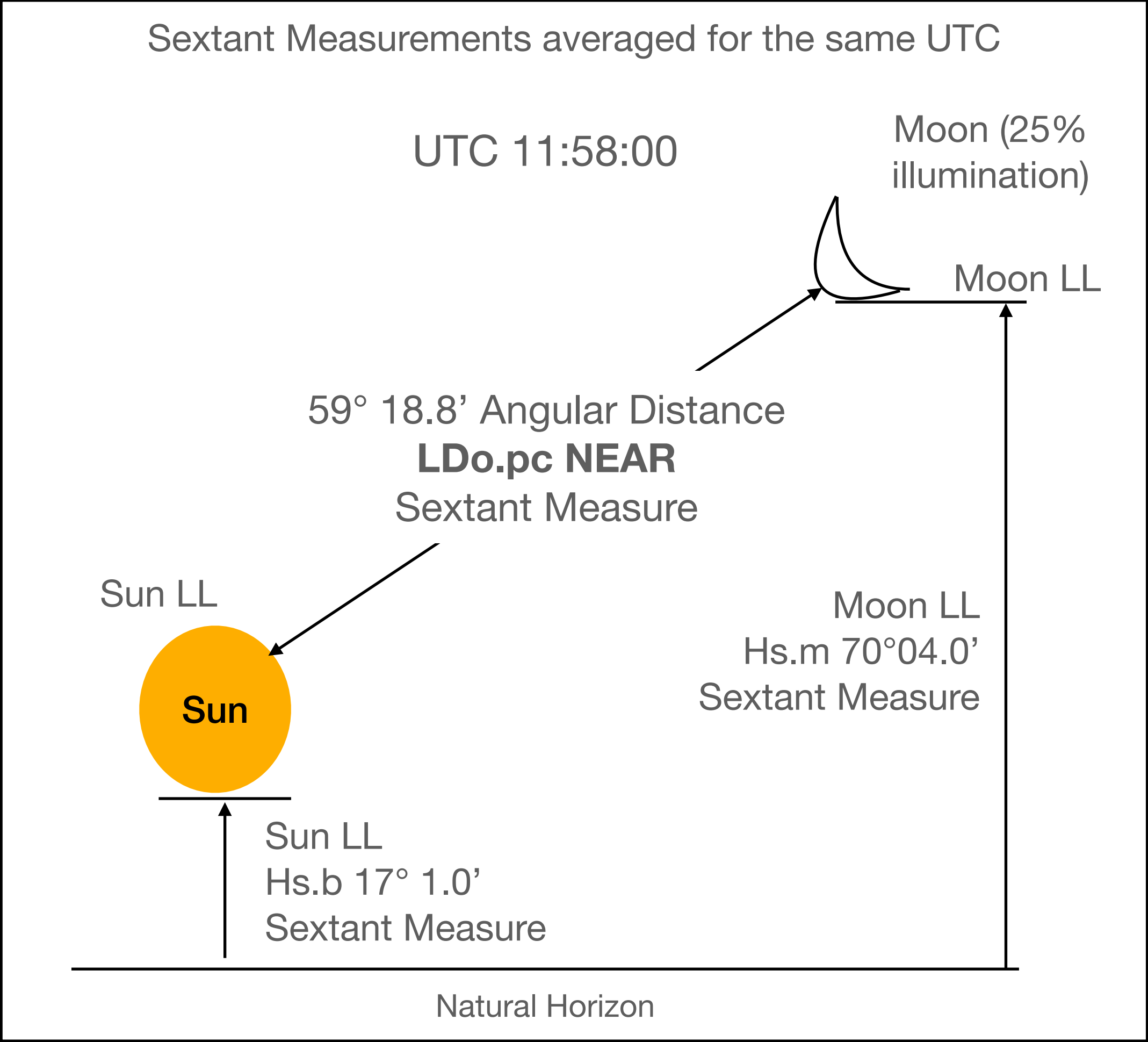
* Lunar Pair: Moon and Jupiter (Celestial Body)	}
* Date: Dec 8, 2025 (at UTC)	
* UTC: 04:30:00 (UTC) estimated sight start time	
Observer Location (DR): N43° 57', W 078° 09'	
Height of Eye (HOE): 2.1m (7 ft).	
IE (Index Error): -1.5'	
Temperature: -12 °C	
Pressure: 1027 in mb	

* The minimum of what you need to know to pre-set your sextant. Lunar Pair and Date & Time

The rest, you'll eventually need to know and input at your convenience

Back to Use Case 1: 18 Aug 2025 11:58 Sun-Moon

14. Theory and Logic - Direct Triangle Method



Observer Location (DR): N43°16.1', W 076° 58.8'

Date: Aug 18, 2025 (at UTC)

UTC: 11:58:00 (at UTC) an estimated time

Height of Eye (HOE): 2.44m (8 ft).

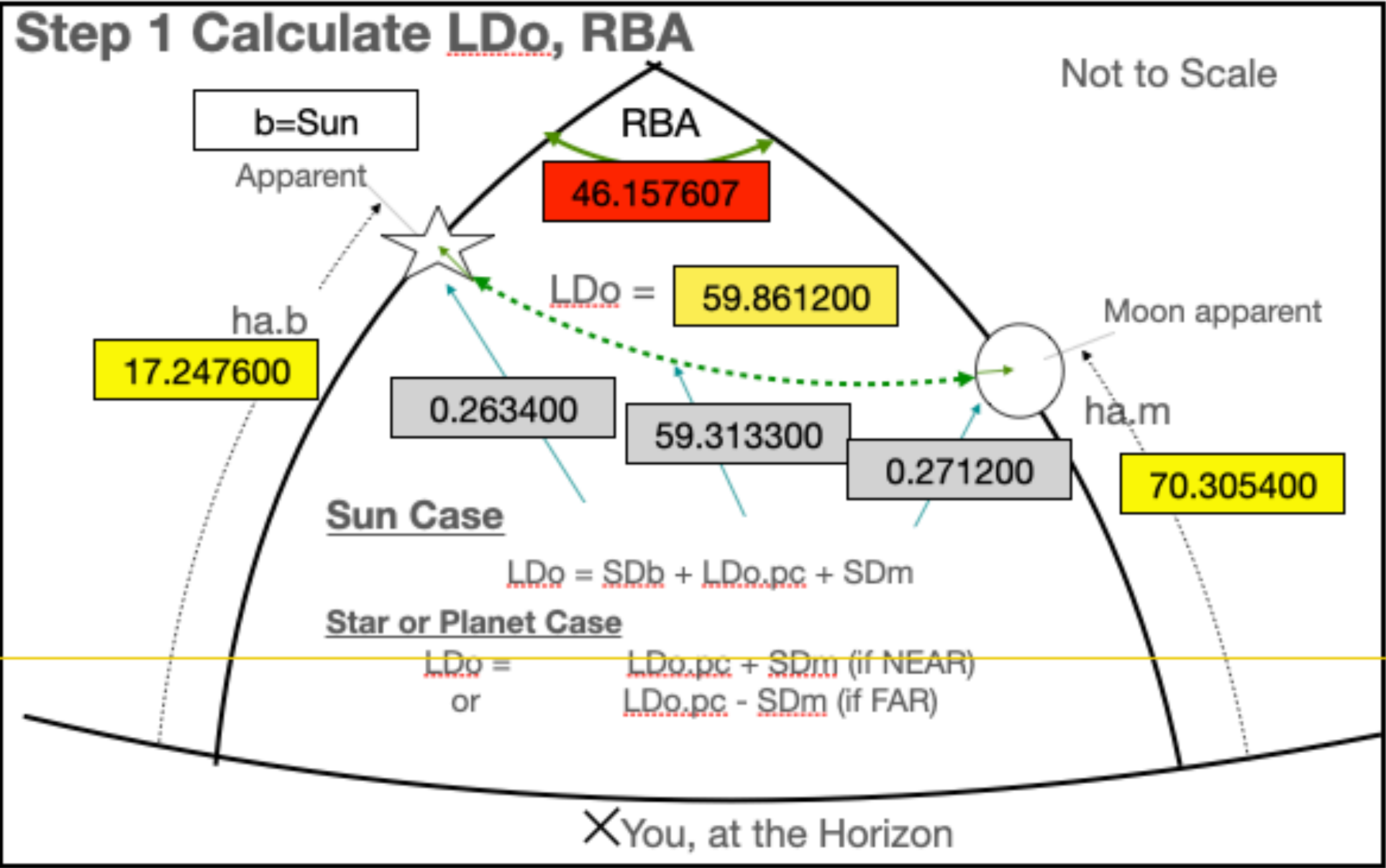
IE (Index Error): -0.8'

Temperature: 17 in °C (63° F)

Pressure: 1013 in mb (29.92" Hg)

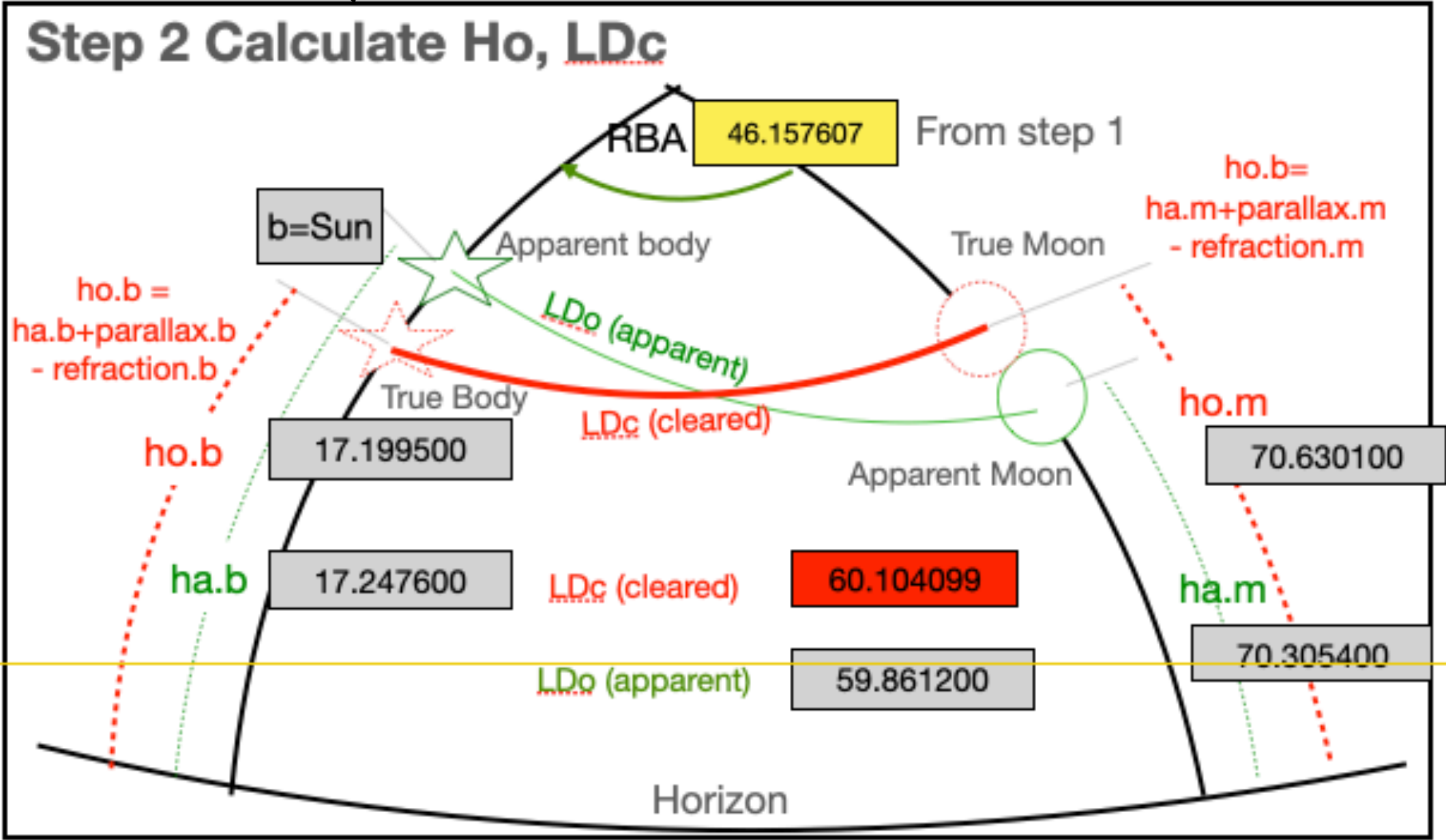
14. Theory and Logic - Direct Triangle Method as reference

Calculations for Lunars is available to view in the Calculations form.



Step 1 illustrates LDo apparent and Ha altitude sights

Step 2 illustrates Ho and LD Cleared of the effect of Parallax and Refraction. Center of the bodies to Center of the Earth



Version 2.4.66, the Interpolation method to derive UTC was used. And then Longitude calculated.

Derived UTC (v 2.4.66)

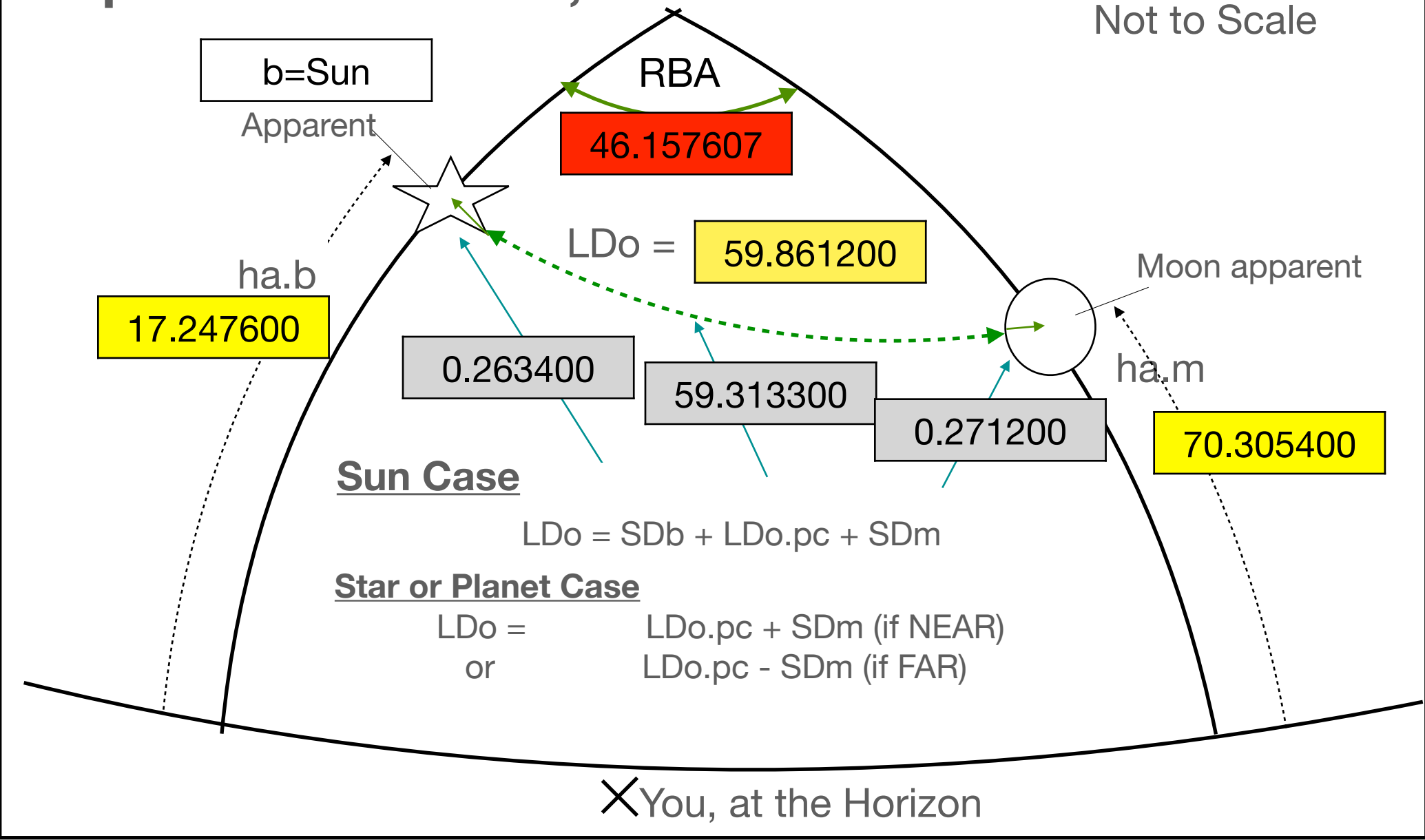
UTC	LDc °
10:28:00	60.9026
11:56:29	60.1041
13:28:00	59.2784

Version 2.8.x, simultaneous - sight result obtained by jointly solving for UTC and observer position on the WGS84 ellipsoid, using the lunar distance cleared as the reference

14. Theory and Logic - Direct Triangle Method as reference

Step 1- Calculate LDo and Relative Bearing Angle

Step 1 Calculate LDo, RBA



$$\cos(LDo) = \sin(ha.m) * \sin(ha.b) + \cos(ha.m) * \cos(ha.b) * \cos(RBA)$$

Or

$$\cos(RBA) = \cos(LDo) / \cos(ha.m) / \cos(ha.b) - \tan(ha.m) * \tan(ha.b)$$

$$\cos(RBA) = 0.502096 / 0.337007 / 0.955032 - 2.793721 * 0.310462$$

$$\cos(RBA) = 0.692677 \quad RBA = 46.157607 \text{ degrees}$$

The Celestial Nav's Calculate Form Shows the calculation

These calculations are shown in the Calculate Form

RBA Bearing Angle between the Moon & body (aka DZ)
LDo.pc LDo pre-cleared. hs Lunar Distance°
Apparent Altitude Corrected
ha.b* Body apparent Alt = hs.b +/- IC - dip +/- limb.b
ha.m* Moon apparent Alt= hs.m +/- IC - dip +/- limb**.m
**Limb incl augmentation
LDo If Body = Sun AND Limb="NEAR"
then LDo.pc +/- IC + SDm + SDb
If Body = star or planet & Body = NEAR
then LDo.pc +/- IC + SDm
If Body = star or planet & Body = FAR
Then LDo.pc +/- IC- SDm
SDm Semi-Diameter of Moon. Positive number.
SDb Semi-Diameter of Sun. Positive number
SD for stars and planets is 0.

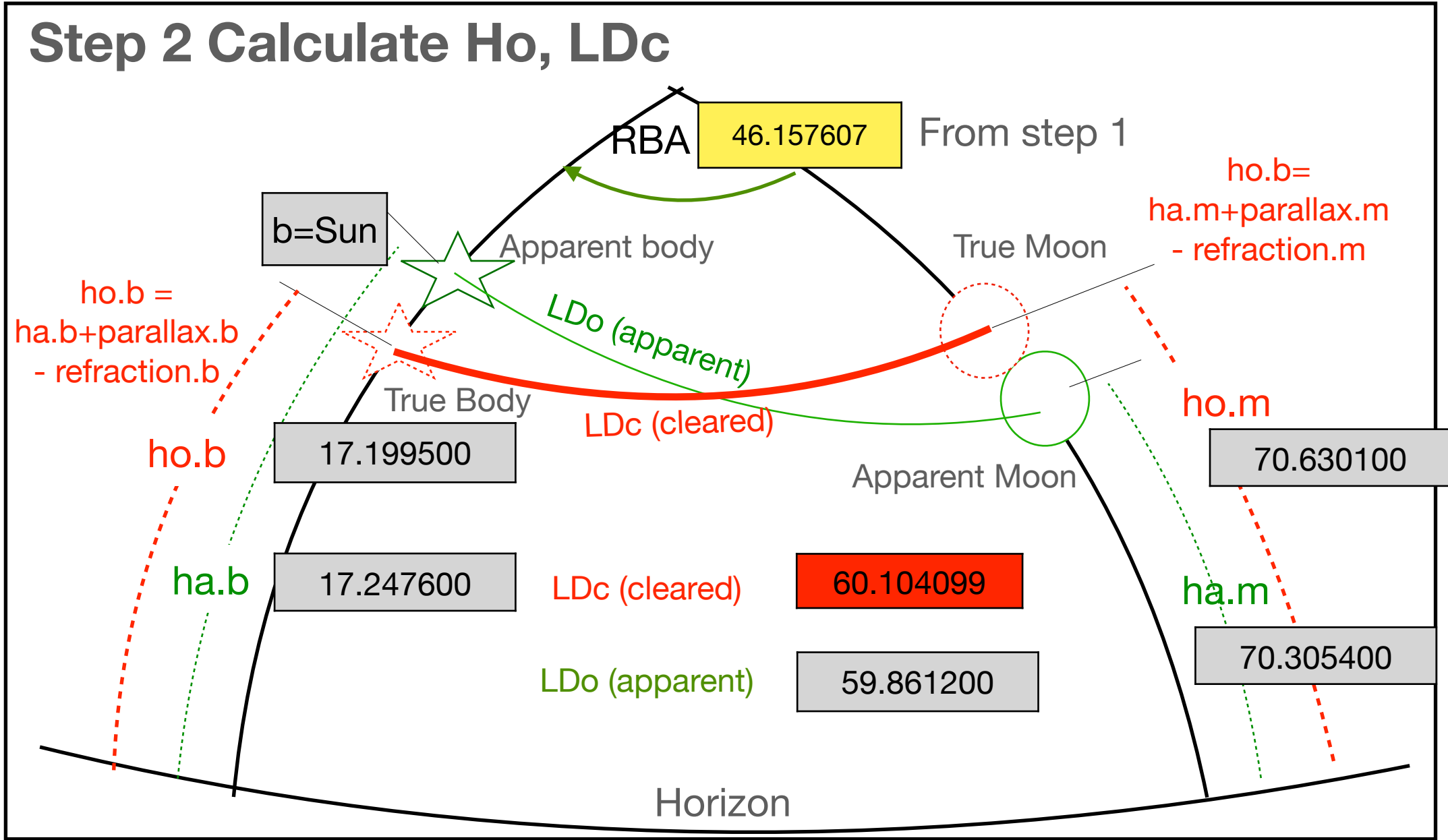
Notes:
* Refraction and Parallax/augmentation is not part of RBA in step 1
** SD for LDo calculation of planets assumed to be 0

INPUTS

	LDo	ha.m	ha.b
			b=Sun
Limb	NEAR	LL	LL
LDO.pc, hs	59.313300	70.066700	17.016700
IC	0.013300	0.013300	0.013300
- dip		-0.045800	-0.045800
Limb (SD)		0.271200	0.263400
LD SDm	0.271200		
LD SDb	0.263400		
LDo/ha	59.861200	70.305400	17.247600
Radians	1.044775	1.227061	0.301027
Cos	0.502096	0.337007	0.955032
Tan	1.722401	2.793721	0.310462

14. Theory and Logic - Direct Triangle Method as reference

Step 2- Calculate LD Cleared using Law of Cosines



RBA	Relative Bearing Angle solved in Step 1
LDo-pc	Pre-cleared Lunar Distance Sextant measure
LDo	Lunar Distance observed
ha.b	Body apparent altitude= hb +/-IC -dip +/-limb
ha.m	Moon apparent altitude= hm +/- IC - dip +/- limb
ho.b	ha.b + parallax.b - refraction.b
ho.m	ha.m + parallax.m* - refraction.m
LDc	Cleared Lunar Distance
* Note: parallax for Moon includes augmentation as an additional altitude correction.	

	LDo	h.m	h.b
Body			b=Sun
Limb	NEAR	LL	LL
LDo.pc/hs	59.313300	70.066700	17.016700
IC	0.013300	0.013300	0.013300
- dip		-0.045800	-0.045800
Limb		0.271200	0.263400
LD Limb.m	0.271200		
LD Limb.b	0.263400		
LDo/ha	59.861200	70.305400	17.247600
Radians	1.044775	1.227061	0.301027
Cos	0.502096	0.337007	0.955032
Tan	1.722401	2.793721	0.310462
Refraction		-0.005600	-0.050400
Parallax		0.330300	0.002300
dh		0.324700	-0.048100
ho		70.630100	17.199500
Radians		1.232728	0.300188
Cos		0.331666	0.955281
Sin		0.943397	0.295700

Recap Step 1

Cos (RBA) 0.502096 / 0.337007 / 0.955032 - 2.793721 X 0.310462

Cos (RBA) 0.6926770 RBA= 46.157607

Step 2 Calculate LD

Cos (LDo) = sin (ho.m) * sin (ho.b) + cos(ho.m) * cos(ho.b) * cos(RBA)

Cos (LDc) = 0.943397 X 0.295700 + 0.331666 X 0.955281 X 0.692677

Cos (LDc) = 0.498426

LDc = 60.104099 Degrees 60 ° 6.2459 ‘

14. Theory and Logic - Direct Triangle Method as reference

Step 1 and 2

Cos (LDo) = sin (ha.m) * sin (ha.b) + cos(ha.m)*cos(ha.b)*cos(RBA)

Or

Cos (RBA) = Cos (LDo)/Cos (ha.m)/cos(ha.b) - tan (ha.m)*tan (ha.b)

Cos (RBA) = 0.502096 / 0.337007 / 0.955032 - 2.793721 * 0.310462

Cos (RBA) = 0.692677

RBA = 46.157607

degrees

The Celestial Nav's Calculate Form Shows the calculation

Recap Step 1

Cos (RBA) 0.502096 / 0.337007 / 0.955032 - 2.793721 X 0.310462

Cos (RBA) 0.6926770

RBA= 46.157607

Step 2 Calculate LD

Cos (LDo) = sin (ho.m) * sin (ho.b) + cos(ho.m) * cos(ho.b) * cos(RBA)

Cos (LDc) = 0.943397 X 0.295700 + 0.331666 X 0.955281 X 0.692677

Cos (LDc) = 0.498426

LDc = 60.104099

Degrees

60 ° 6.2459 '

Calculations Form show the Direct Triangle Method Calculations

Lunar-distance UTC recovery

Results

Calculations

Auditable WGS84 solution with a Direct Triangle comparison. This page shows the observation reductions, formula substitutions, ephemeris scan, root refinement, position intersections and warnings used to produce Results.

Sun: HE = 0.00291201, PAIC = 0.00230414 deg
Ho.m = Htopo.m + Palt.m = 70.63017600 deg
Ho.b = Htopo.b + Palt.b = 17.20028831 deg

E. DIRECT SPHERICAL-TRIANGLE CLEARING

cos (DZ) = [cos (LDo) - sin (Ha.m) sin (Ha.b)] / [cos (Ha.m) cos (Ha.b)]
cos (DZ) = [cos (59.86122881) - sin (70.30543830) sin (17.24758772)] / [cos (70.30543830) cos (17.24758772)]
cos (DZ) = 0.692677298914
DZ = acos[cos (DZ)] = 46.15758428 deg

cos (LDc) = sin (Ho.m) sin (Ho.b) + cos (Ho.m) cos (Ho.b) cos (DZ)
cos (LDc) = sin (70.63017600) sin (17.20028831) + cos (70.63017600) cos (17.20028831) (0.692677298914)
cos (LDc) = 0.498436576787
LDc = acos[cos (LDc)] = 60.10338145 deg (60° 06.2029')

*** Note RBA (Relative Bearing Angle) is same as DZ

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RMB 21 Sept, 2026

14 Calculations Form show the calculations for UTC and Position.

Version 2.8.x

```
F. UTC/POSITION MATCHING TRAIL
-----
Residual F(t) = forward-modelled raw LD(t) - observed raw LD.
Closest coarse evaluation: offset -90.000 s, residual +0.047744 arcmin.

COARSE SCAN (every evaluated point)
step  branch  offset(s)  model(deg)  target(deg)  residual(')  UTC
0      1      -2700.000  59.71379325  59.31333333  +24.027595  2025-08-18 11:13:00.000
0      2      -2700.000  59.71402807  59.31333333  +24.041684  2025-08-18 11:13:00.000
1      1      -2670.000  59.70919900  59.31333333  +23.751940  2025-08-18 11:13:30.000
1      2      -2670.000  59.70943377  59.31333333  +23.766026  2025-08-18 11:13:30.000
2      1      -2640.000  59.70460477  59.31333333  +23.476286  2025-08-18 11:14:00.000
2      2      -2640.000  59.70483947  59.31333333  +23.490368  2025-08-18 11:14:00.000
3      1      -2610.000  59.70001054  59.31333333  +23.200632  2025-08-18 11:14:30.000
3      2      -2610.000  59.70024519  59.31333333  +23.214711  2025-08-18 11:14:30.000
4      1      -2580.000  59.69541632  59.31333333  +22.924979  2025-08-18 11:15:00.000
4      2      -2580.000  59.69565091  59.31333333  +22.939055  2025-08-18 11:15:00.000
5      1      -2550.000  59.69082212  59.31333333  +22.649327  2025-08-18 11:15:30.000
5      2      -2550.000  59.69105665  59.31333333  +22.663399  2025-08-18 11:15:30.000
6      1      -2520.000  59.68622792  59.31333333  +22.373675  2025-08-18 11:16:00.000
6      2      -2520.000  59.68646239  59.31333333  +22.387743  2025-08-18 11:16:00.000
...
173    1      +2490.000  58.91912852  59.31333333  -23.652289  2025-08-18 12:39:30.000
173    2      +2490.000  58.91935289  59.31333333  -23.638826  2025-08-18 12:39:30.000
174    1      +2520.000  58.91453590  59.31333333  -23.927846  2025-08-18 12:40:00.000
174    2      +2520.000  58.91476021  59.31333333  -23.914388  2025-08-18 12:40:00.000
175    1      +2550.000  58.90994328  59.31333333  -24.203403  2025-08-18 12:40:30.000
175    2      +2550.000  58.91016753  59.31333333  -24.189948  2025-08-18 12:40:30.000
176    1      +2580.000  58.90535068  59.31333333  -24.478959  2025-08-18 12:41:00.000
176    2      +2580.000  58.90557487  59.31333333  -24.465508  2025-08-18 12:41:00.000
177    1      +2610.000  58.90075809  59.31333333  -24.754515  2025-08-18 12:41:30.000
177    2      +2610.000  58.90098221  59.31333333  -24.741067  2025-08-18 12:41:30.000
178    1      +2640.000  58.89616550  59.31333333  -25.030070  2025-08-18 12:42:00.000
178    2      +2640.000  58.89638956  59.31333333  -25.016626  2025-08-18 12:42:00.000
179    1      +2670.000  58.89157293  59.31333333  -25.305624  2025-08-18 12:42:30.000
179    2      +2670.000  58.89179692  59.31333333  -25.292185  2025-08-18 12:42:30.000
180    1      +2700.000  58.88698036  59.31333333  -25.581178  2025-08-18 12:43:00.000
180    2      +2700.000  58.88720429  59.31333333  -25.567742  2025-08-18 12:43:00.000

ROOT REFINEMENT (every bisection evaluation)
bracket iter  left(s)  right(s)  midpoint(s)  model(deg)  target(deg)  residual(')
1      1      -90.000  -60.000  -75.000  59.31183235  59.31333333  -0.090059
1      2      -90.000  -75.000  -82.500  59.31298071  59.31333333  -0.021158
1      3      -90.000  -82.500  -86.250  59.31355489  59.31333333  +0.013293
1      4      -86.250  -82.500  -84.375  59.31326779  59.31333333  -0.003932
1      5      -86.250  -84.375  -85.312  59.31341141  59.31333333  +0.004685
1      6      -85.312  -84.375  -84.844  59.31333960  59.31333333  +0.000376
1      7      -84.844  -84.375  -84.609  59.31330362  59.31333333  -0.001783
1      8      -84.844  -84.609  -84.727  59.31332169  59.31333333  -0.000699
1      9      -84.844  -84.727  -84.785  59.31333057  59.31333333  -0.000166
1     10      -84.844  -84.785  -84.814  59.31333501  59.31333333  +0.000101
2      1      -90.000  -60.000  -75.000  59.31206197  59.31333333  -0.076282
2      2      -90.000  -75.000  -82.500  59.31321034  59.31333333  -0.007379
2      3      -90.000  -82.500  -86.250  59.31378453  59.31333333  +0.027072
2      4      -86.250  -82.500  -84.375  59.31349744  59.31333333  +0.009846
2      5      -84.375  -82.500  -83.438  59.31335397  59.31333333  +0.001238
2      6      -83.438  -82.500  -82.969  59.31328216  59.31333333  -0.003071
2      7      -83.438  -82.969  -83.203  59.31331798  59.31333333  -0.000921
2      8      -83.438  -83.203  -83.320  59.31333590  59.31333333  +0.000154
2      9      -83.320  -83.203  -83.262  59.31332702  59.31333333  -0.000379
2     10      -83.320  -83.262  -83.291  59.31333146  59.31333333  -0.000113

CONVERGED ROOT EVALUATIONS
root  offset(s)  final bracket(s)  model(deg)  target(deg)  residual(')
1      -84.800  [-84.814, -84.785]  59.31333287  59.31333333  -0.000028
2      -83.306  [-83.320, -83.291]  59.31333375  59.31333333  +0.000025
```

```
G. UTC CANDIDATES AND UNCERTAINTY

Matching UTC candidates:
* 2025-08-18 11:56:35.200
  watch correction      = -84.800 s
  LD cleared (matched geocentric distance) = 60.10296746 deg
  local residual rate   = -33.072700 arcmin/hour (-0.551212 arcmin/minute)
  input angle sigma RSS = 0.734847 arcmin
  covariance sigma UTC  = 21.804 s
  covariance sigma pos   = 4.053 NM
2025-08-18 11:56:36.694
  watch correction      = -83.306 s
  LD cleared (matched geocentric distance) = 60.10274277 deg
  local residual rate   = -33.073131 arcmin/hour (-0.551219 arcmin/minute)
  input angle sigma RSS = 0.734847 arcmin
  covariance sigma UTC  = 21.804 s
  covariance sigma pos   = 5.401 NM
Warning: More than one joint time/position solution exists; use DR, hemisphere, another altitude or a second

H. POSITION INTERSECTION AND WARNINGS

The joint solver intersects the Moon and body altitude constraints at their individual corrected UTCs, advancing DR used to rank branches: 43° 16.1000' N, 076° 58.8002' W
Position candidate 1: 43.314367°, -76.616249° (nearest saved DR)
Selected branch true azimuths: Moon Zn 134.45°; Sun Zn 88.29°.
Azimuth separation (delta Zn): 46.16°; effective COP crossing: 46.16°.
Notice: moderate position geometry; uncertainty is amplified along one direction.
Estimated joint position uncertainty: 4.05 NM (1-sigma; local linear model).
```

The recovered correction is a constant watch offset. It may be applied to other sightings recorded while the

**If you have a Sextant available,
Give it a try the next chance you get.**

Thank you